SkyRoof v.1.51

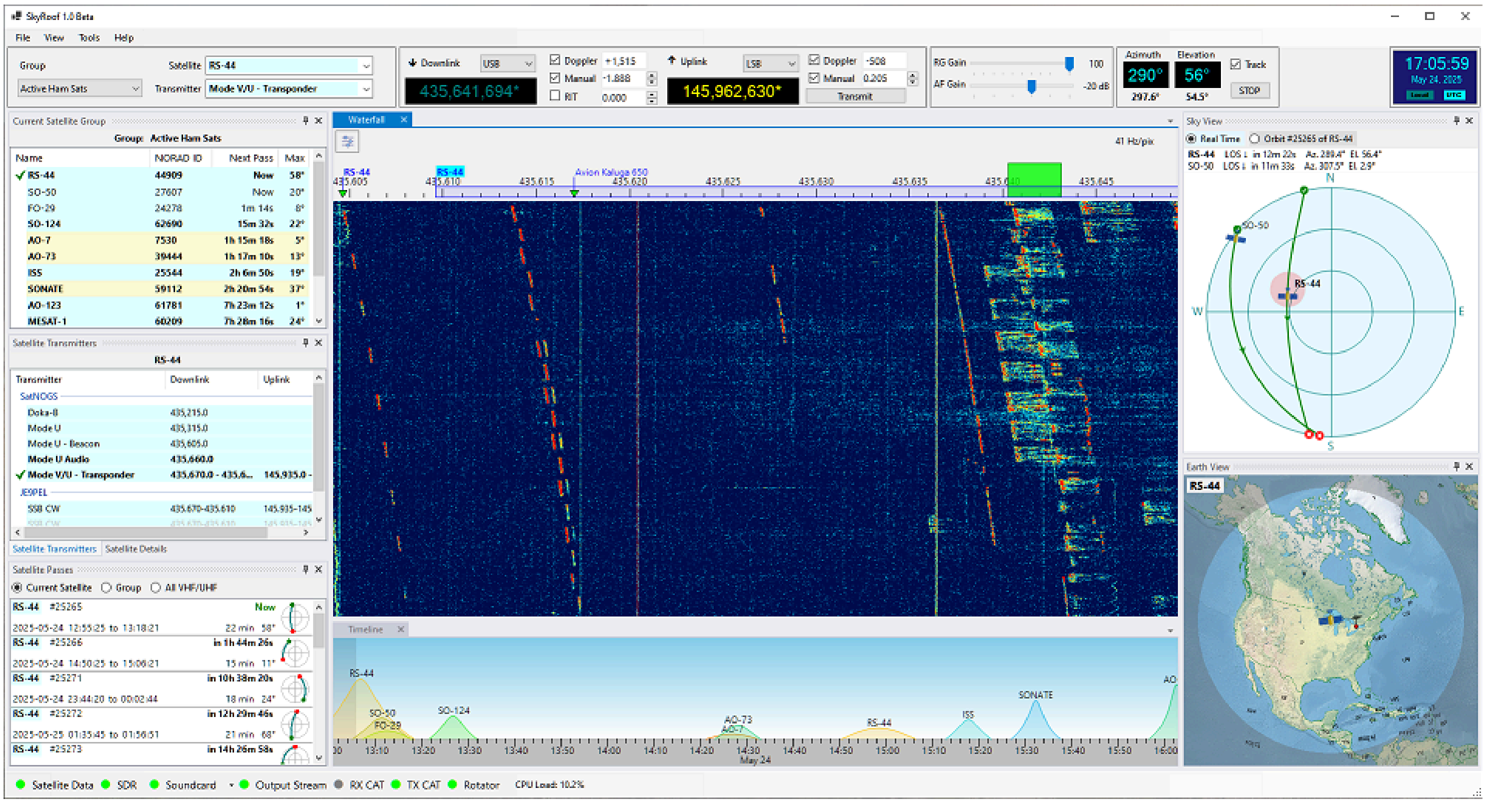
software for Hams and satellite enthusiasts

User's Guide

Table of Contents

Overview

SkyRoof is an open source, 64-bit Windows application for Hams and satellite enthusiasts, available on the terms of the GPL v.3 license. It combines satellite tracking and SDR functions in one program, which opens some interesting possibilities. For example, all satellite traces on the waterfall are labeled with satellite names, the boundaries of the transponder segments follow the Doppler shift, and all frequency tuning is done visually, with a mouse.



Features

The main features of SkyRoof are:

- detailed information about all satellites that transmit in the Ham bands;
- satellite tracking in real time;
- pass prediction for the selected satellites;
- visual representation of the current satellite position and future passes, using:
  - Sky View - the view of the sky from your location;
  - Earth View - the view of the Earth from the satellite;
  - Time Line - the satellite passes on the time scale;
  - Pass List - the details of the predicted passes;
- SDR-based waterfall display that covers the whole satellite segments on the VHF and UHF bands, with zoom and pan;
- SDR-based SSB/CW/FM receiver with RIT and Doppler tracking;

- audio and I/Q output to external programs via VAC or UDP;
- frequency scale with satellite names and transponder segments, Doppler-corrected;
- built-in telemetry decoder for FSK, AFSK, GFSK, GMSK and BPSK downlinks, with automatic frame submission to the SatNOGS network;
- built-in SSTV image decoder for the Robot and PD modes, with no external program required;
- built-in SSDV image decoder that rebuilds the pictures sent as data in the telemetry stream, and can combine the receptions of several passes into one picture;
- decoding and playback of the Codec2 voice messages that satellites transmit in their telemetry frames;
- FT4 Console for making FT4 contacts through linear-transponder satellites, without an external program;
- QSO logging with ADIF export, and a QSO Scheduler;
- audio recording of satellite passes;
- CAT control of an external transceiver;
- antenna rotator control.

The program can work without an SDR, or even without any radio at all, but many useful functions are not available in this mode.

System Requirements

Hardware

- **Computer**: 64-bit PC. 3-GHz Quad-core CPU is recommended;
- **Video Card**: OpenGL 3.3 or higher, 512 Mb of texture memory;
- **Monitor**: screen resolution 1900x1280 or higher, 4K recommended;
- **Internet**: required, to download satellite data;
- **SDR**: optional, but highly recommended. The drivers included in the setup support:
  - Airspy;
  - AirspyHF+;
  - SDRplay;
  - RTL-SDR;
  - HackRF;

  Other radios can be used if a SoapySDR driver module is available for them, see Setting Up SDR;
- **Transceiver**: optional. Currently the command definition files are available for:
  - FT-817;
  - FT-818;
  - FT-847;
  - FT-991a;
  - IC-705;
  - IC-706MKIIG;
  - IC-915;
  - IC-910H;
  - IC-9700;
  - TS-2000;

  Many other radios are supported via HamLib;
- **Antenna rotator** - optional, any rotator supported by HamLib.

Software

- **OS**: Windows 10 or Windows 11, 64-bit only;
- **SkyCAT**: optional, for CAT control;
- **HamLib**: optional, for CAT and rotator control.

Quick Start

Installation

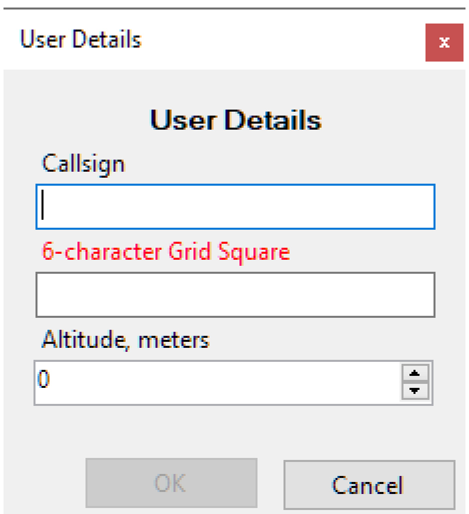
To install SkyRoof, download the installer from the [Download] page, run it, and follow the on-screen instructions.

First Run

When you run SkyRoof for the first time, the program performs several important, but somewhat lengthy steps. Fortunately, they need to be done only once.

User Information Input

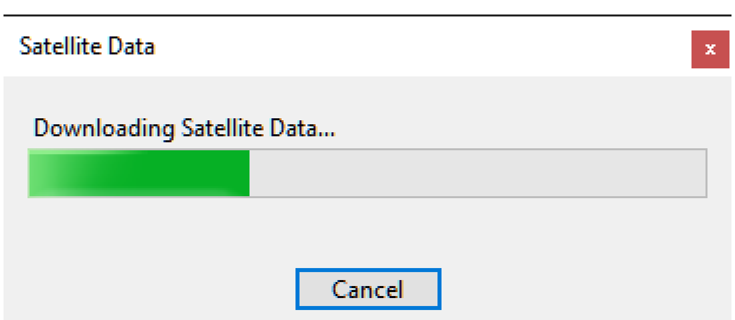
You will be presented with the User Details dialog:



Enter your callsign, 6-character grid square and your altitude above the sea level. The grid square is required, the program cannot proceed without that information. The other two values are optional.

Satellite Data Download

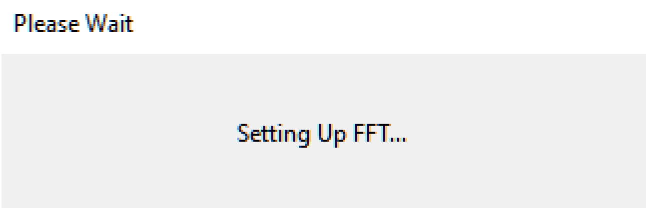
Then SkyRoof downloads the satellite data: make sure that your computer is connected to the Internet.



Wait until the data are downloaded and imported. Again, SkyRoof cannot proceed without this data downloaded at least once, so if you click on Cancel, the program terminates.

FFT Setup

Wait for SkyRoof to try different ways of computing the FFT transform and to find the one that works best on your system. This may take quite some time!

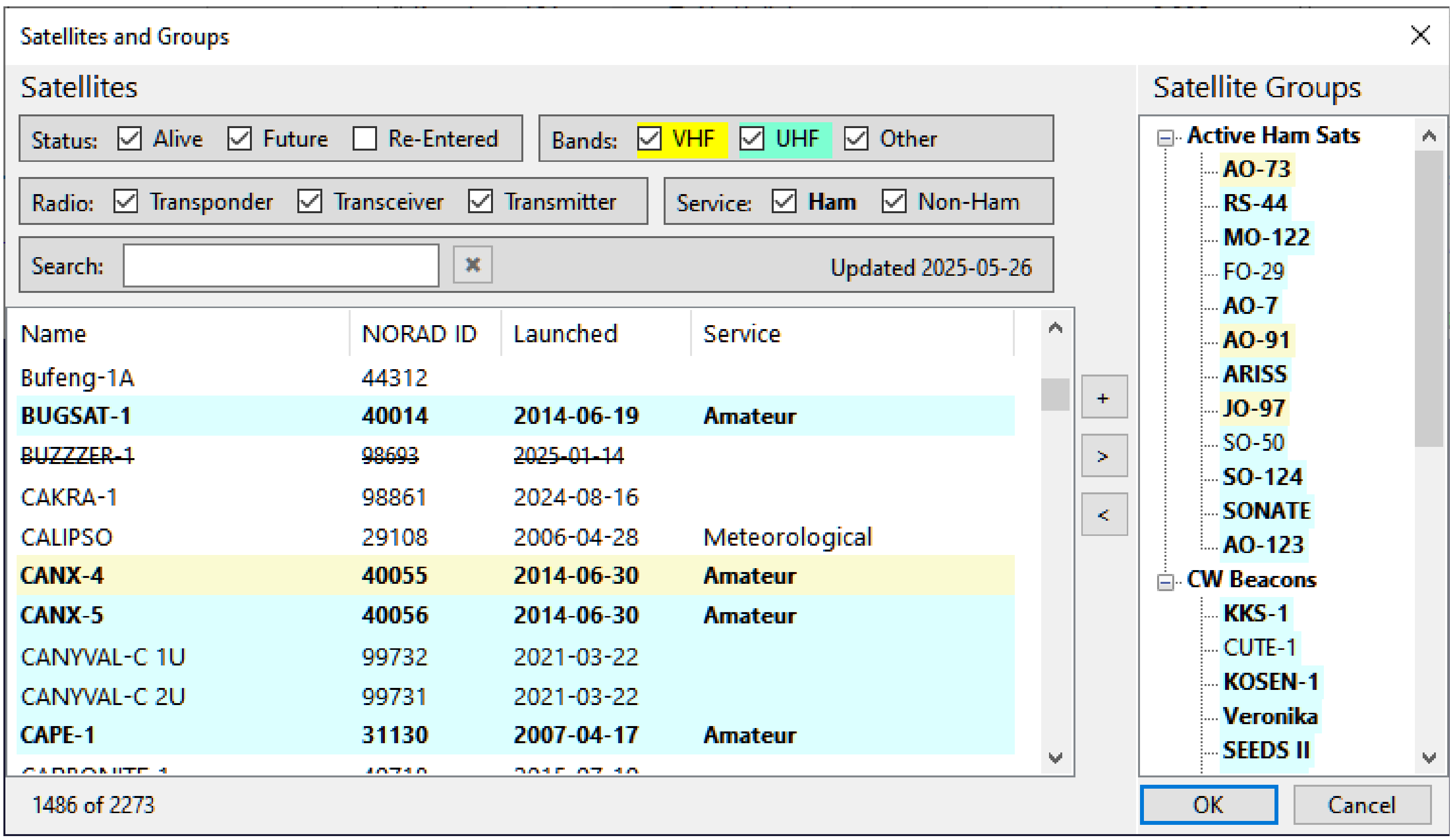


That's all for the quick start! Now you can use the program for tracking the satellites in frequency and space, and for predicting the satellite passes over your location. To do more than that, you have to perform the rest of the setup steps described in the next sections.

Creating Satellite Groups

SkyRoof comes with two pre-defined groups of satellites created for your convenience, **Active Ham Sats** and **CW Beacons**. The first group lists the satellites carrying the linear transponders, FM repeaters or digital systems that were available to Hams at the time of this writing. The second group includes the satellites that send beacon signals or telemetry in Morse Code, or just transmit an unmodulated carrier (a.k.a. Continuous Wave, CW). Most likely, you will want to modify or delete these groups and add your own ones. Here is how to do this.

Click on **Tools / Satellites and Groups** in the main menu to open the **Satellites and Groups window**:



The left panel lists [all satellites](#) known to SkyRoof, the right panel shows the groups.

- to create a group, click on the **[+]** button, then enter the group name;
- to add a satellite to the group, drag it from the satellite list onto the group, or click on the **[>]** button;
- to delete a group, or a satellite from the group, select it in the right panel and press the Delete key, or click on the **[<]** button.
- click on OK to save the changes.

The **Satellites and Groups window** has many commands to filter and search satellites, to rename them, and to view detailed information about the satellites and their transmitters. These commands are described in the Satellites and Groups Window section of this document.

Configuring the Window Layout

The layout of SkyRoof's main window is under your full control. Any panel may be shown or hidden, docked anywhere in the window, or left floating.

Show and Hide

Show the panels using the menu commands in the **View** section, hide them using the same command again, or by clicking on the Close button on panel's caption bar.

Docking

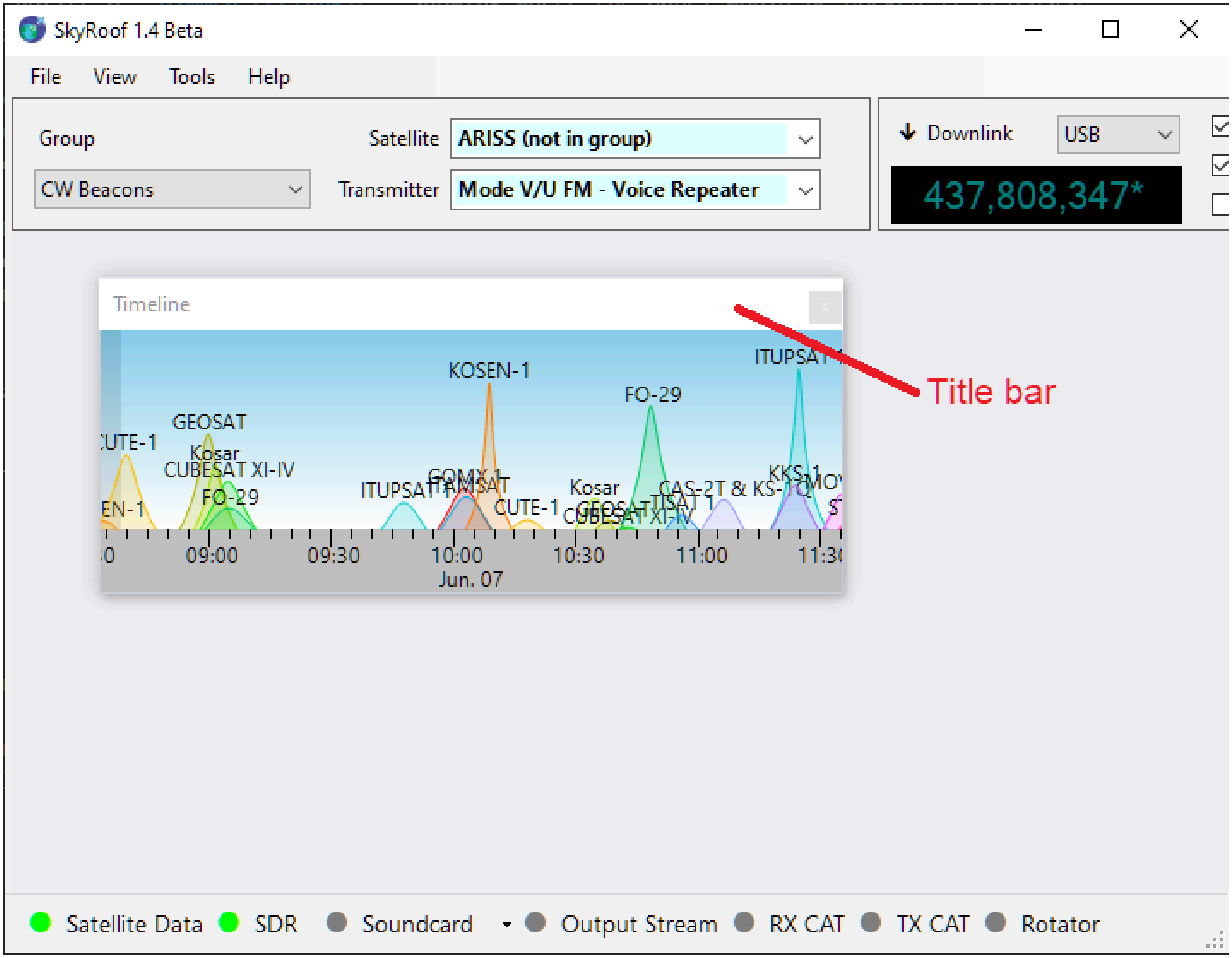
The panels you open are initially in the floating state. This is not very convenient as you cannot move or resize the main window without breaking your panel arrangement. You can dock the panels so that they move and resize when you move/resize the main window.

How to Dock a Floating Panel

The Timeline panel in the screenshot below is floating. Let us dock it to the bottom of the main window.

1. **Find the Panel's Title Bar**

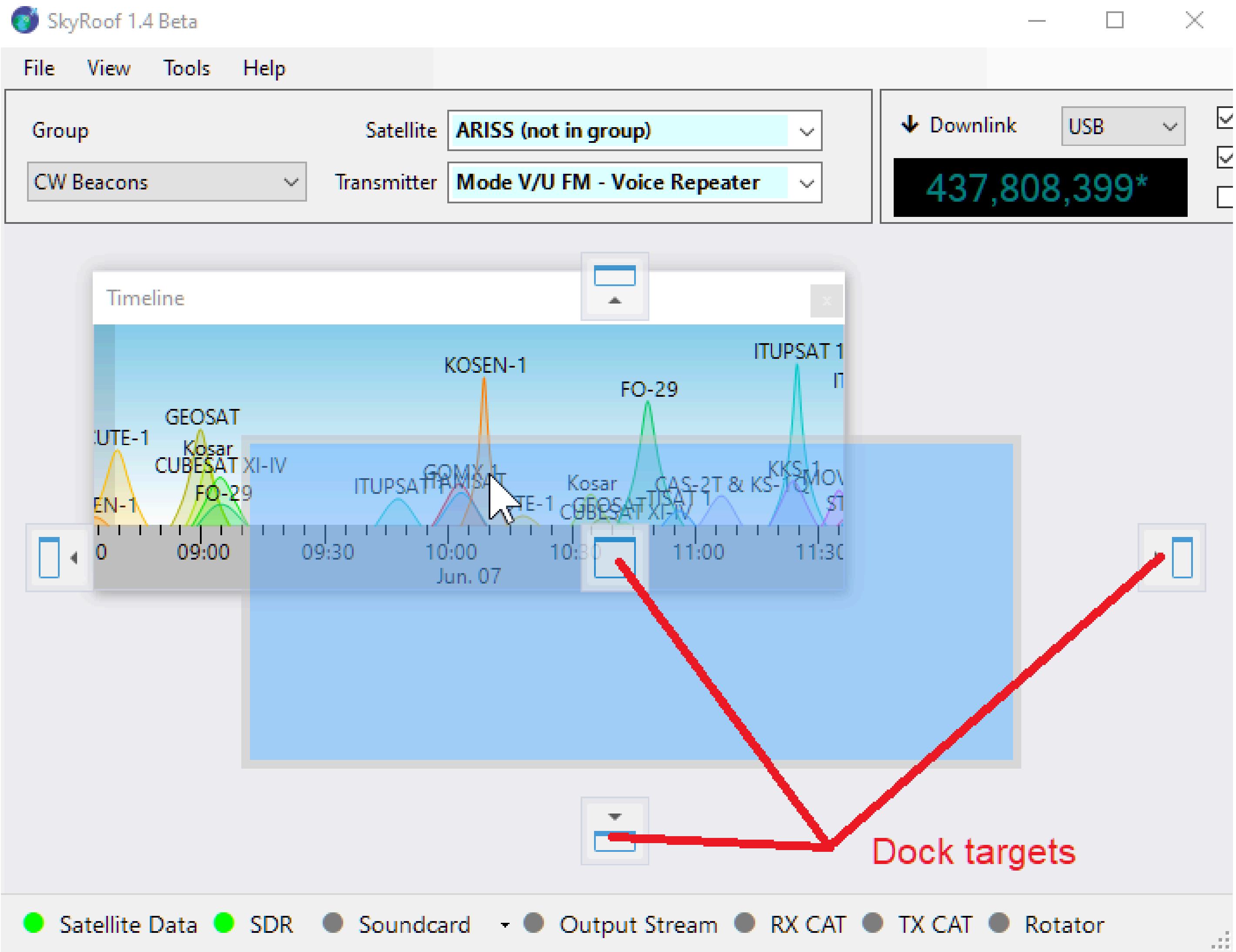
   Locate the top bar of the floating panel labeled "*Timeline*". This is known as the **title bar**.



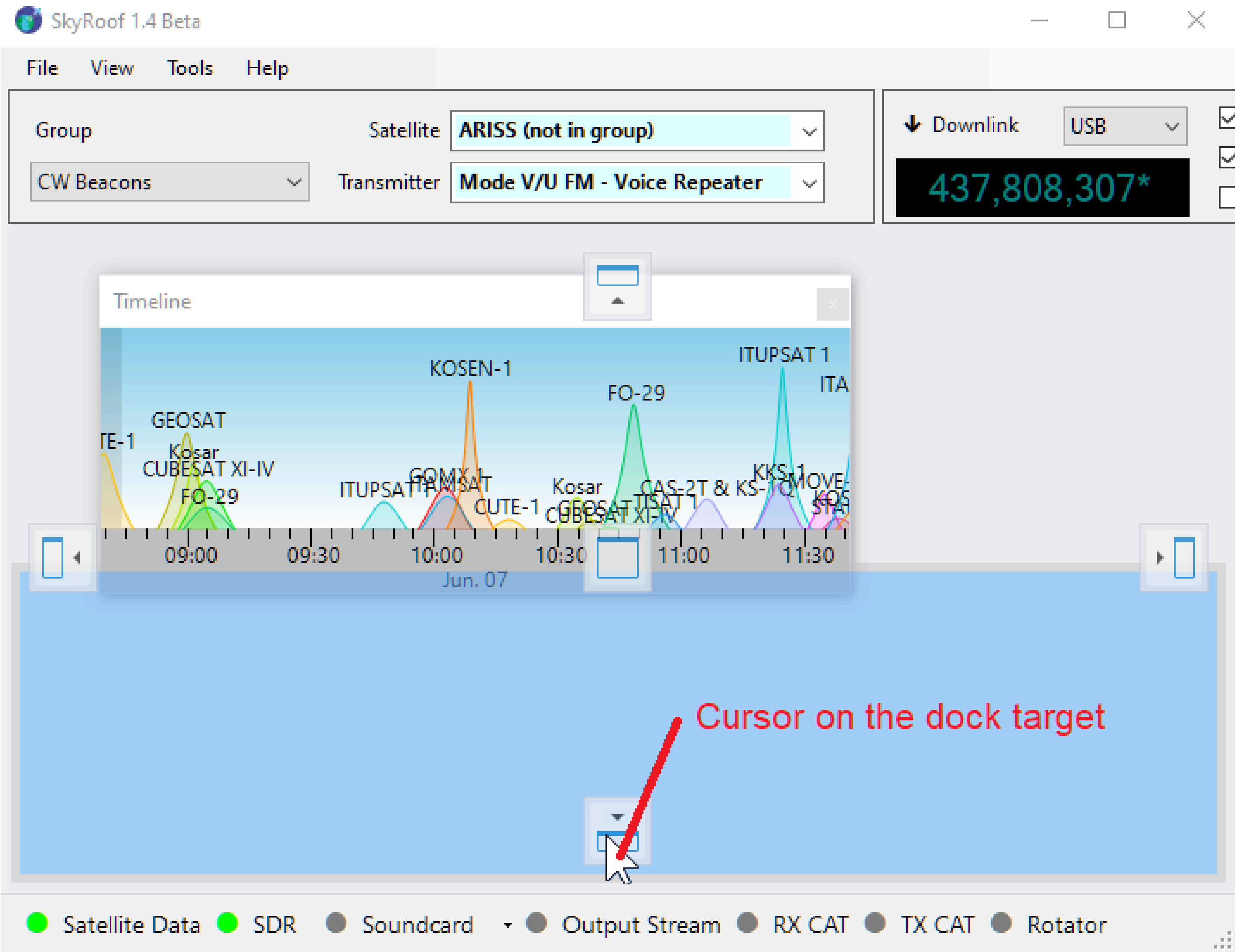
2. **Click and Hold**
   Move your mouse cursor to the title bar and press the **left mouse button**. Keep holding the button down.

3. **Drag the Panel**
   While holding the mouse button, move the panel by dragging it with the mouse. As you begin to drag, you will see **dock target icons** appear in the main window—these icons represent the available docking positions.
   You will also notice a **dimmed rectangle** showing where the panel would be docked if released.

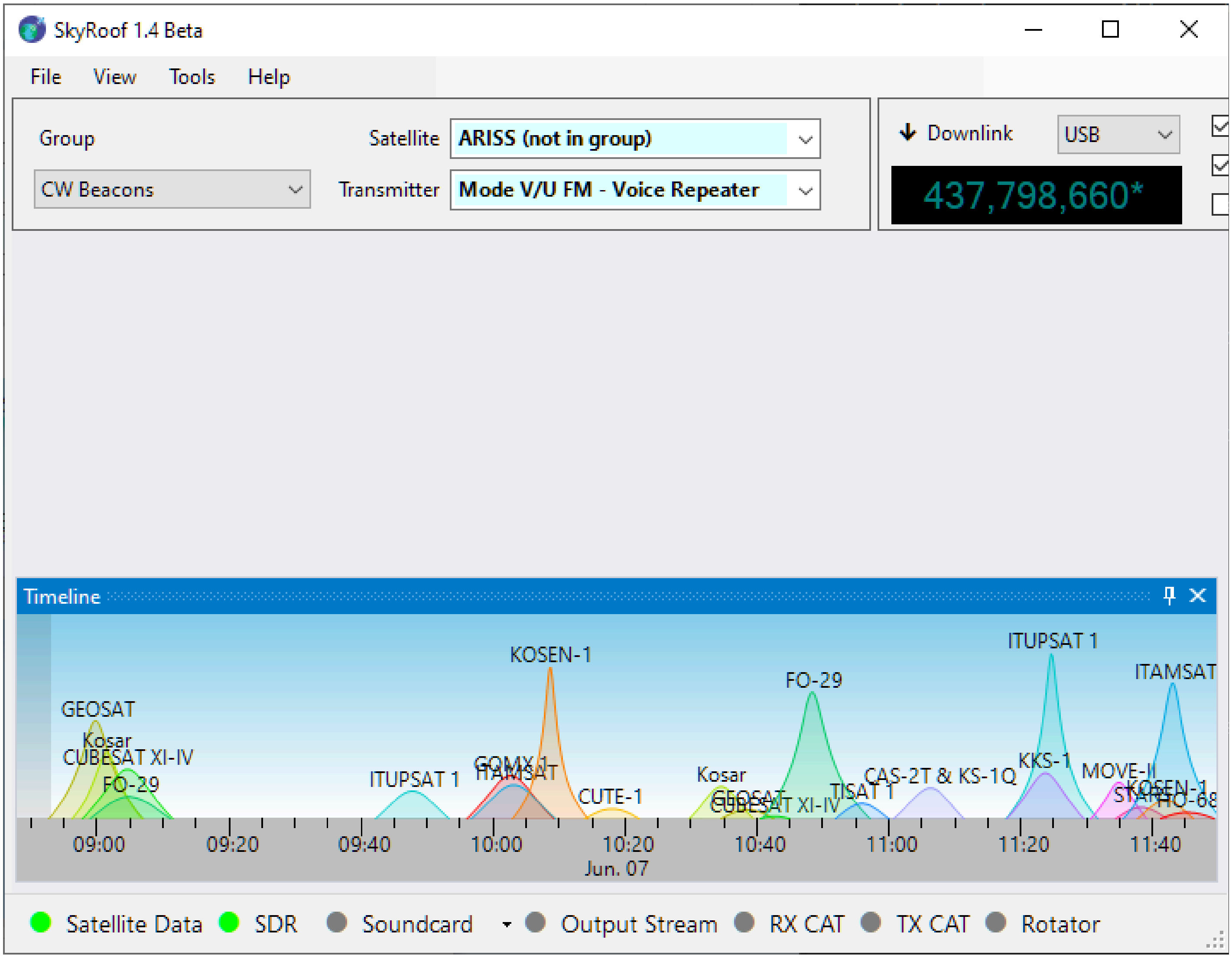
> ⚠️ **Important:** This rectangle is only a visual preview. Do **not** try to align it with the dock target icon. Instead, focus on where your mouse cursor is.



4. **Position the Mouse Cursor Over a Dock Target Icon**
   Move your **mouse cursor** over one of the dock target icons (not the rectangle). The docking location will update automatically. The panel will only dock if the cursor is directly over the icon.

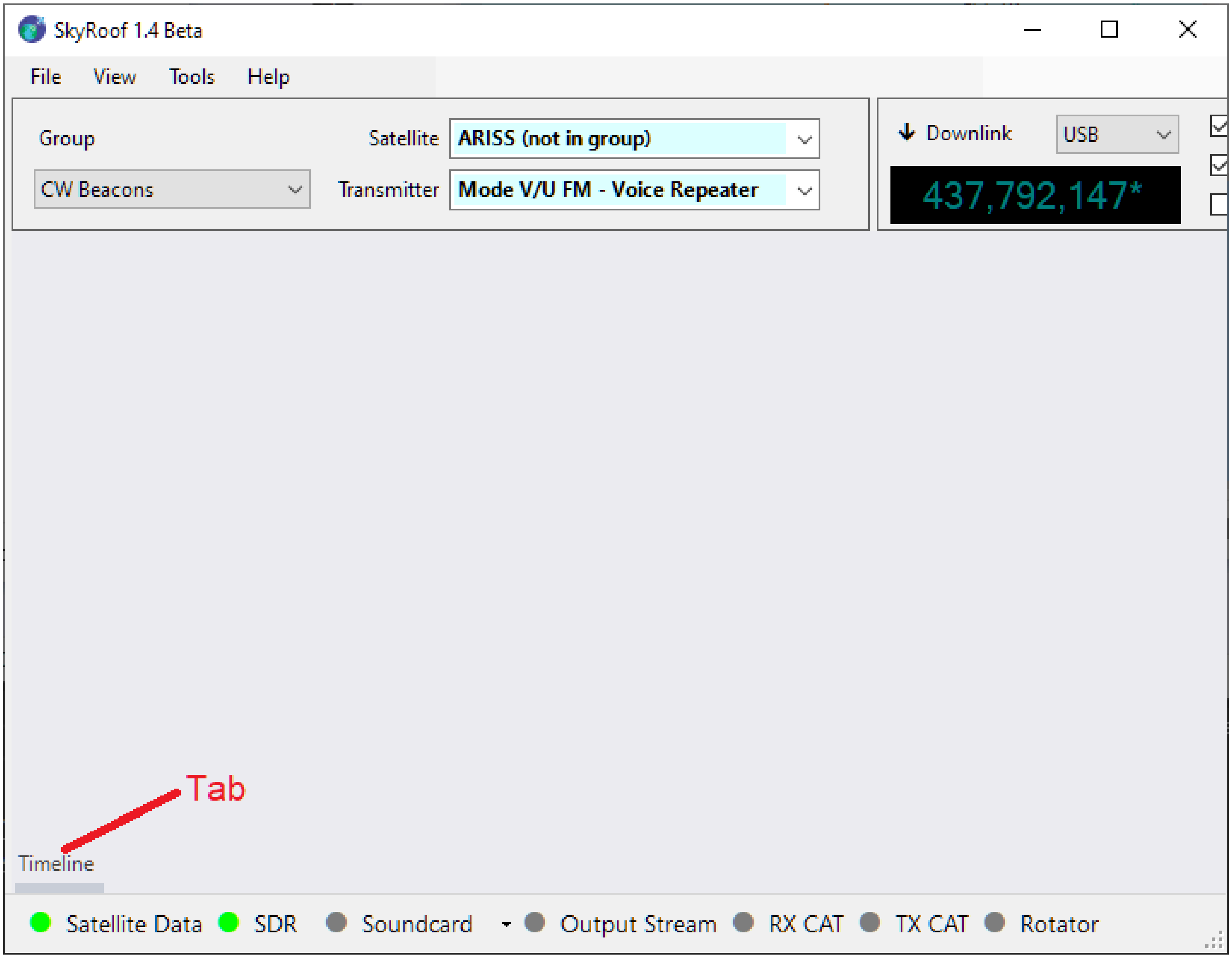


5. **Release to Dock**
   Once the cursor is over your desired dock target icon, **release the mouse button**. The panel will snap into place — either at the side, in the center, or nested inside another panel, depending on the selected icon. Once docked, the panel becomes part of the main window layout, helping keep your workspace clean and organized.



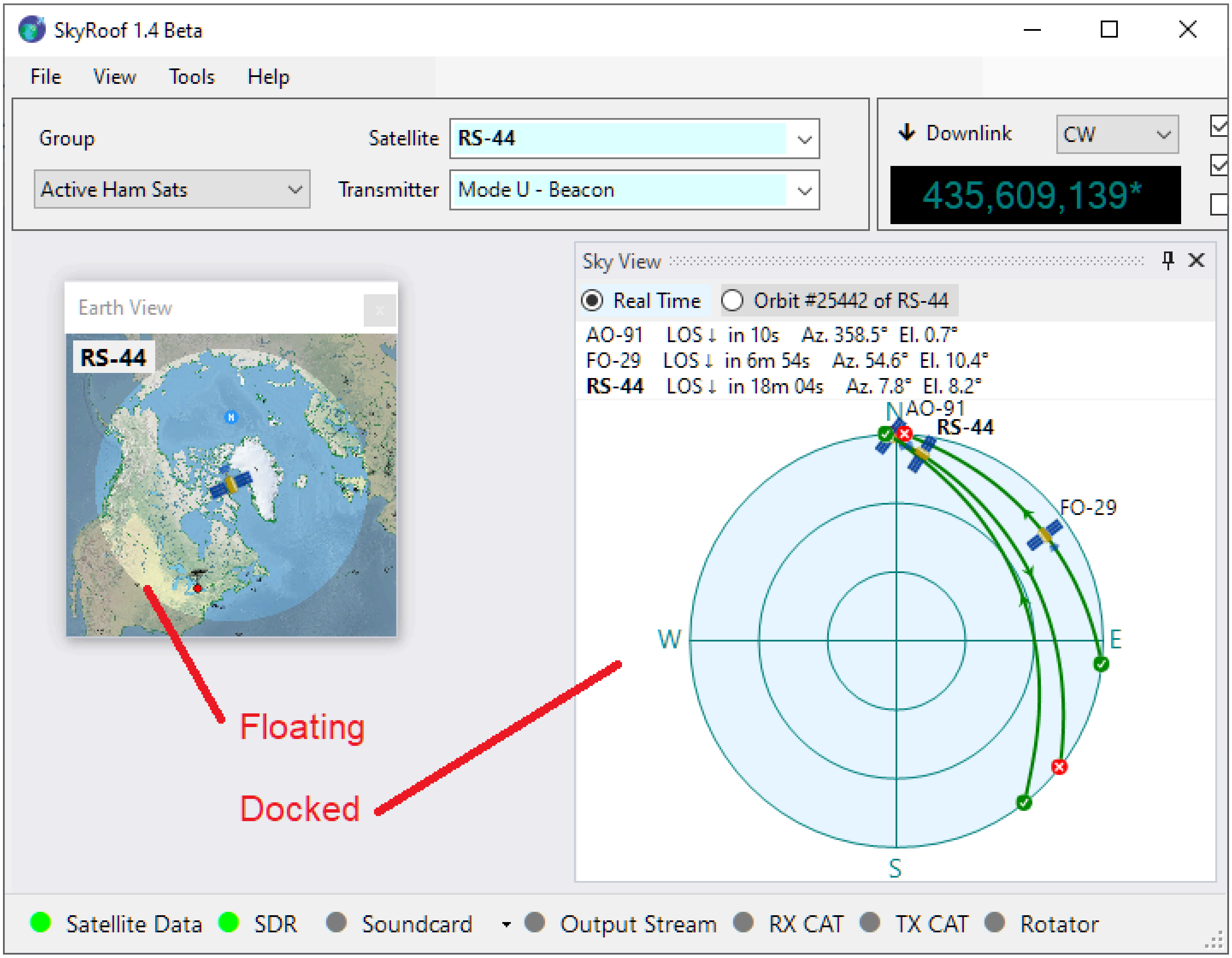
Auto-Hide

A panel can be switched to the **auto-hide mode**, which allows it to stay hidden until you click on its tab. In the screenshot below, notice the **"Timeline" tab** in the bottom-left corner. Clicking this tab temporarily slides the **Timeline** panel into view. To enable auto-hide mode, click the **Auto-Hide** button located on the panel's title bar.

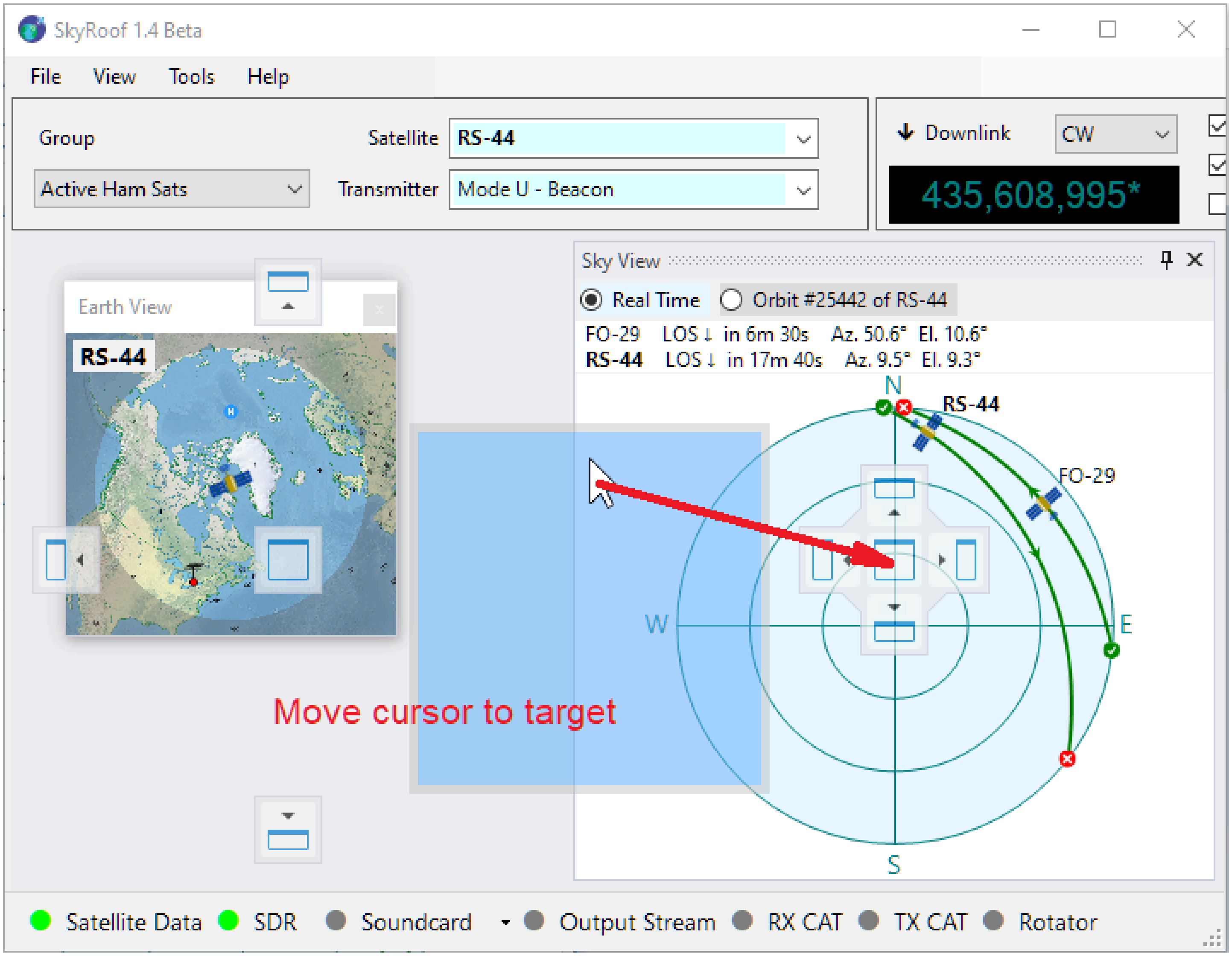


Tabbing

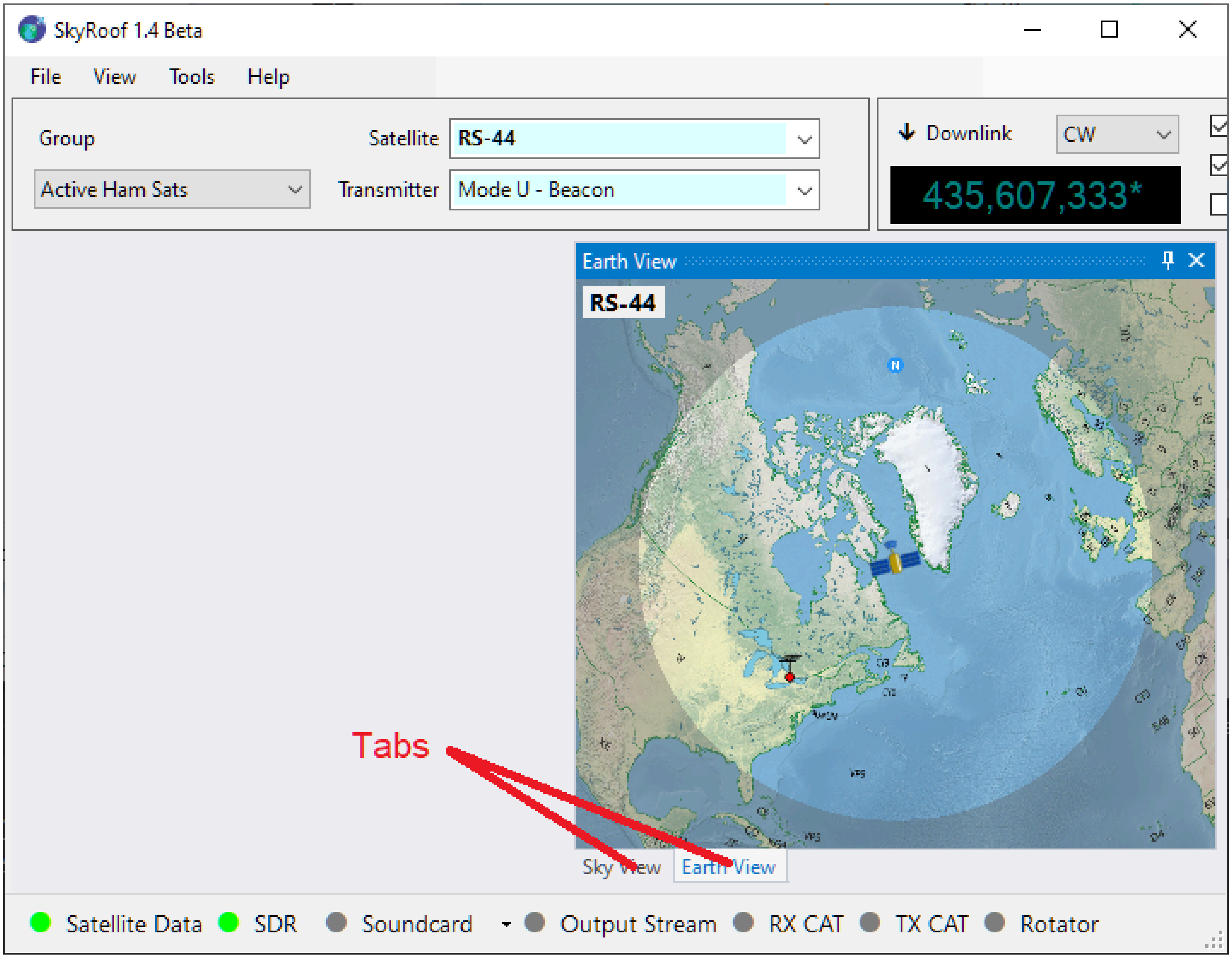
You can also organize panels as **tabs**.



If you drop one panel directly onto another, they will be grouped into a tabbed interface.



In the screenshot below the [Sky View Panel] and [Earth View Panel] have been combined into a tabbed view. You can switch between these panels by clicking on their respective tabs.



Theme

Click on **Tools / Theme** in the main menu to select the color theme. **System** follows the Windows app color setting, while **Light** and **Dark** force the theme regardless of it. The selected theme is applied the next time SkyRoof starts.

Setting Up SDR

Supported Radios

SkyRoof uses the Soapy SDR engine to interface with the SDR radios. The drivers for the following radios are included in the SkyRoof setup:

- Airspy;
- AirspyHF+;
- SDRplay;
- RTL-SDR;
- HackRF;

Radios that are not on this list can also be used, see Adding Support of Other Radios below.

Installing The Drivers

Most of the SDR devices require the driver to be installed before you can start using them. Check the manufacturer's web site, or search on Google, for the driver installation instructions. At the time of this writing, the following instructions were available on the Web: Airspy, RTL-SDR, SDRplay, HackRF, PlutoSDR.

Once you install the drivers and make your radio work with its native software, proceed to the next step.

Adding Support of Other Radios

SkyRoof can work with any SDR that has a SoapySDR driver module, even if the module does not come with SkyRoof. To add such a radio:

- obtain the SoapySDR driver module for your radio. Pre-built modules for Windows are included in PothosSDR, where they are located in the `lib\SoapySDR\modules0.8` folder. The complete list of the SoapySDR drivers, with links to their source code, is available on the SoapySDR wiki. The module must be a 64-bit build for the SoapySDR ABI version 0.8;

- copy the module dll, and any dlls that it depends on, to the `lib\SoapySDR\modules0.8` folder of the SkyRoof installation, usually `C:\Program Files\Afreet\SkyRoof\lib\SoapySDR\modules0.8`;

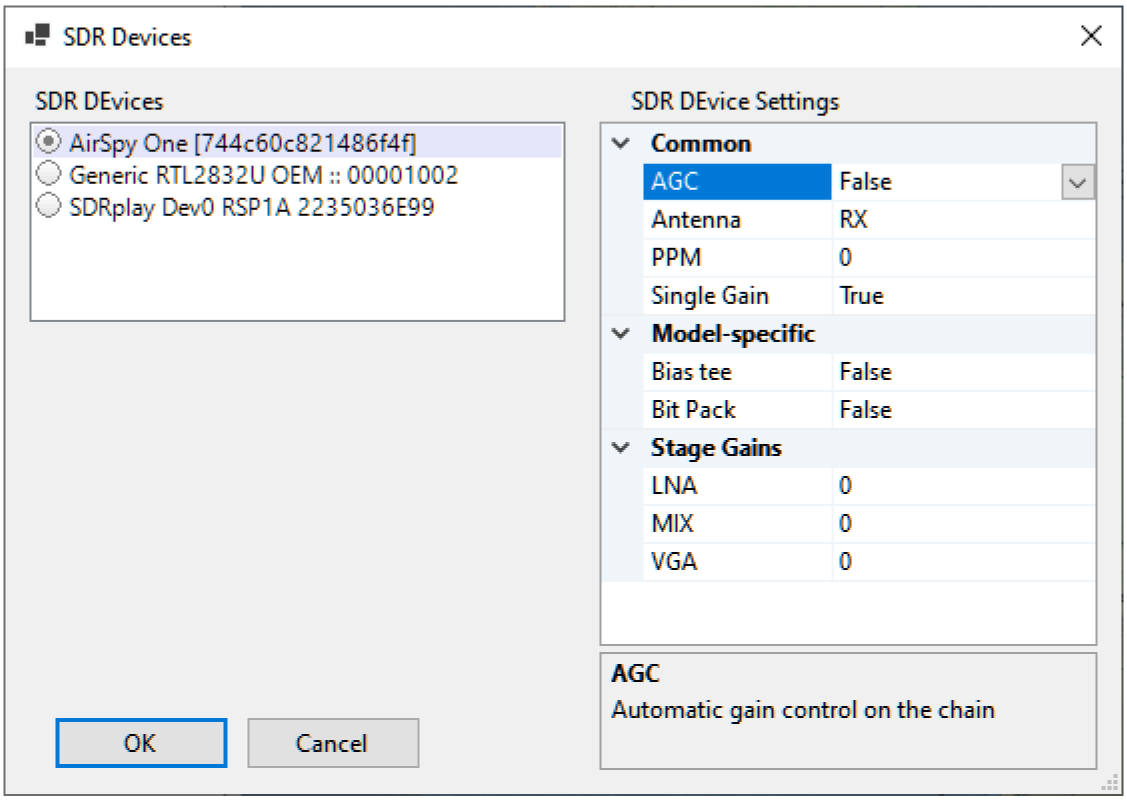
- install the manufacturer's driver for the radio as described in the previous section;

- restart SkyRoof and open the **SDR Devices** dialog. The radio should appear in the list of devices.

> **NOTE**
> Some driver modules interfere with the detection of other SDR devices. The PlutoSDR module is one such example, and for this reason it is no longer included in the SkyRoof setup. If your radios are no longer detected after you add a new module, delete that module from the `modules0.8` folder.

Selecting an SDR device

Connect your SDR device to the computer, then click on **Tools / SDR Devices** in the main menu. This will open the **SDR Devices dialog**:



All active SDR devices are listed on the left panel. Click on the one that you want to use.

Configuring the device

The right panel shows all settings that the device driver understands. The setting names and descriptions (shown on the bottom panel) come from the driver, with two exceptions described below. For information about these settings see the documentation that comes with the radio.

The two settings, common to all radios, are:

- **PPM** - the correction factor for the SDR clock frequency, expressed in parts per million. This setting is important for the correct operation of the Doppler tracking algorithm, see the Calibrating PPM Correction section for details;
- **Single Gain** - when set to true (**and AGC is off**), the SDR gain is controlled by the **RF Gain** slider on the toolbar. This is the recommended setting. When it is set to false, the settings in the **Stage Gains** are applied to the individual stages of the SDR, and the gain slider is disabled.

Using an SDR with a Transverter

If you connect your SDR to the IF output of a VHF/UHF transverter (so the SDR is tuned to 28–30 MHz while it actually receives 144/432 MHz signals), enable the **SDR Offset** in the **Transverter** section of the Settings. SkyRoof will tune the SDR to the IF band while the waterfall scale and the frequency display continue to show the actual satellite RF. See Setting Up Transverter.

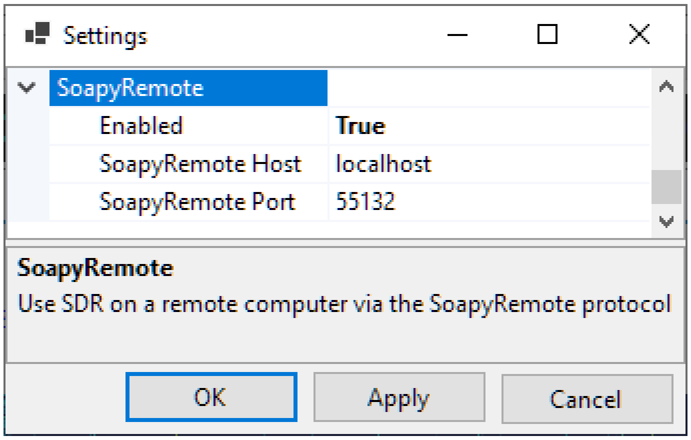
Using Remote SDR

SkyRoof can use SDR devices connected to a remote (or local) computer via the SoapyRemote driver. To enable remote access to SDR:

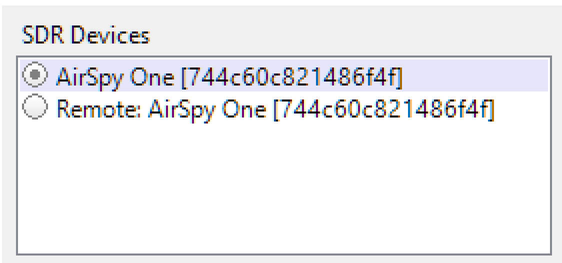
- **On the remote computer:**
  - install SoapySDR which is available as part of PothosSDR;
  - run the remote server that comes with SoapySDR:

    ```
    SoapySDRServer.exe --bind
    ```

- **In SkyRoof:**
  - enable **SoapyRemote** in the **Settings** dialog;
  - enter the **host** name of the remote computer, or leave "localhost" if the radio is on the same computer.



- Open the **SDR Devices** dialog and select the remote SDR device from the list. If the device and remote server are on the localhost, you will have a choice between a direct connection to the radio and a connection via the server.



Calibrating PPM Correction

Motivation

The clock frequency of an SDR, as it comes from the factory, is rarely accurate. Typical errors are in the range of a few PPM (parts per million), which translates to a tuning error of 1-2 Khz on the 70 cm band. For accurate tracking of the satellite signals this error must be calibrated out. The calibration process is simple, we just find a signal of known frequency, check on what frequecy it appears on the waterfall, and compute the PPM correction factor from the difference between the two.

There is plenty of signals on the air that may be used for calibration, if one knows what to look for. One of such signals is the [FCCH channel](#) of a [GSM](#) downlink. This channel is located 67,708 Hz above the center frequency of a GSM channel, and the accuracy of its frequency is claimed to be better than 0.05 PPM.

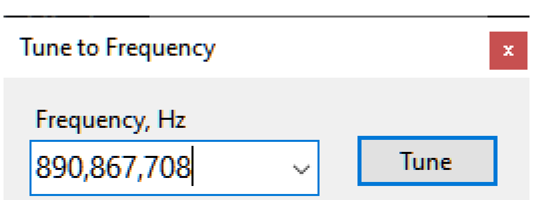
3-rd Party Software

For the RTL-SDR dongles you can use the [Kalibrate](#) utility that performs such calibration automatically. For other radios follow the steps below.

Steps

1. Find a strong GSM signal, or any other signal of known frequency. In my area one of such signals is present on 890.8 MHz.

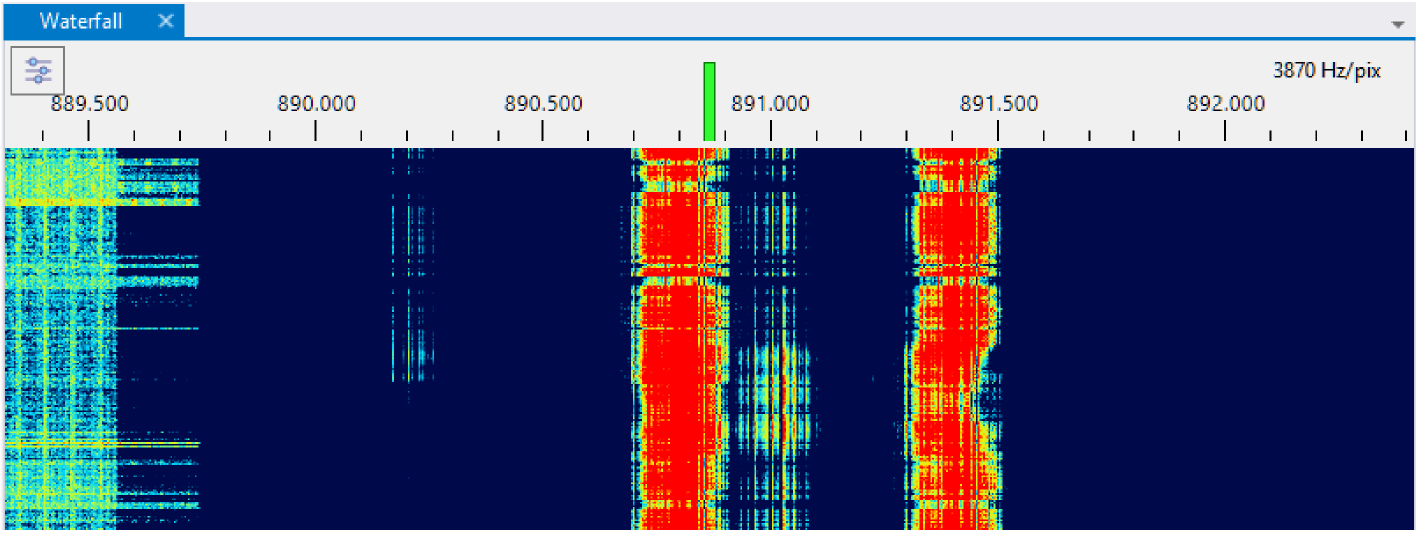
2. Click on the Downlink frequency display in the [Frequency Control](#) panel on the toolbar to open the frequency entry dialog:



3. Enter the frequency of the channel plus the FCCH offset:

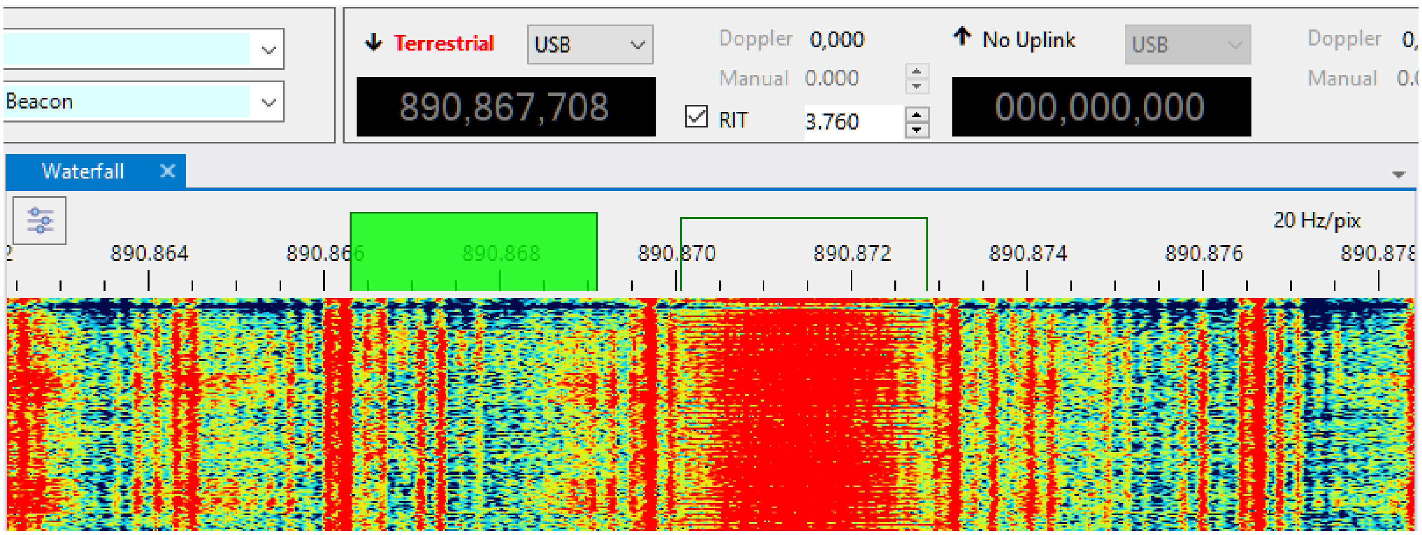
```
890,800,000 + 67,708 = 890,867,708 Hz
```

4. Click on the Tune button in the dialog and verify that the SDR is tuned to the desired frequency:



5. Zoom in by spinning the mouse wheel over the waterfall display:

6. Find the FCCH signal. In the screenshot below it is about 4 kHz above the expected frequency:



7. Now let us measure the offset between the receiver frequency (the center of the green rectangle that represents the receiver passband) and the FCCH frequency. Tick the **RIT** checkbox on the **Frequency Control panel** and adjust the RIT offset until the RIT passband (the clear rectangle) aligns with the signal. You can tune RIT in many different ways, as described in the Frequency Control and Frequency Scale sections. For now, just use the up/down buttons in the RIT offset box, or spin the mouse wheel over that box.

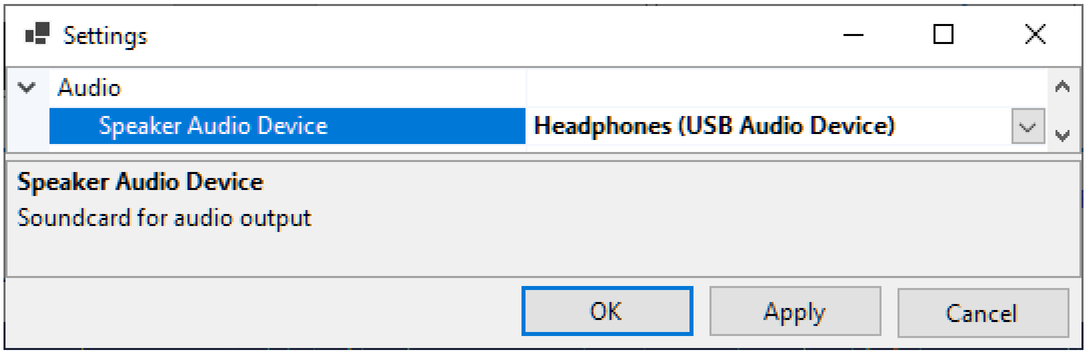
8. Compute the PPM correction. The frequency error measured in the previous step is 3,760 Hz, so the PPM is:

```
3,760 / 890,867,708 * 1e6 = 4.22 PPM
```

9. Now enter this value in the SDR Devices dialog, and you are done.

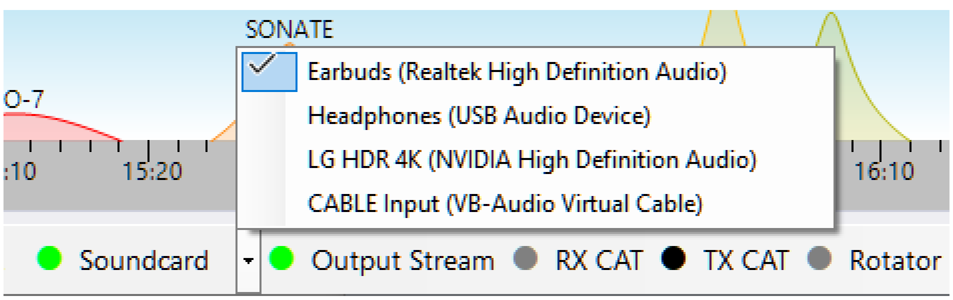
Setting Up Audio

Click on **Tools / Settings** in the main menu to open the Settings window:



- **Speaker Audio Device** - select the audio device that will be used to output the audio received with SDR;
- **FM Squelch** enable or disable squelch in the FM mode.

The output to the soundcard can be toggled by clicking on the **Soundcard** label on the status bar. A drop-down list next to **Soundcard** allows switching between the audio devices:

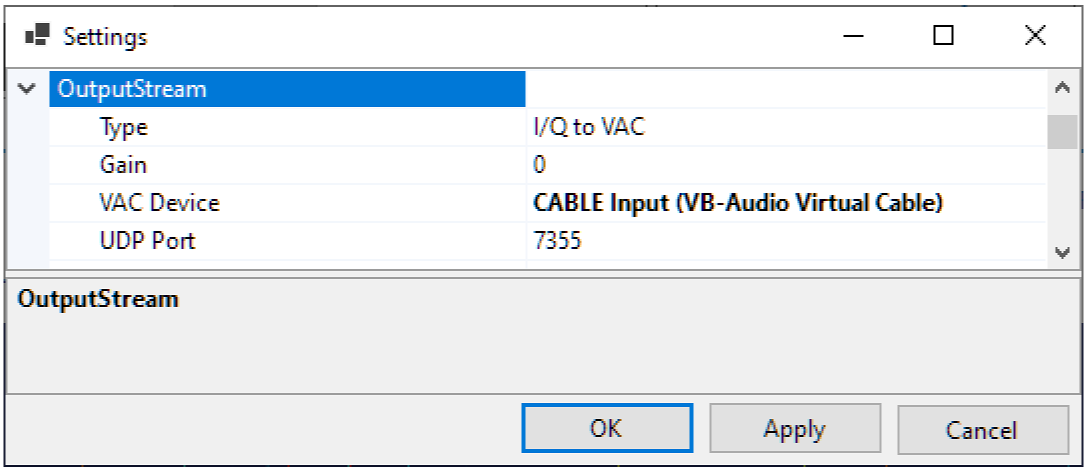


Setting Up Output Stream

SkyRoof can optionally send the raw I/Q data or demodulated audio, either to a Virtual Audio Cable (VAC) or as a stream of UDP packets. The data are sent as 32-bit floating point values in the IEEE 754 format. The sampling rate is 48 kHz in all streaming modes.

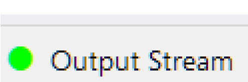
Configuring

Click on **Tools / Settings** in the main menu to open the Settings window:



- **Type** - select the stream type:
    - I/Q to VAC;
    - Audio to VAC;
    - I/Q to UDP;
    - Audio to UDP.
- **Gain** - gain or attenuation, in dB, that will be applied to the streamed data;
- **VAC Device** - the Virtual Audio Cable device to use;
- **UDP Port** - the UDP port number to use.

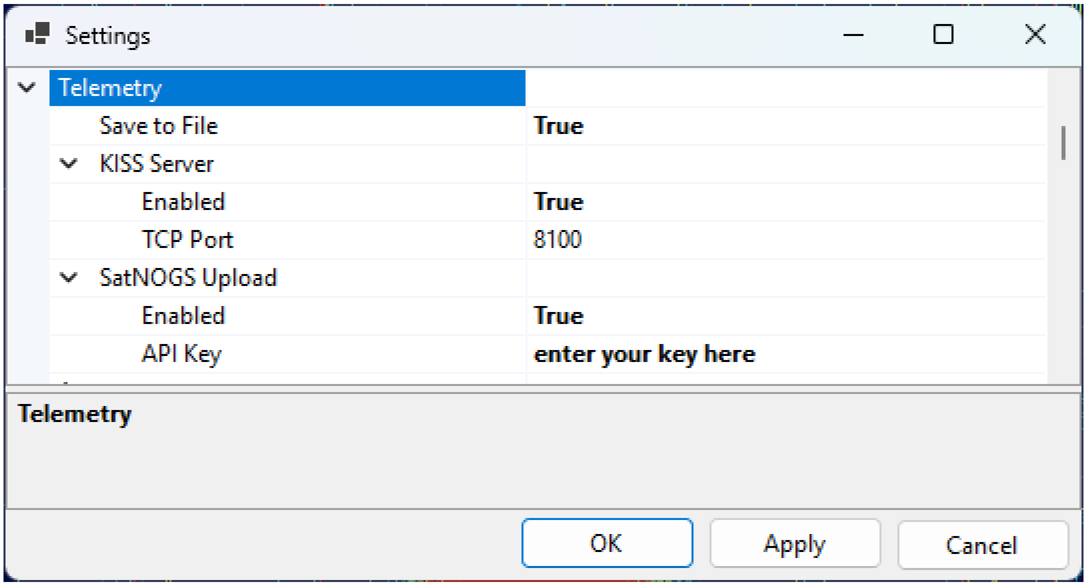
Output streaming can be turned on and off by clicking on the **Output Stream** label on the status bar:



Setting Up Telemetry Decoding

These settings control the optional outputs of the built-in telemetry decoder used by the Telemetry panel — saving decoded frames to a file, sharing them over a KISS server, and uploading them to the SatNOGS database.

Decoder Settings



The outputs are configured on the **Telemetry** page of the Settings window (**Tools / Settings**):

- **Save to File** — when enabled, every decoded frame is appended to a daily log file in the `TelemetryDecodes` subfolder of the data folder. The file contains the same information that is shown in the panel: the time, satellite, transmitter, frame length and address, the decoded telemetry, and the ASCII, HEX and META views of the frame.
- **KISS Server** — makes the decoded frames available to other telemetry software over a KISS-over-TCP server:
  - **Enabled** — turns the server on;
  - **TCP Port** — the TCP port to listen on (8100 by default).

  Point any program that reads KISS frames over TCP — for example a satellite telemetry decoder or a custom logging script — at `localhost` and this port to receive the frames as SkyRoof decodes them.
- **SatNOGS Upload** — forwards each decoded frame to the SatNOGS DB, contributing your telemetry to the community database:
  - **Enabled** — turns uploading on;
  - **API Key** — your personal SatNOGS DB API key (see below).

Signals the Database Does Not Describe

The Telemetry panel normally reads a transmitter's modulation, baud rate and framing from the satellite database. Three cases are not covered there, and all three are handled the same way — by entering the parameters yourself in the **Signal Details** dialog (the gear button in the panel header), or by pressing **Discover** and letting the search work them out from the signal:

- **an unsupported transmitter** — the panel says *telemetry format not supported*, *CW decoding not supported* or *FM decoding not supported*. Its database description either is wrong or names something SkyRoof cannot decode as it stands; overriding the modulation and framing is what makes it decodable.

- **a transmitter with no parameters at all** — the dialog opens on an empty form, every field blank. Fill in what you know, or press **Discover** and let the search fill it in.

- **a terrestrial signal** — tune the radio to a terrestrial frequency (see Frequency Control) and the panel decodes whatever is on it, once you have given it the parameters. Nothing about a terrestrial signal is in any database, so this is the only way in.

SSTV is one of the choices in the **Modulation** field. Selecting it starts the SSTV decoder on the tuned frequency, terrestrial or not, and needs no baud rate, framing or deviation — the mode is read from the image's own VIS header. See Receive SSTV Images.

Terrestrial Parameters

Terrestrial decoding differs from satellite decoding in a few ways, all of which follow from there being no transmitter record behind the signal:

- the parameters last as long as terrestrial mode does. Moving the dial keeps them — you are decoding whatever is at the tuned frequency — while leaving terrestrial mode for a satellite discards them;

- moving the dial starts a **new pass entry** in the tree, headed by the tuned frequency instead of by a satellite and transmitter name. The frame log records it the same way;

- the frames appear in the tree, are written to the frame log and are served over the KISS server like any others, but they are **never uploaded to SatNOGS** — that database records frames against a satellite, and there is none here;

- the parameters cannot be saved to `transmitters-override.json`. That file is keyed by transmitter, so **Save** is unavailable while terrestrial; the parameters are a tool for the session, not a correction to the database.

Uploading to SatNOGS

The SatNOGS DB is a community database of satellite telemetry. To contribute the frames that SkyRoof decodes:

1. **Create an account.** Go to db.satnogs.org, click **Login**, and register (the SatNOGS network uses a single account across all of its sites). Confirm your e-mail address and log in.

2. **Copy your API key.** Open your account page (your callsign or name in the top-right corner) and go to **Settings**. Your **API Key** is shown there as a long string of letters and digits. Copy it.

3. **Configure SkyRoof.** In **Tools / Settings**:

   - on the **User** page, fill in your **Callsign** and your Maidenhead **Grid Square** — both are required, because each uploaded frame is tagged with the observer's call and location;
   - on the **Telemetry** page, expand **SatNOGS Upload**, set **Enabled** to true, and paste the API key into the **API Key** field.
4. **Decode some frames.** Open the Telemetry panel and decode a pass of a supported satellite. Each frame that passes its integrity (CRC) check is uploaded automatically; frames that fail the check are not sent.

5. **Verify on the server.** Confirm that your frames arrived, either way:

   - **Web:** open the satellite's page on db.satnogs.org (for example, by clicking on the **SatNOGS** link in the Satellite Details panel), click on **Data** and see if you are listed under **Most Recent Observers**.

   - **API:** query the telemetry endpoint for that satellite and look for your callsign in the `source` field of the most recent frames, for example:

     ```
     curl -H "Authorization: Token YOUR_API_KEY"
     "https://db.satnogs.org/api/telemetry/?satellite=NORAD_ID"
     ```

     Replace `YOUR_API_KEY` with your key and `NORAD_ID` with the satellite's NORAD catalog number.

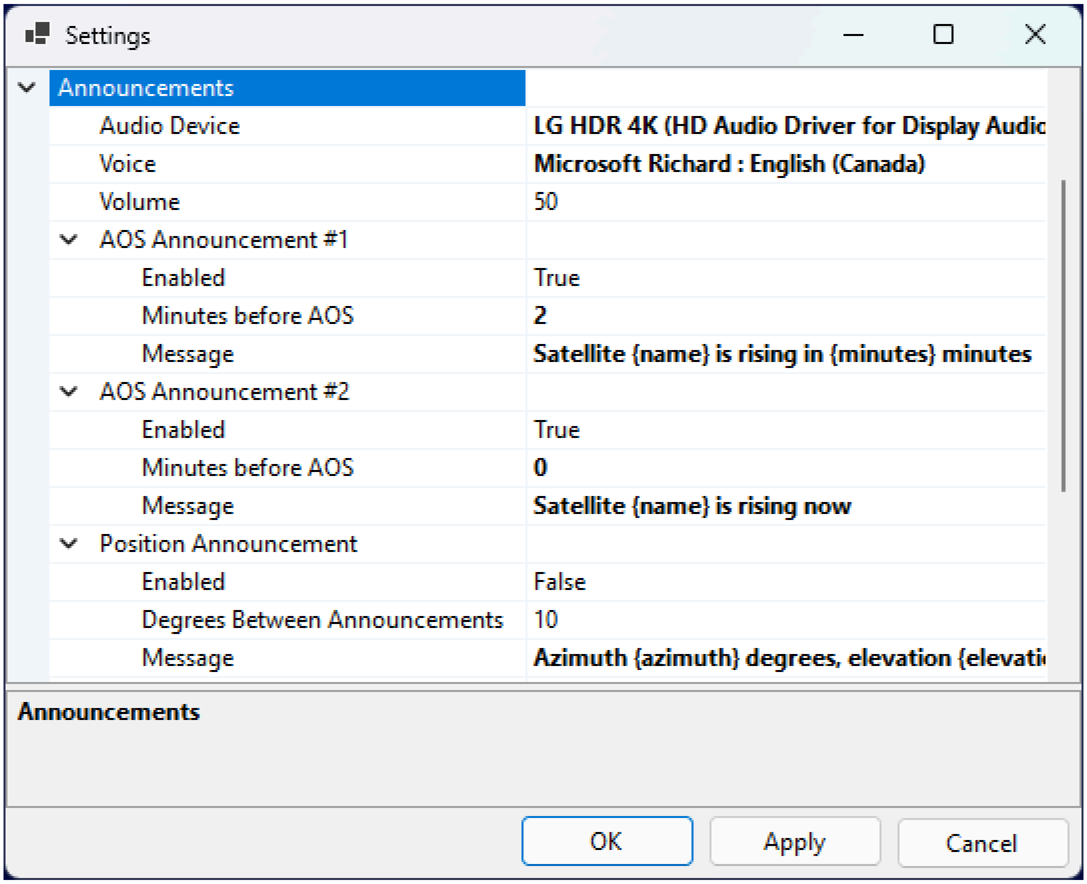
See Also

- Telemetry Panel

Setting Up Voice Announcements

SkyRoof can make voice announcements of the the satellite AOS events and position changes. Up to two AOS announcements may be enabled.

Click on **Tools / Settings** in the main menu to open the Settings window:



- **Audio Device** - select the audio device that will be used to play the voice announcements;
- **Voice** - select one of the voices available on your system. To install a new voice package in Windows, go to **Settings > Time & language > Speech** and then select **Add voices** to download and install the desired voice package.
- **Volume** - set the volume between 1 and 100;
- **Enable** - enable or disable the announcement;
- **Minutes Before AOS**: enter 0 to 5 minutes;
- **Degrees Between Announcements**: 1° to 30°. Satellite position is announced when the angular distance between the previous and current positions exceeds this value;
- **Message** - enter the announcement message. For the satellite name enter `{name}`, for the number of minutes before AOS enter `{minutes}`, for the azimuth and elevation enter `{azimuth}` and `{elevation}` respectively.

Setting Up CAT Control

SkyRoof uses an external program, either skyctld.exe from SkyCAT or rigctld.exe from HamLib, to control the transceiver. In both cases the CAT control commands are sent using the TCP protocol, so the radio may be moved to a remote computer and controlled via the network if desired.

SkyCAT

SkyCAT is specifically designed for the best satellite tracking experience. It has an open architecture where support of the new radio models is added by creating the command definition files with the corresponding commands. Check the SkyCAT web site for the list of currently supported radios. If your radio model is not yet supported, either create a command definition file for it using the instructions on the SkyCAT web site, or use rigctld.exe instead (see below).

Using skycat.exe

If your radio is supported by SkyCAT, use skycatd.exe, a command line program that comes as part of the SkyCAT package. To start using it, follow the setup instructions on the SkyCAT web site.

The latest command definition files are available here. Update the file for your radio before you proceed.

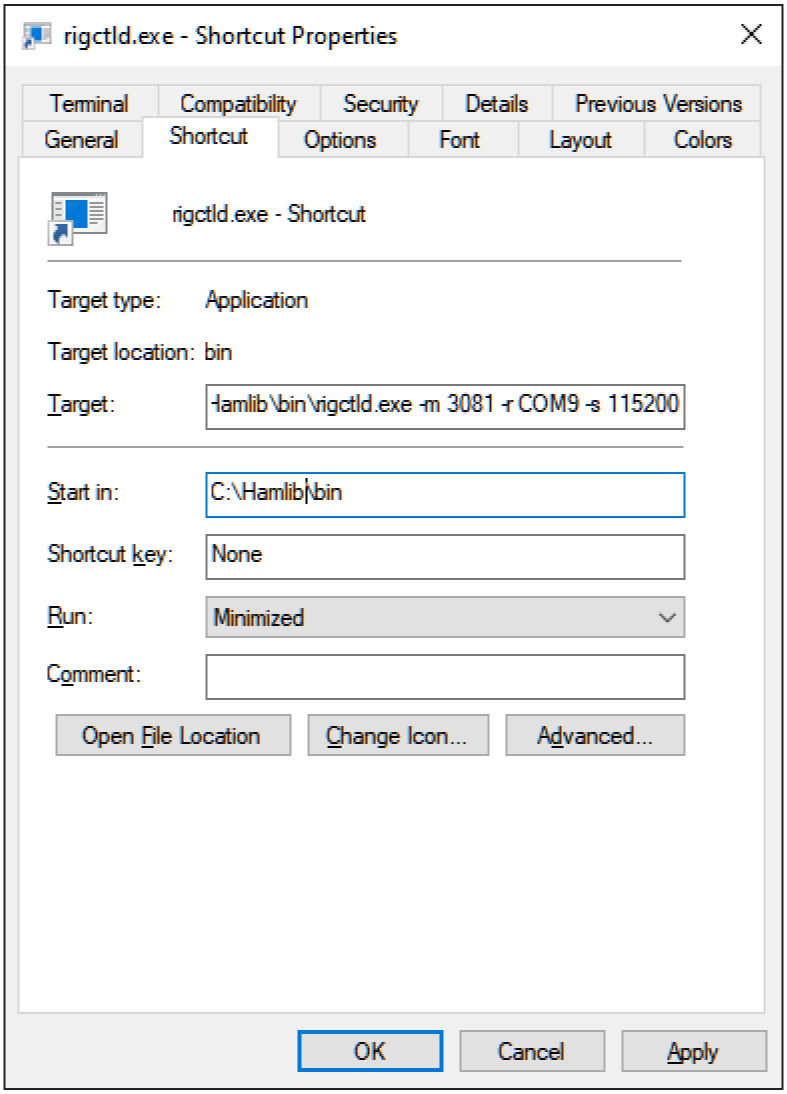
Using rigctld.exe

If a SkyCAT command definition file for your transceiver is not yet available, use **rigctld.exe**, a HamLib-based CAT control daemon. Note, however, that some commands may not work properly with rigctld.exe.

1. Download **hamlib-w64-4.5.5.exe** from GitHub. Other versions may not work correctly.

2. Run the downloaded file to install HamLib, note the folder where it is installed.

3. Create a shortcut to start *rigctld.exe*, with command line arguments:



The arguments on the command line must be tailored for your specific radio and COM port settings. Refer to the [rigctld documentation] for a complete description of the arguments.

Assuming that HamLib is installed in the default location, here is an example string for the shortcut:

```
"C:\Program Files\hamlib-w64-4.5.5\bin\rigctld.exe" -m 3081 -r COM9 -s 115200
```

In the string above the following arguments are used:

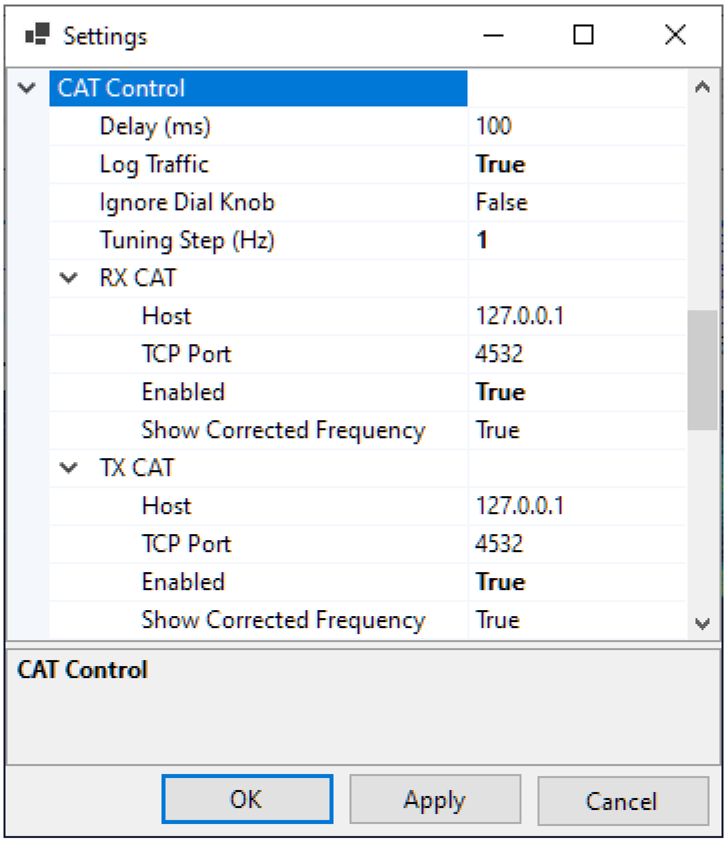
- **-m 3081** - the radio model; 3081 is the Id of IC-9700 (see the [list of id's]);
- **-r COM9** - the COM port used by the radio. In this case, the USB connection to IC-9700 creates two virtual COM ports, COM9 and COM10. The port with the lower number is used for CAT;
- **-s 115200** - use the highest available COM port speed;

    - **-vvvvv** - optional, writes detailed information to the console window. Useful for troubleshooting.
4. Run rigctld.exe using this shortcut before you enable CAT control in SkyRoof.

Settings

The CAT Control settings in the Settings dialog are the same for skycatd.exe and rigctld.exe.

Click on **Tools / Settings** in the main menu to open the **Settings dialog**:



- **Delay** determines how often SkyRoof sends commands to the radio. The default delay of 100 ms is good in most cases. Increase the delay if your radio's CAT interface is slow;
- **Log Traffic** should be set to False and enabled only for debugging;
- **Ignore Dial Knob** - by default, CAT control allows you to change the frequency both in the program and by spinning the dial knob. If for some reason this causes trouble, change this setting to True, so that the dial knob rotation is ignored.

The two sections in the Settings, **RX CAT** and **TX CAT**, allow you to use either the same radio for RX and TX, or two different radios. You can also enable only one of those, or disable both. The recommended configuration is to use an SDR for reception and a transceiver for transmission, in this case RX CAT should be disabled.

To use the same radio for RX and TX, set **Host** and **TCP Port** to the same values in both sections.

To use two different radios, create a second shortcut for the second radio, and specify a different port number on the command line. Enter this port number in the settings as well, and run two instances of **rigctld.exe** using both shortcuts.

The settings in the RX and TX sections are:

- **Host** - should be "127.0.0.1" or "localhost" if skycatd or rigctld is running on the same computer as SkyRoof. It may be changed to a different address for remote control;
- **TCP Port** - 4532 is the default port used by skycatd and rigctld. Use a different port in one of the sections to control different radios for RX and TX;
- **Enabled** - enable or disable CAT. Another way to toggle CAT is to click on the CAT labels on the status bar:

- **Show Corrected Frequency** - The SkyRoof can display either the nominal frequency of the satellite transmitter, or the frequency with all corrections applied. Another way to toggle this setting is via the right-click menu on the frequency display widget on the toolbar.

Using CAT with a Transverter

If your transceiver does not understand VHF/UHF frequencies (e.g., a "dumb" HF rig connected via a transverter, or an HF rig such as the IC-7300), enable the **RX CAT Offset** and / or **TX CAT Offset** in the **Transverter** section of the Settings. SkyRoof will then send the IF frequency (e.g., 29.950 MHz) to the radio instead of the satellite RF (e.g., 145.950 MHz). Modern rigs with built-in XVRT support (e.g., the K3S) do their own conversion internally and should leave these offsets **disabled**.

When a CAT offset is enabled but no transverter band covers the current frequency, the **RX CAT** or **TX CAT** label on the status bar turns **yellow** and SkyRoof skips the CAT command rather than send a frequency the radio cannot tune. Hover over the label to see the explanation.

See [Setting Up Transverter](#).

Model-Specific Notes

IC-9700

- set **CI-V USB Port** in the transceiver menu to **Unlink from REMOTE**;
- note that the radio is used in the Dual Watch mode, not in the Sat mode, so the upper frequency on the transceiver screen is uplink, the lower one is downlink.

IC-991A

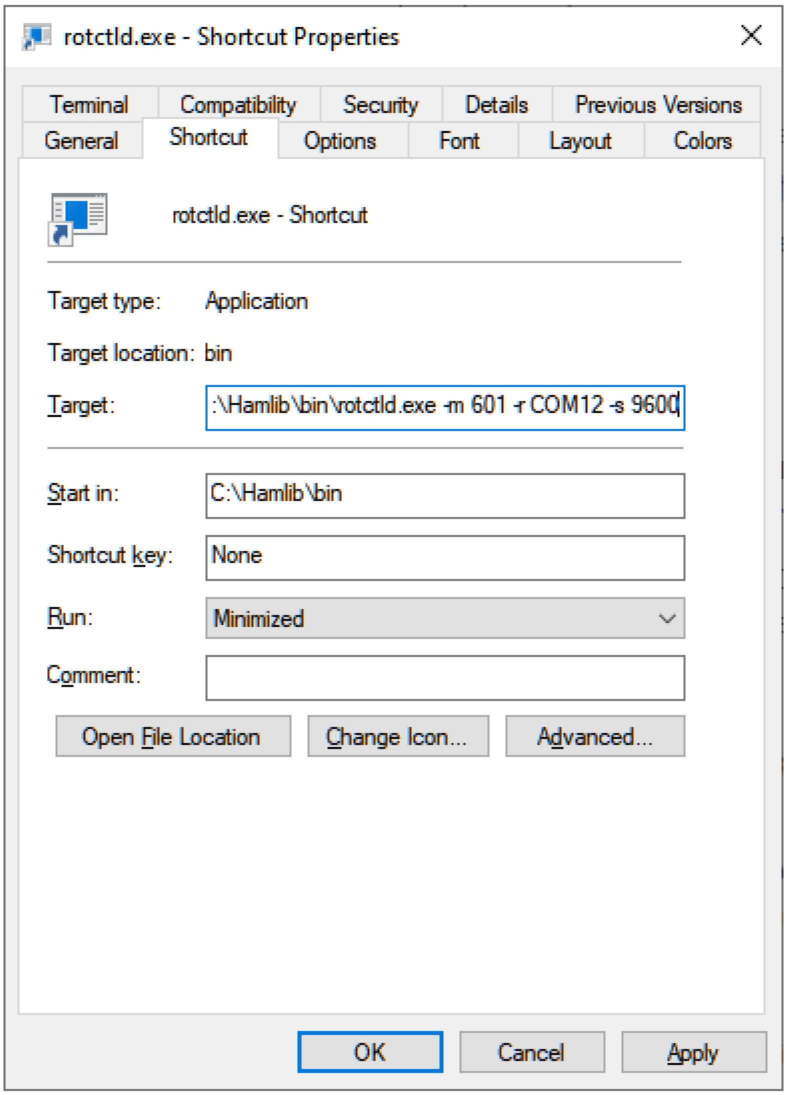
- set **CAT RTS** in the transceiver menu to Disabled.

Setting Up Rotator Control

rotctld.exe

SkyRoof uses **rotctld.exe**, a HamLib-based rotator control daemon, to control the antenna rotator. See the [Setting Up CAT Control](#) section for the instructions how to download and install HamLib.

Create a shortcut to start *_rotctld.exe_, with command line arguments:



The arguments on the command line must be tailored for your specific rotator and COM port settings. Refer to the [rotctld documentation](#) for a complete description of the arguments.

Assuming that HamLib is installed in the default location, here is an example string for the shortcut:

```
"C:\Program Files\hamlib-w64-4.5.5\bin\rotctld.exe" -m 601 -r COM12 -s 9600
```

In the string above the following arguments are used:

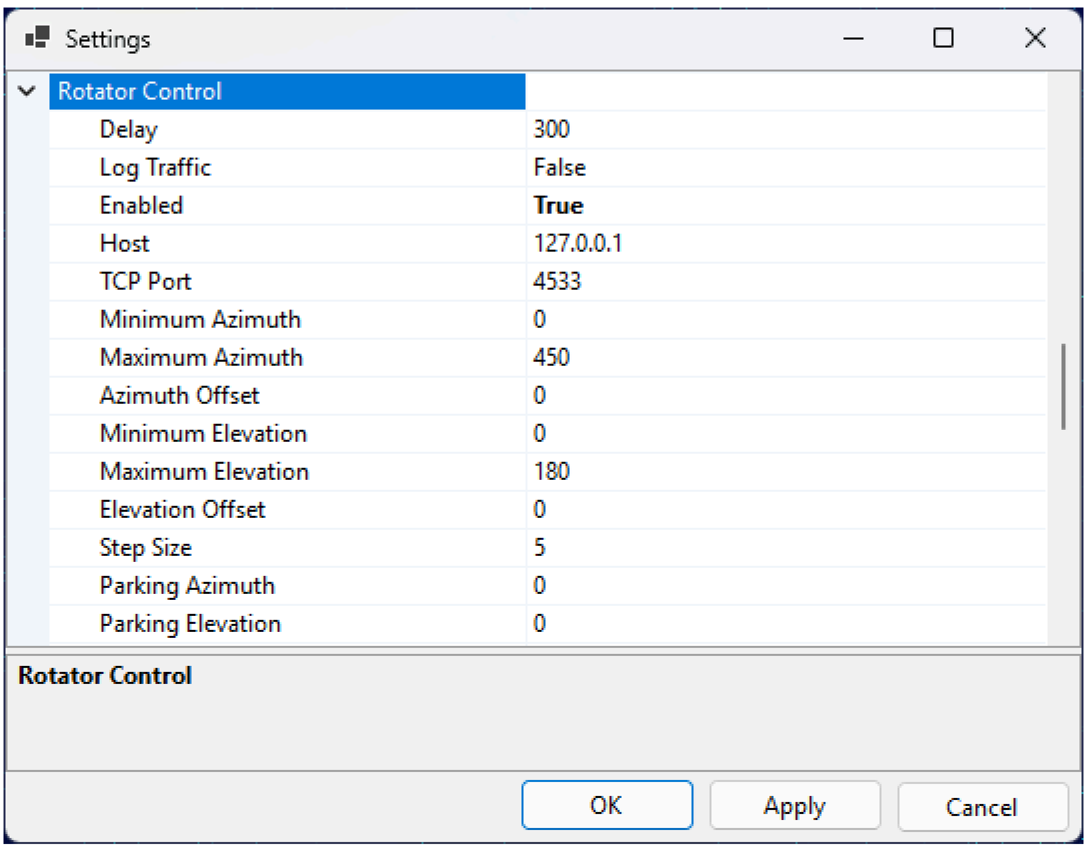
- **-m 601** - the rotator model is Yaesu GS-232A;
- **-r COM12** - the COM port used by the rotator;
- **-s 9600** - the COM port speed.

Run rotctld.exe using this shortcut before you enable rotator control in SkyRoof.

The list of rotator models supported by rotctld.exe is available on the [HamLib web site](). Note that one of the models is **PSTRotator**. When this model is selected, rotctld.exe just passes the rotation commands to the PST Rotator software.

Settings

Click on **Tools / Settings** in the main menu to open the [Settings dialog]():



- **Delay** determines how often SkyRoof sends commands to the rotator. The default delay is 300 ms, but you can set it to a lower value, such as 100 ms, without adverse effects;
- **Log Traffic** should be set to False and enabled only for debugging;
- **Enabled** - enable or disable rotator control. Another way to toggle the rotator control is to click on the Rotator label on the status bar;
- **Host** - should be "127.0.0.1" or "localhost" if rotctld is running on the same computer as SkyRoof. It may be changed to a different address for remote control;
- **TCP Port** - 4533 is the default port used by rotctld;

- **Minimum Azimuth**, **Maximum Azimuth**, **Minimum Elevation**, **Maximum Elevation** - specify the range of azimuth and elevation values your rotator accepts;
- **Azimuth Offset**, **Elevation Offset** - if your rotator is not perfectly calibrated, these settings allow you to apply a correction;
- **Step Size** - to prevent the rotator from starting and stopping too often, change the bearing only when the required change is greater than the step size. The default value is 5 degrees;
- **Parking Azimuth**, **Parking Elevation** - specify the parking position for your antenna.

If your rotator does not control elevation, set the MinimumElevation and MaximumElevation to the same value. With such settings, wrong elevation will not be considered a bearing error. Note that the bearing error is indicated with a pink color on the Rotator Control panel.

See Also

- Smart Antenna Rotation

Setting Up Transverter

If you receive on an HF SDR connected to a VHF/UHF transverter, or transmit with an HF transceiver connected to a transverter, the SDR and the rig "see" the satellite signal at the transverter's **IF frequency** (typically 28–30 MHz), not at the actual RF frequency (144 MHz, 432 MHz). SkyRoof can apply the LO offset for you so that:

- the **SDR** is tuned to the IF that the transverter outputs, while the waterfall scale and the frequency display continue to show the RF frequency of the satellite;
- the **transceiver** (via CAT) receives the IF frequency that it can actually tune, instead of the VHF/UHF frequency that it cannot.

The three offsets — for the SDR, for the RX CAT channel and for the TX CAT channel — are independently enabled, so you can mix-and-match: a smart VHF/UHF rig with an SDR-on-transverter setup, a dumb HF rig with the SDR on the antenna directly, or any other combination.

> **NOTE**
>
> Frequencies are **always** displayed at the actual RF (144/432 MHz). The IF translation is applied internally by SkyRoof when it tunes the SDR or sends a CAT command.

When You Need This

- **HF transceiver + transverter, no built-in XVRT support** (e.g., IC-735, IC-7300): enable **RX CAT** and **TX CAT** offsets. SkyRoof sends 29.950 MHz to the radio when the satellite is on 145.950 MHz.

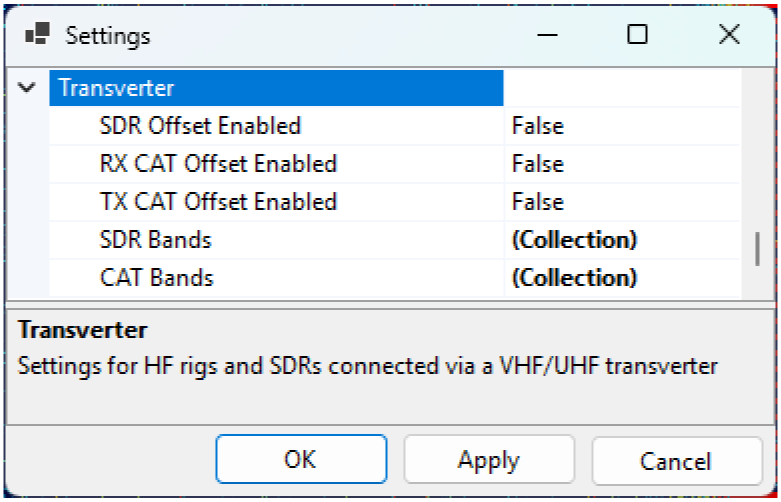
- **Modern transceiver with built-in XVRT offset** (e.g., K3S + Elecraft XV): the radio already converts internally. Leave the **CAT** offsets **disabled** so SkyRoof sends the RF frequency, which the rig converts itself.

- **HF SDR on the transverter IF output** (e.g., Airspy HF+, RTL-SDR with an upconverter on HF range, IC-R7000 IF tap): enable the **SDR** offset. The SDR is tuned to the IF band, but the waterfall and frequency display continue to show the actual satellite RF.

- **VHF/UHF SDR on the antenna directly, no transverter**: leave all three offsets disabled; SkyRoof behaves exactly as without this feature.

Settings

Click on **Tools / Settings** in the main menu to open the Settings dialog, then expand the **Transverter** section:

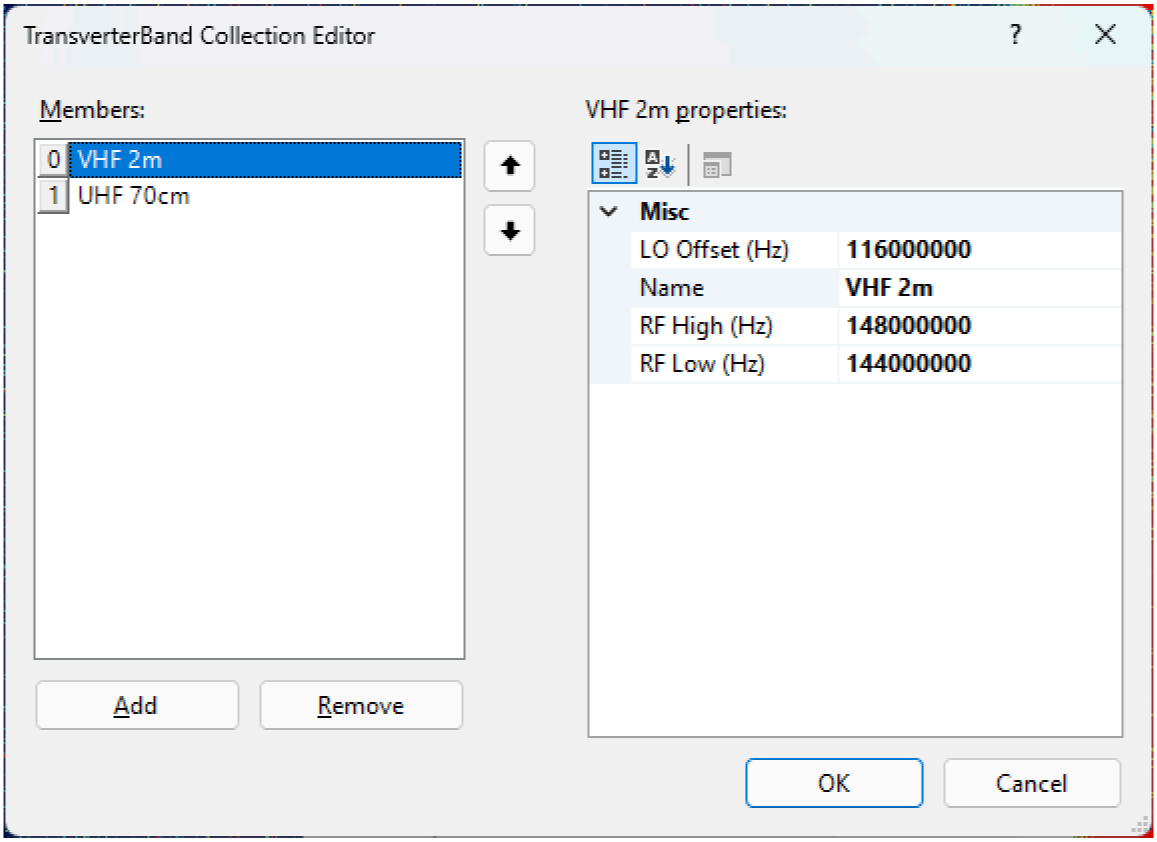


- **SDR Offset Enabled** — when on, the SDR is tuned to the IF of the matching SDR transverter band, not to the satellite RF;
- **RX CAT Offset Enabled** — when on, the RX frequency sent to the radio is offset by the LO of the matching CAT band;
- **TX CAT Offset Enabled** — same, for the TX frequency.
- **SDR Bands** — list of bands for the SDR path;
- **CAT Bands** — list of bands for the CAT path.

The two band lists are separate because the SDR transverter and the transmit transverter often use different local oscillators, and because the CAT path's RF range may be narrower than the SDR path (the SDR sees the entire transverter passband, but the radio's HF receiver can only tune a slice of it).

Editing Bands

Click the **...** button in the **SDR Bands** or **CAT Bands** row to open the collection editor. You can **add new bands, edit the existing ones, or delete bands**:



Each band has:

- **Name** — a free-text label (e.g., `VHF 2m`, `UHF 70cm`, `23 cm`);
- **RfLow / RfHigh** — RF frequency range in Hz, inclusive;
- **LoOffset** — local oscillator offset in Hz. IF = RF − LO.

When SkyRoof needs the IF for an active RF, it looks up the first band whose `[RfLow, RfHigh]` range covers that RF and subtracts its `LoOffset`. If no band matches, the offset is zero (SDR case) or the CAT command is skipped with a warning (CAT case — see below).

Restoring the Defaults

If you delete entries from a band list and want to bring back the seeded defaults, right-click on the **SDR Bands** or **CAT Bands** row and choose **Reset**. This replaces the entire list with the default entries. An empty list, on its own, is preserved across sessions — SkyRoof will not "helpfully" re-seed it on the next launch.

Default Bands

Default SDR Bands

Name	RF Low (MHz)	RF High (MHz)	LO (MHz)	IF Range
VHF 2m	144.000	148.000	116.000	28.0–32.0 MHz

Name	RF Low (MHz)	RF High (MHz)	LO (MHz)	IF Range
UHF 70cm	432.000	438.000	407.000	25.0–31.0 MHz

The SDR band table covers the full ham VHF allocation and the satellite portion of the UHF allocation. The IF spans (4 MHz and 6 MHz) exceed the SDR's maximum bandwidth of 3.1 MHz, so the SDR cannot show the entire transverter passband at once — see Behavior at Runtime below for how SkyRoof handles this.

Alternative SDR Bands (10.7 MHz IF)

For receivers that use 10.7 MHz as the IF output (e.g., the IC-R7000 wideband communications receiver), edit the SDR bands to use these LO values instead:

Name	RF Low (MHz)	RF High (MHz)	LO (MHz)	IF Range
VHF 2m	144.000	148.000	135.300	8.7–12.7 MHz
UHF 70cm	432.000	438.000	424.300	7.7–13.7 MHz

Default CAT Bands

Name	RF Low (MHz)	RF High (MHz)	LO (MHz)	IF Range
VHF 2m	145.800	146.000	116.000	29.8–30.0 MHz
UHF 70cm AO-7 uplink	432.000	434.000	404.000	28.0–30.0 MHz
UHF 70cm	435.000	437.000	407.000	28.0–30.0 MHz

The CAT RF ranges are narrower than the SDR ranges so the resulting IF lands inside a typical HF rig's tuning range (28–30 MHz). The AO-7 mode U-V uplink (432–434 MHz) is split into its own entry because it uses a different LO than the rest of the 70 cm band.

Behavior at Runtime

When **SDR Offset Enabled** is on:

- The SDR is retuned to the IF center of the active satellite's band on the first activation and whenever the satellite crosses to a different band. The waterfall history resets on each such retune;
- Within a band, only the audio slicer offset moves as the Doppler-corrected signal drifts; the SDR center stays put and the waterfall keeps scrolling;

- If the transverter band is wider than the SDR's bandwidth (the default 4 MHz VHF and 6 MHz UHF bands are), the SDR center is repositioned as needed to keep the Doppler-corrected signal visible, while keeping the SDR passband within the transverter band limits;
- The frequency scale, the satellite labels and the pass-visibility check always work in RF, so what you see on screen is the satellite's actual frequency regardless of where the SDR is physically tuned.

When **RX CAT** or **TX CAT** offset is enabled but the active RF does not fall into any CAT band, SkyRoof **skips the CAT frequency command** rather than sending a frequency the radio cannot tune. A warning is written to the log, and:

- the corresponding **RX CAT** or **TX CAT** label on the status bar turns **yellow**;
- the tooltip on that label explains the out-of-band condition, naming the offending frequency.

This is the signal that you need to either disable the CAT offset (if you are listening to a non-ham satellite) or add a band entry to the CAT list (if you want to extend coverage).

Frequency Display Tooltips

When any of the three offsets is enabled, the tooltip on the downlink and uplink frequency display in the Frequency Control panel includes extra lines:

- `SDR Transverter IF: <value> MHz` — only on the downlink, when the SDR offset is enabled and an SDR band matches;
- `CAT Transverter IF: <value> MHz` — on the downlink when RX CAT offset is enabled, and on the uplink when TX CAT offset is enabled, when a CAT band matches.

This lets you confirm at a glance what IF frequency is being sent to the SDR and to the radio.

Common Configurations

K3S with XVRT + SDR on the transverter IF

- `SdrOffsetEnabled = true`, both CAT offsets `false`.
- SkyRoof tunes the SDR to the IF band and keeps the satellite visible there.
- SkyRoof sends 435.835 MHz to the K3S (unchanged); the rig applies its own XVRT conversion internally.

IC-7300 (no transverter awareness) + SDR on the transverter IF

- `SdrOffsetEnabled = true`, `RxCatOffsetEnabled = true`, `TxCatOffsetEnabled = true`.
- Downlink 435.835 MHz → SDR shows 28.835 MHz IF; IC-7300 commanded to 28.835 MHz.
- Uplink 145.965 MHz → IC-7300 commanded to 29.965 MHz.
- The IC-7300 covers 1.8–54 MHz continuously, so both IFs are inside its tuning range.

SDR on the antenna, smart rig

- All offsets disabled. Behaviour is identical to a SkyRoof setup without any transverter.

AO-7 mode U-V

- Uplink 432.xxx MHz → matches the **UHF 70cm AO-7 uplink** CAT band → CAT LO = 404 MHz → IF ≈ 28.xxx MHz.
- Downlink 145.xxx MHz → matches the SDR's **VHF 2m** band → SDR LO = 116 MHz → IF ≈ 29.xxx MHz.

Non-ham telemetry satellite (e.g., 400.550 MHz)

- No CAT or SDR band covers 400 MHz → both offsets resolve to zero → SDR tunes directly to 400.550 MHz, RF is sent to the radio (if it can tune there) → no transverter behaviour.

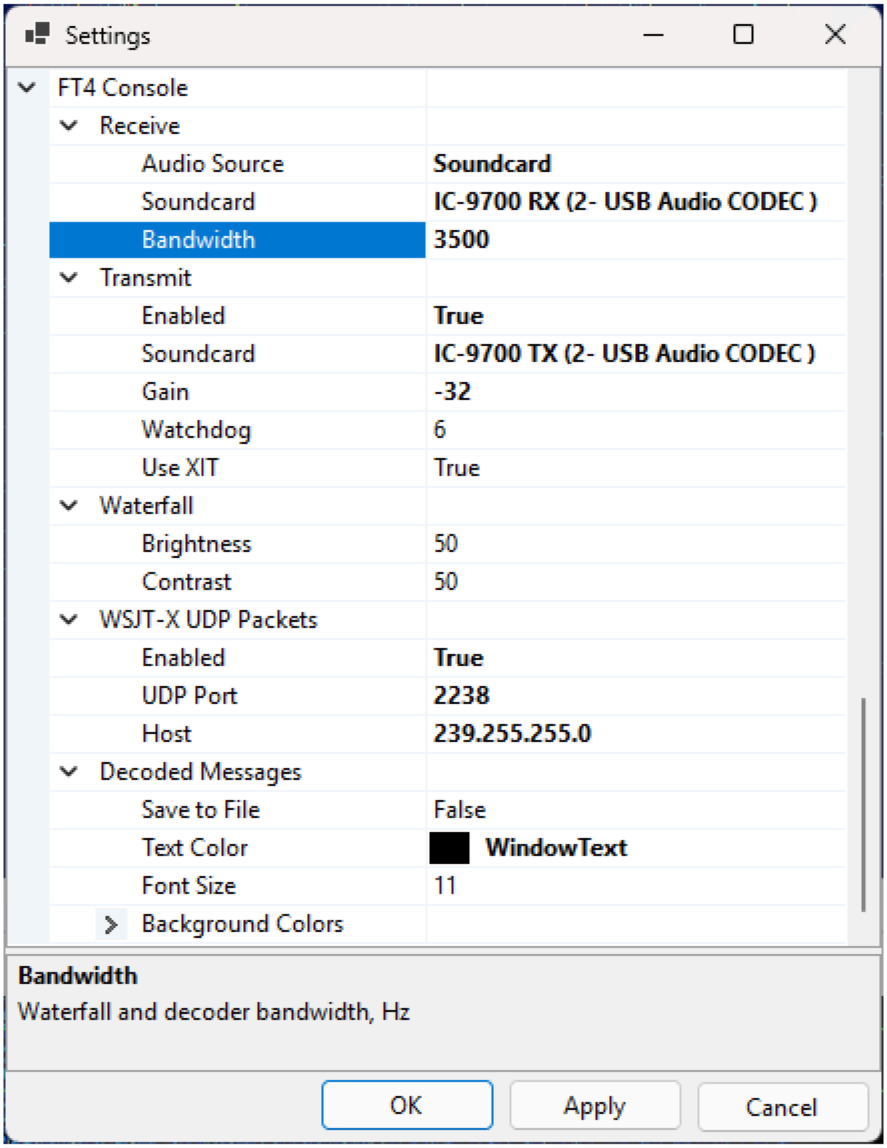
See Also

- Setting Up SDR
- Setting Up CAT Control
- Frequency Control

Setting Up FT4 Console

FT4 Console Settings

To configure FT4 Console, click on Tools / Settings to open the Settings dialog and scroll down to the FT4 Console section:



Receive

- **Audio Source** - either SDR or Soundcard. SDR is the default, but it is also possible to receive FT4 from the transceiver via a soundcard;

- **Soundcard** - if Soundcard is selected as audio source, this field should contain the soundcard to use. Windows allows you to rename the soundcards, so you can change the default name, such as "USB Audio CODEC" to something that is easy to recognize, "IC-9700 RX" in my case;

- **Bandwidth** - the receiver bandwidth, in Hertz. Affects both the waterfall display and decoding. See **Receiver Bandwidth** below.

Transmit

- **Enabled** - by default, the transmit function is disabled. Set this field to true if you want to transmit FT4;

- **Soundcard** - the soundcard that sends the FT4 audio to the transmitter;

- **Gain** - the FT4 output audio is mutliplied by this factor (in dB). The default is -30 dB, see the **TX Gain** section below for the instructions how to find the correct value;

- **Watchdog** - the maximum transmit time. After the specified number of minutes, if the user has not performed any actions, the transmissions automatically stop;

- **Use XIT** - offset the transmitter frequency relative to the receiver depending on the transmitted audio tone. See **Using XIT** below for details.

Waterfall

- **Brightness** - waterfall brightness, 0 to 100;

- **Contrast** - waterfall contrast, 0 to 100;

WSJT-X UDP Packets

- **Enabled** - enable sending the UDP packets with information about decoded messages and QSO. The format of the messages is the same as in WSTT-X;

- **UDP Port** - the port to which the messages will be sent;

- **Host** - this could be a Unicast or Multicast address. See **UDP Packets** below.

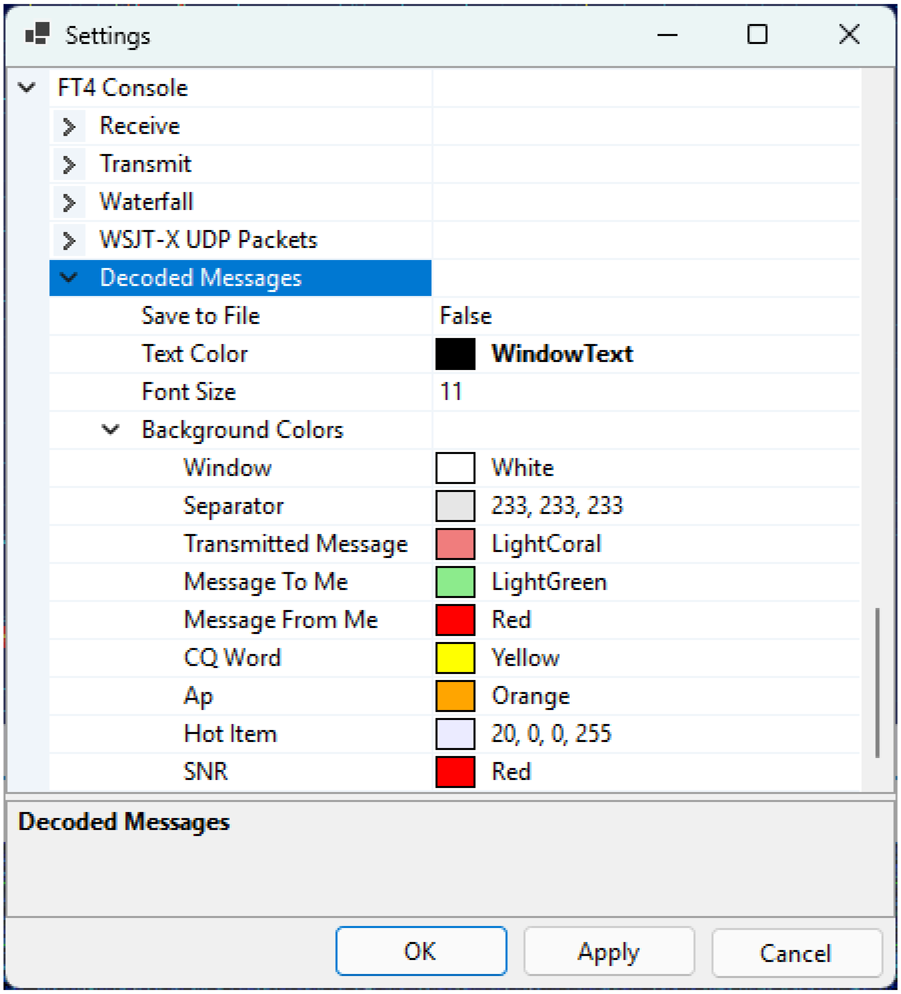
Decoded Messages

- **Save to File** - when set to true, all received and transmitted messages will be saved to a file in the [Data folder](), FT4 sub-folder;

- **Text Color** - the text color in the Decoded Messages window;

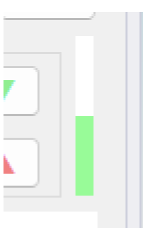
- **Font Size** - the font size in the Decoded Messages window;

- **Background Colors** - the background colors of the messages of different types:



RX Signal Strength

When FT4 is decoded from the SDR, the signal strength is adjusted automatically, but when the signals are coming from a soundcard, the amplitude of the input audio must be adjusted so that the signal strength bar is at about 50%:



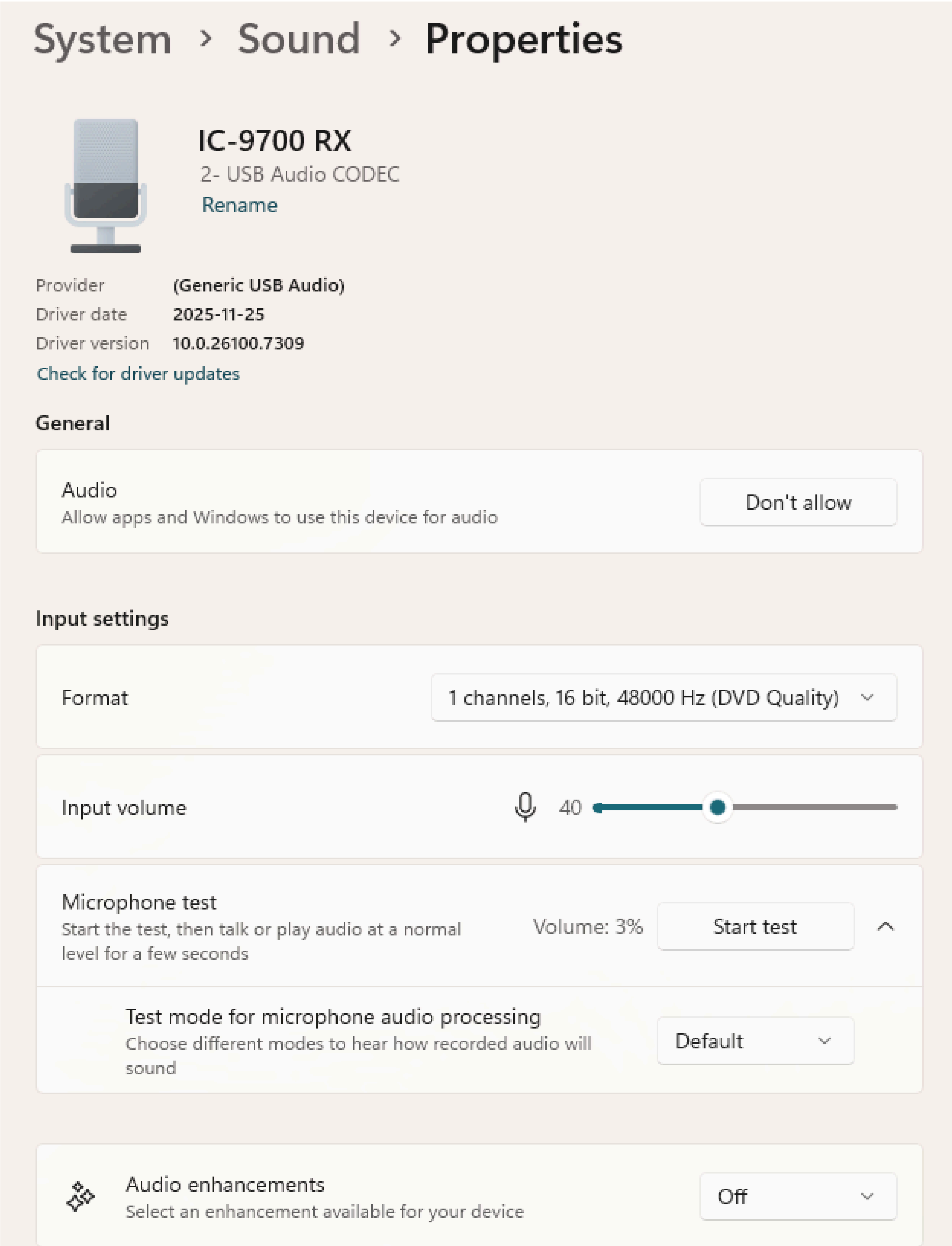
This could be done in several ways.

Option 1: Transceiver Menu

In IC-9700 press **Menu / Set / Connectors / USB AF/IF Output** and adjust **AF Output Level**.

Option 2: Windows Audio Settings

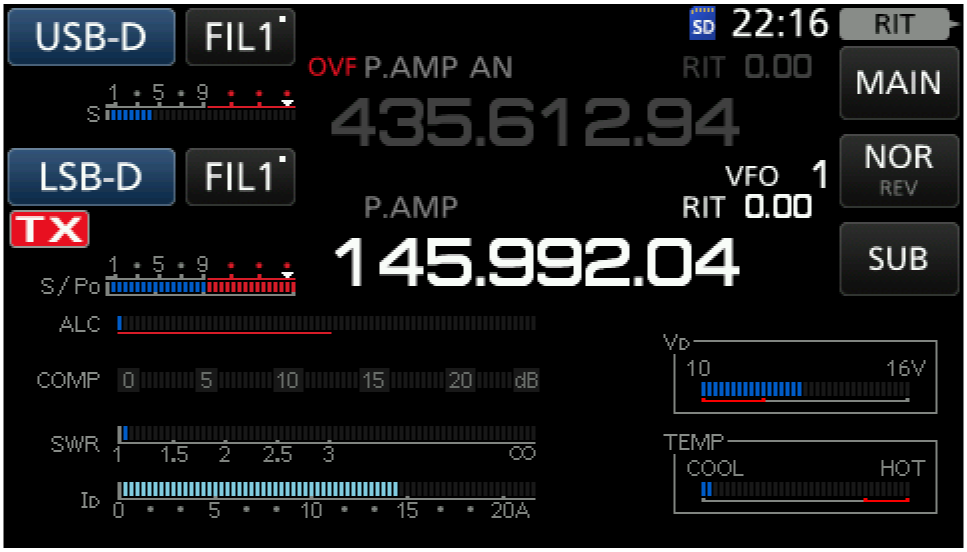
In Windows 11 open **Settings / System / Sound**, select your soundcard, and adjust the **Input Volume** setting:



TX Gain

The amplitude of FT4 audio output must be set carefully to ensure the desired output power and to prevent transmission of audio harmonics. To adjust the output amplitude, click on the **Tune** button in FT4 Console to start transmission and adjust **TX Gain slider**. Use the meters in the transceiver to ensure that the radio delivers full output power, and that the ALC level is around zero.

In IC-9700 you can display all meters in the satellite mode by pressing **Menu / Meter**, as shown in the screenshot below. **S/Po** must be at the maximum, and **ALC** at the minimum. For IC-9700 the optimal TX gain is about -32 dB.



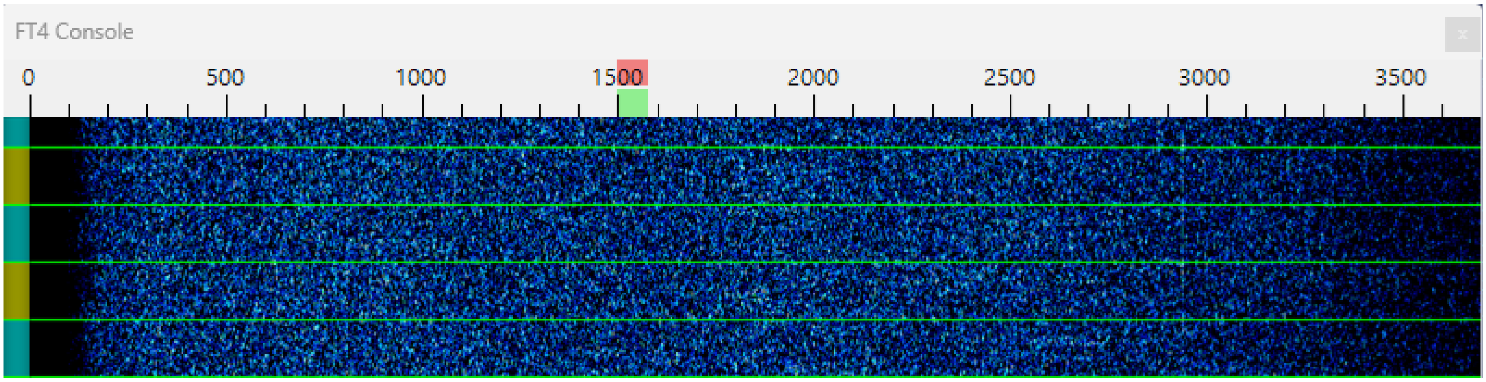
Once the audio output level is set, adjust the output power of your radio according to your operating needs. In most cases only a few watts are required for satellite contacts if a good antenna is used.

Receiver Bandwidth

The **FT4 Console / Receive / Bandwidth** parameter in the Settings dialog controls the bandwidth of the FT4 decoder, as well as the frequency segment shown on the waterfall display.

When decoding from an SDR, set it to 5000 Hz, the bandwidth of the SDR receiver in the USB-D and LSB-D modes;

When decoding from a soundcard, set this parameter to match the receiver bandwidth of your radio. Make sure that the dark area on the waterfall is not wider than 200-300 Hz at each side, as in the screenshot below:



In most transceivers the RX bandwidth is around 3000 Hz, even in the Data mode. IC-9700 may we configured for a bandwidth of 3600 Hz in USB-D and LSB-D modes.

Using XIT

Keep the **Transmit / Use XIT** option set to true unless unless your radio is unable change the TX frequency when transmitting.

When this option is enabled, the audio tone sent to the transmitter is always in the range of 1000 Hz to 2000 Hz. To transmit signals outside of this range, the transmitter frequency is changed up or down. For example, to transmit on 3200 Hz, the transmitter is tuned 2000 Hz up (or down for inverting transponders), and the audio tone sent to the radio is 1200 Hz.

The XIT mode serves two purposes:

- it prevents the odd audio harmonics from being transmitted since they are outside of the transmitter passband;
- it ensures that you can transmit on any audio frequency, even outside of the transmitter passband.

UDP Packets

SkyRoof can send UDP packets with information about decoded messages and completed QSO, in the same format as WSJT-X. There is a number of programs that listen to such messages and use them to write QSO to the log or to update Worked statistics. Some even reply to these messages with callsign status information, so that SkyRoof can color-highlight the callsigns according to their Needed status.

This option is disabled by default. You can enable it in the Settings dialog.

Both Unicast and Multicast messaging is supported. In the Unicast mode the messages can be sent to only one program. If you start another program that listens on the same UDP port, that program fails with the Port in Use error. In the Multicust mode multiple programs can receive messages on the same port. Use the multicast mode if possible, or use unicast with older programs that do not support multicast.

UDP packets are sent in the multicast mode if the IP address is in the range of 239.0.0.0 - 239.255.255.255, otherwise they are sent as unicast.

The default settings are:

Unicast:

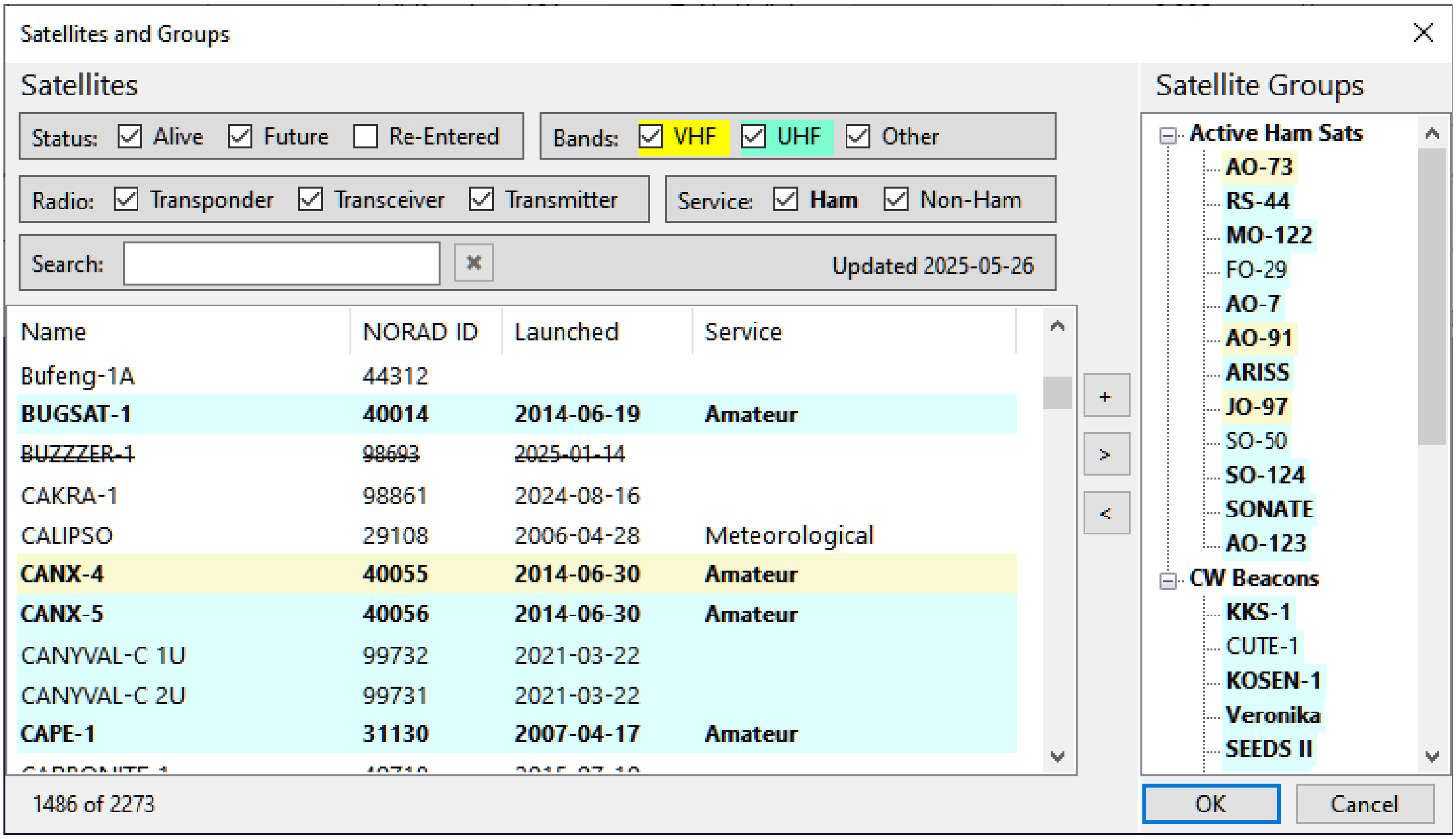
- IP Address: 127.0.0.1
- Port: 7311

Multicast:

- IP Address: 239.255.255.0
- Port: 2238

Satellites and Groups

Click on **Tools / Satellites and Groups** in the main menu to open the **Satellites and Groups** window:



Exploring the satellite data

This window is a great tool for browsing the information about the satellites that is available in SkyRoof. For a detailed description of the data see the Satellite Data section.

Highlighting

The satellites are highlighted based on their properties as described in the Satellite Highlighting section.

Filtering

Using the checkboxes at the top of the left panel, the satellites may be filtered by:

- **Status** - Alive, Future or Re-Entered. If you cannot find some satellite in the list, tick the Re-Entered checkbox, maybe this satellite has already re-entered the atmosphere;
- **Bands** - show only the satellites that have at least one transmitter working in the VHF (2m), UHF (70cm) or in any other band;

- **Radio** - the radio type: linear transponder, FM transceiver or a telemetry/beacon transmitter;
- **Service** - Ham or non-Ham, as marked in the database.

Searching

Use the **Search** box to search the satellites by name, callsign or NORAD Id. The search is case-insensitive, punctuation is ignored. If a satellite has multiple names, any name may be used. For example, the RS-44 satellite is found by entering "RS-44", "rs44" or "DOSAAF-85" in the search box.

The numbers on the bottom bar show the total number of satellites in the database and the number of those that match the filters and search string.

Viewing Details

Right-click on a satellite and click on **Satellite Details**, or press **Ctrl-D**, to open the Satellite Details window.

Renaming Satellites

To rename a satellite, press **F2**, or right-click on the satellite and click on **Rename** in the popup menu.

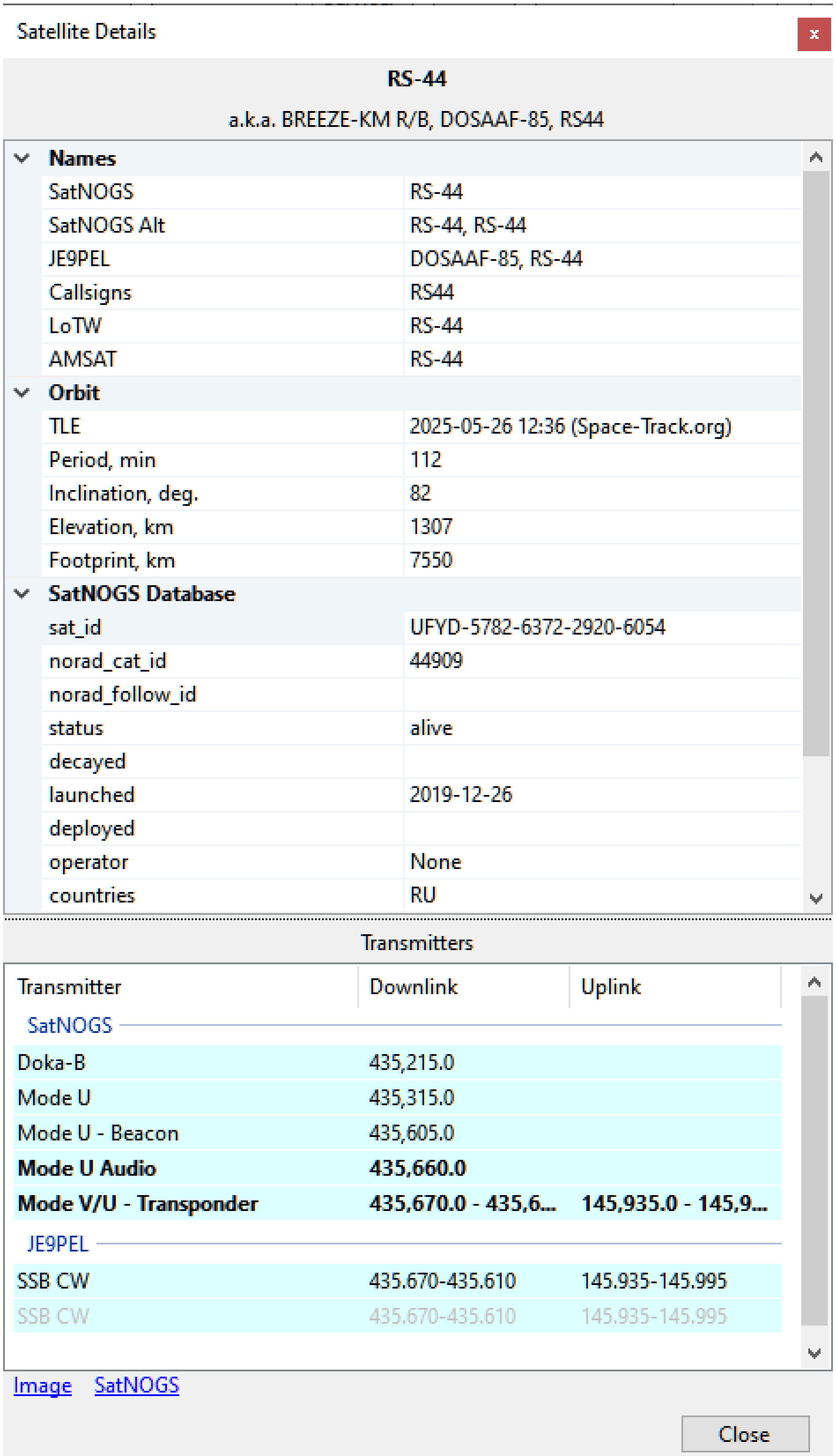
Editing Satellite Groups

The right panel of the window shows the satellite groups. See the Creating Satellite Groups section for information about creating and editing the satellite groups. The following editing commands are available:

- the **[+]** button creates a new group. Press **F2**, or use the popup menu, to rename the group;
- the **[>]** and **[<]** buttons add or remove the satellites to/from the group;
- drad-and-drop from the satellite list to the group adds the satellite to the group;
- drag-and-drop of a satellite between the groups moves it to another group;
- drag-and-drop with the **Ctrl** key down adds a copy of the satellite to another group;
- drad-and-drop re-orders the groups, or the satellites in the group;
- the Delete key deletes the group, or the satellite from the group;
- the popup menu of the group or satellite is an alternative way of accomplishing the same tasks.

Satellite Details Window

The Satellite Details window is available via the right-click menu in many panels, including the Satellites and Groups window, Current Group panel and Frequency Scale.



The transmitters are highlighted according to the downlink band. Ham transmitters are bold, inactive ones are grayed.

The blue links at the bottom of the window open the web pages with extra information about the satellite.

See the Satellite Data section for the description of the satellite data available in SkyRoof.

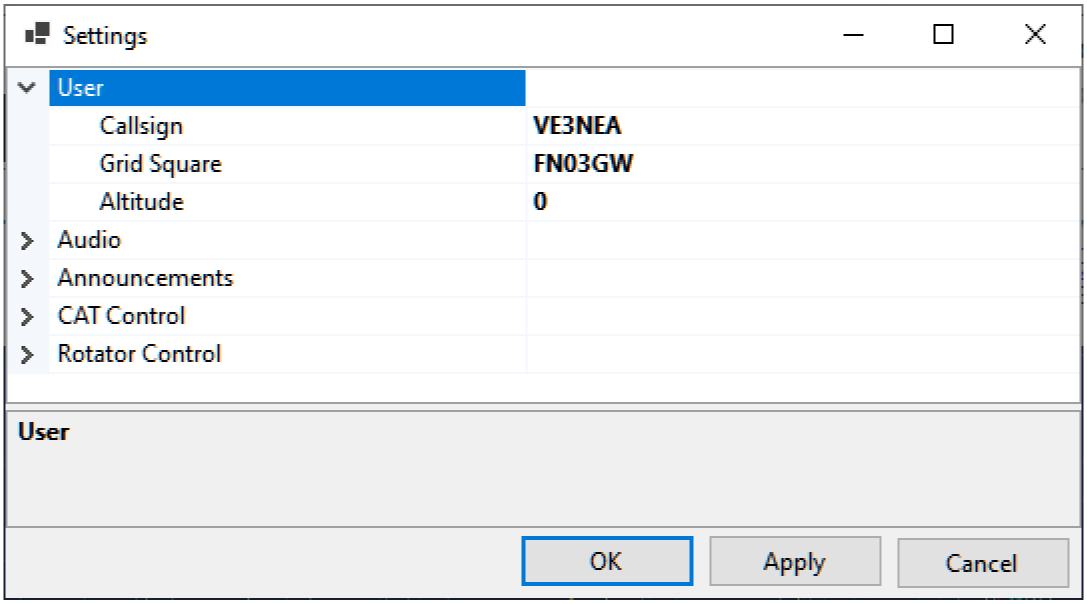
Satellite Highlighting

The satellite entries in various windows and panels, such as Satellites and Groups window, Satellites Details window and Current Group panel, are highlighted according to their properties:

- **Cyan background** - the satellite has at least one transmitter in the UHF band;
- **Yellow background** - the satellite has at least one transmitter in the VHF band;
- **Bold text** - the satellite has the Amateur Service (Ham) flag in the database;
- **Grayed text** - the satellite is not Alive;
- **Striked-out** - the orbit elements (TLE) are not available for this satellite.

Settings window

Click on **Tools / Settings** in the main menu to open the Settings window.



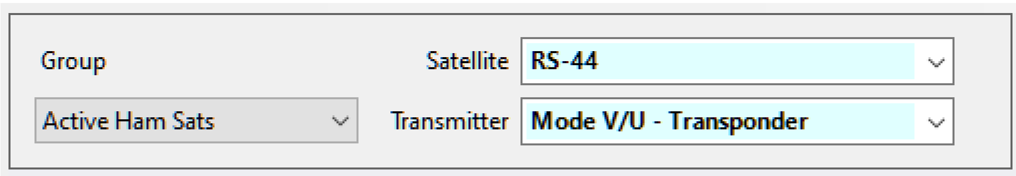
This dialog allows you to edit the settings of SkyRoof. In particular, here you can change the **Callsign**, **Grid Square** and **Altitude** that you entered on the first run of the program.

For the description of other settings see these sections:

- Setting Up Audio
- Setting Up Voice Announcements
- Setting Up CAT Control
- Setting Up Rotator Control
- Setting Up Transverter

Satellite Selector

Satellite Selector is the panel on the toolbar where you can select the satellite group, the satellite within the group, and the transmitter of the satellite:



Move the mouse cursor over the **Satellite** or **Transmitter** drop-down box to see detailed information about the selected item on the mouse tooltip.

See the Creating Satellite Groups section for the instructions how to create and edit the groups.

Satellite Photo

The **Satellite Photo** widget on the toolbar shows a thumbnail image of the currently selected satellite:

Satellite Photo

The image is taken from the SatNOGS satellite database. Hover the mouse cursor over the thumbnail to see the satellite name as a tooltip.

The widget is empty when the selected satellite has no image available in the database.

Image Cache

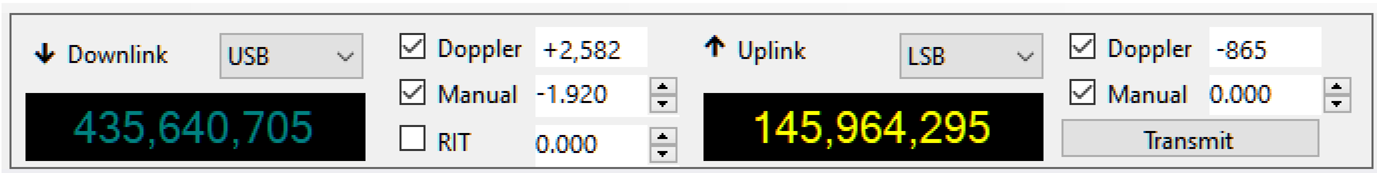
Downloaded thumbnails are cached on disk so they only need to be fetched once. The cache is kept in the `sat_images` sub-folder of the Data Folder. You may delete this folder at any time to force SkyRoof to re-download the images.

Visibility

The widget hides itself automatically when the main window is too narrow to fit all of the other toolbar widgets. Resize the main window wider to bring it back into view.

Frequency Control

Frequency Control is the panel on the toolbar that allows you to read and control the frequencies of the SDR receiver, external receiver and external transmitter:

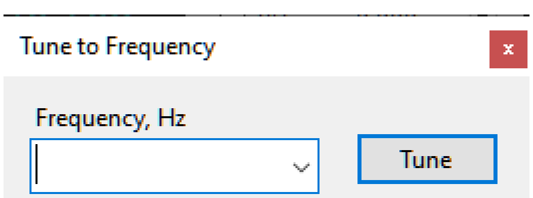


Downlink

The left hand part of the panel represents the receiver settings that apply to both SDR and external radio.

Label

When the receiver is tuned to a downlink transmitter of some satellite, the label "Downlink" appears; when it is tuned to a terrestrial station, the label "Terrestrial" is displayed. To tune to a downlink, select some satellite in the Satellite Selector panel, or select a different transmitter from the drop-down list, or click on the satellite name in any panel. To tune to a terrestrial signal, click on it on the Waterfall Display or on the Frequency Scale, or click on the downlink frequency display and enter the frequency in the **Tune to Frequency** window:



Mode

Select the mode manually for every satellite transmitter that you are using. Your selection is remembered and restored when the transmitter is selected again.

The **Mode** selected in the drop-down box applies to the SDR receiver, if it is enabled, and to the external receiver, if RX CAT is enabled. To enable or disable the SDR or RX CAT, click on the corresponding label on the status bar.

Frequency Display

The frequency display shows either the nominal frequency of the downlink, or the frequency with all corrections applied. Right-click on the display to switch between the two frequencies.

The mouse tooltip of the frequency display shows both frequencies and some other details.

When RX CAT is enabled and working properly, the frequency is shown in a bright color, otherwise the display is dimmed. The color depends on the band: yellow/olive for VHF, cyan/teal for UHF, white/gray for all other bands.

Doppler

The **Doppler** box shows the current Doppler offset of the downlink signal. This value is not editable, but Doppler correction may be enabled or disabled using the checkbox. See the Doppler Tracking section for a detailed discussion of Doppler offset calculation and tracking.

Manual

The manual correction of the downlink frequency. The frequencies of the satellite downlink signals usually differ from the nominal values in the database, for different reasons, by a few hundred Hertz and up to a couple of kilohertz. This difference is pretty stable, so it is enough to enter the correction once to have the receiver accurately tuned. SkyRoof remembers the manual correction for each satellite.

The value of the manual correction may be entered in the **Manual** box by clicking on the up/down buttons, or by spinning the mouse wheel over the box, or by typing the value directly. However, it is more convenient to adjust the correction visually, using the mouse on the Frequency Scale.

The checkbox allows you to disable the manual correction if necessary.

RIT

The RIT function is useful when listening to a conversation of two stations that are not exactly on the same frequency, or when your CQ is answered off the frequency.

The RIT offset may be entered in the RIT box, but it is more convenient to control it on the Frequency Scale.

Use the checkbox, or the commands on the **Frequency Scale**, to toggle RIT.

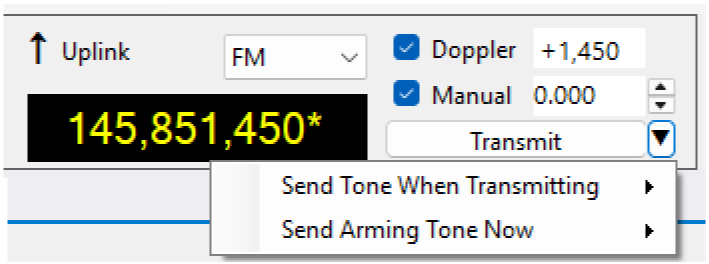
Uplink

The Uplink part of the panel is similar to the Downlink part described above. It is enabled only if the selected satellite transmitter has an uplink. The bright color of the frequency display means that TX CAT is enabled and working properly. The **Transmit** button switches the external radio between the RX and TX modes.

The **Manual Correction** setting of the uplink allows you to align your transmit and receive frequencies. See the [Frequency Scale](#) section for details.

CTCSS Tone

The FM repeater satellites relay your signal only if a sub-audible CTCSS (PL) tone is present on the uplink. The small button with a triangle, to the right of the **Transmit** button, opens the menu with two tone commands:



The button is shown only if the uplink mode is FM and the radio can switch the CTCSS encoder over CAT.

Send Tone When Transmitting

The tone that the radio sends during every transmission. Check **Enabled** to turn the encoder on and off, and select the tone below the separator, for example 67.0 Hz for SO-50 and 141.3 Hz for PO-101. The menu lists the 38 tones that all supported radios can send, which includes every tone used by the FM satellites.

The tone and the on/off state are saved separately for each satellite transmitter, so switching to another satellite applies the tone of that satellite, and turning the tone off does not lose the tone that you selected.

If the radio has no CAT command for the tone frequency, for example the IC-706MKIIG, the tone entries are disabled, and only **Enabled** is available. Select the tone frequency on the radio in that case.

Send Arming Tone Now

SO-50 has a 10-minute on-board timer that must be armed before the satellite passes traffic. Select 74.4 in this menu to send a 2-second carrier with the 74.4 Hz arming tone. Sending it again within the 10-minute window restarts the timer. When the burst ends, SkyRoof restores the tone and the on/off state of the **Send Tone When Transmitting** command.

The command is disabled if the radio cannot select the tone or cannot be keyed over CAT. Key the radio manually for about 2 seconds with the arming tone selected in that case.

> **NOTE**
>
> On Icom radios, the **Auto Repeater** function turns CTCSS off when the frequency changes, and the Doppler correction changes the frequency continuously. SkyCAT turns Auto Repeater off automatically on the radios whose CI-V supports it. On the IC-910 and IC-706MKIIG, and when the radio is controlled via rigctld, turn Auto Repeater off in the menu of the radio.

Dial Knob

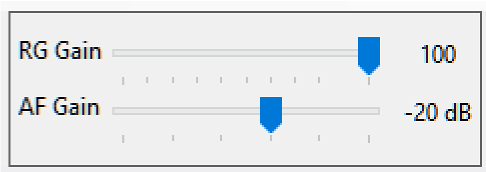
The dial knob of the transceiver can be used to tune the frequency when CAT control is enabled and the **Ignore Dial Knob** option is set to **false**.

When both RX CAT and TX CAT are enabled, the dial knob controls the receiver frequency.

> **NOTE**
>
> When the radio is in the SAT mode, the NOR/REV switch should be in the NOR position for correct tuning with the dial knob.

Gain Control

The Gain Control panel on the toolbar has two sliders to control the RF and AF gain:

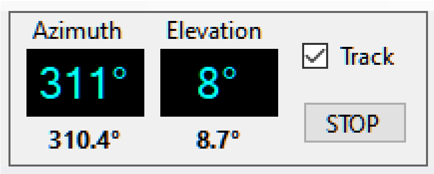


The **RF Gain** slider is enabled only if the **Single Gain** setting is set to **true** in SDR settings.

To adjust the gain, click on the slider, or spin the mouse wheel over it, or drag the thumb control.

Rotator Control

The Rotator Control panel on the status bar shows the current position of the selected satellite and the antenna bearing, if the rotator control function is enabled:

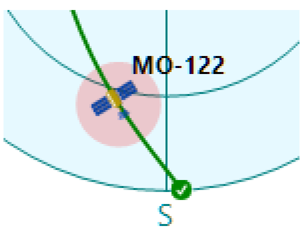


Display

The large Azimuth and Elevation display shows the satellite location, the small numbers below it show the antenna bearing.

The satellite location is dimmed when the rotator control function is disabled. Click on **Rotator** on the status bar to enable or disable this function.

When rotator control is enabled, the current antenna bearing is marked on the Sky View panel with a red spot:



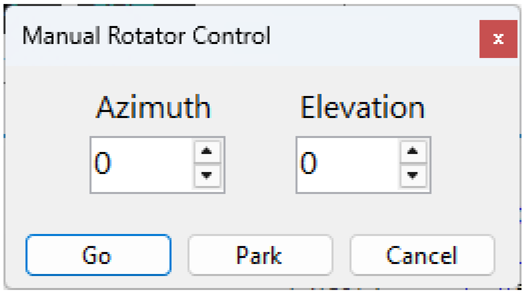
Tracking

When rotator control is enabled but the **Track** checkbox is not ticked, the panel only displays the antenna bearing but does not attempt to change it. Tick the **Track** checkbox to start tracking. Note that the check box is cleared when you switch to another satellite.

In the satellite tracking mode, the antenna bearing turns pink if it differs from the satellite position by more than 1.5 the **Step Size** setting entered in the rotator settings.

Manual Control

Click on the satellite position display to open the **Manual Rotator Control** window:



- click on **Go** to rotate the antenna to the specified azimuth and elevation;
- click on **Park** to rotate the antenna to the parking position.

Stopping

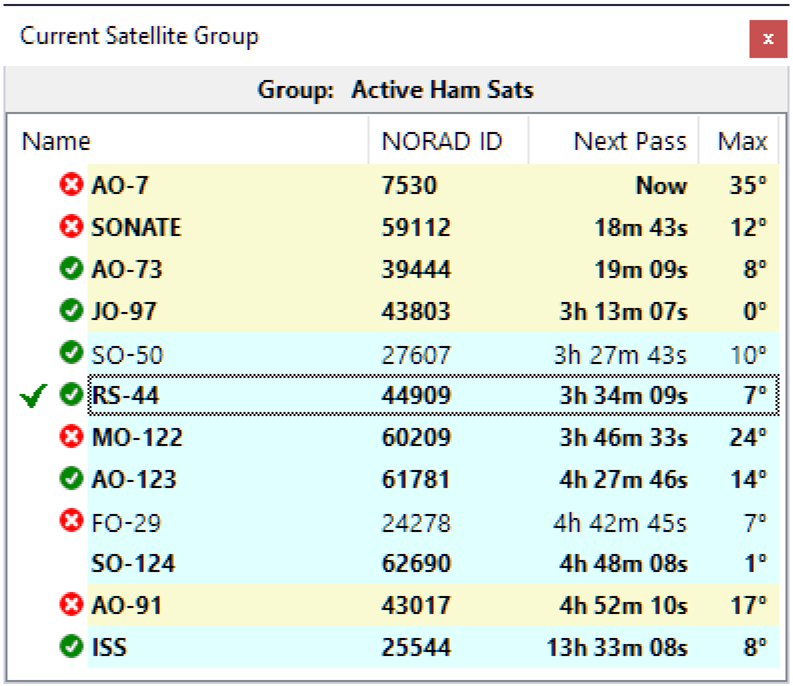
To stop antenna rotation, either manual or due to the satellite tracking, click on the **Stop** button.

See Also

- Smart Antenna Rotation
- Auto Selection Panel — tracking the antenna automatically on scheduled passes

Current Group

The Current Satellite Group panel shows the list of the satellites in the currently selected group:


Current Satellite Group

Group: Active Ham Sats

Name	NORAD ID	Next Pass	Max
AO-7	7530	Now	35°
SONATE	59112	18m 43s	12°
AO-73	39444	19m 09s	8°
JO-97	43803	3h 13m 07s	0°
SO-50	27607	3h 27m 43s	10°
RS-44	44909	3h 34m 09s	7°
MO-122	60209	3h 46m 33s	24°
AO-123	61781	4h 27m 46s	14°
FO-29	24278	4h 42m 45s	7°
SO-124	62690	4h 48m 08s	1°
AO-91	43017	4h 52m 10s	17°
ISS	25544	13h 33m 08s	8°

The green and red icons indicate real-time satellite status (active or inactive) according to the AMSAT web site. See [Satellite Data] for details.

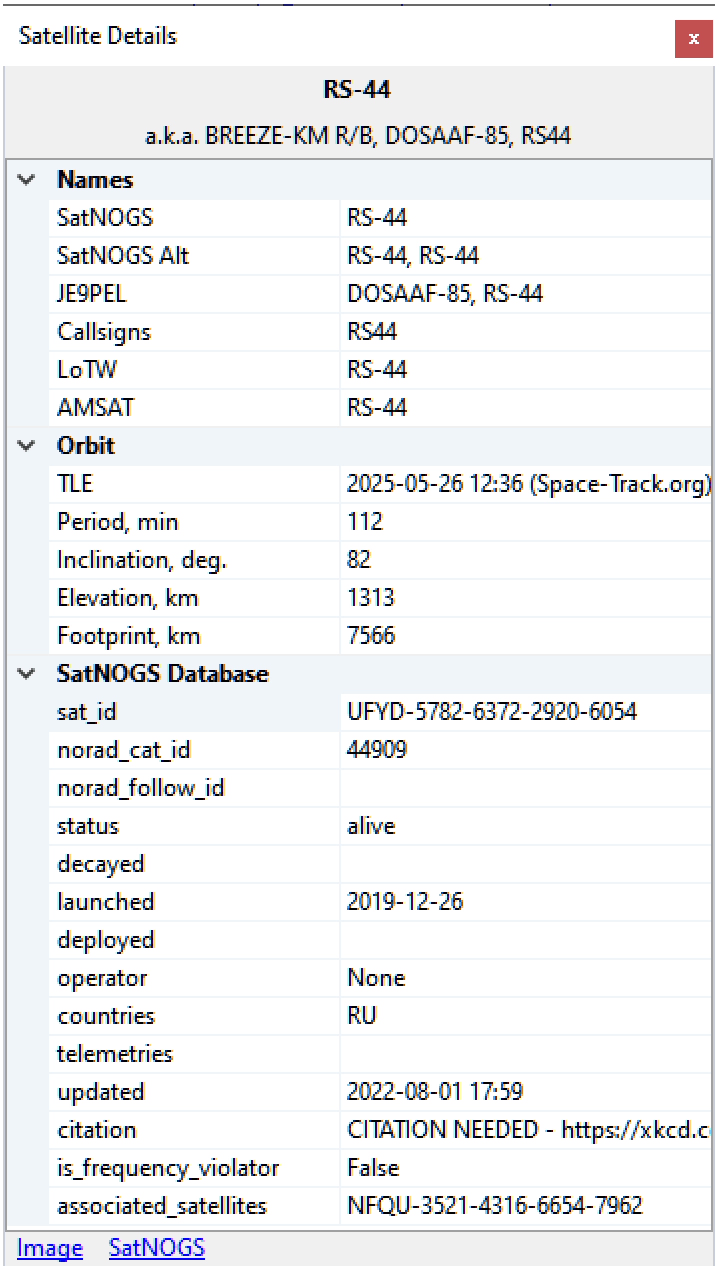
Click on a satellite to select it.

Move the mouse cursor over the satellite name to see the mouse tooltip.

Right-click on the satellite and click on **Satellite Details** in the popup menu to open the [Satellite Details Wndow].

Satellite Details

The Satellite Details panel shows information about the currently selected satellite:

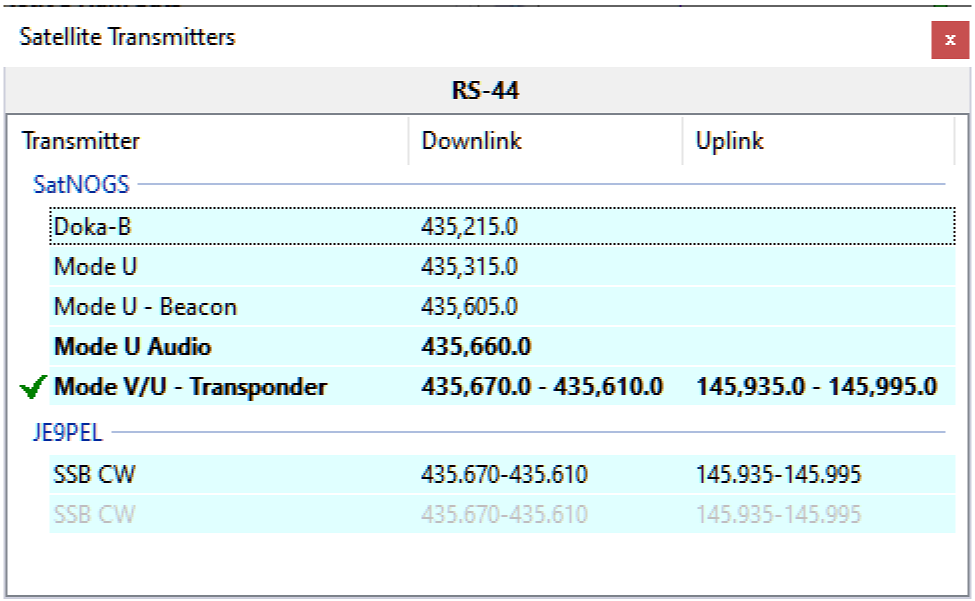


This panel displays the same information as the upper portion of the Satellite Details window. The difference is that, unlike the window, the panel may be docked anywhere in the user interface.

See the Satellite Data section for information about available data.

Satellite Transmitters

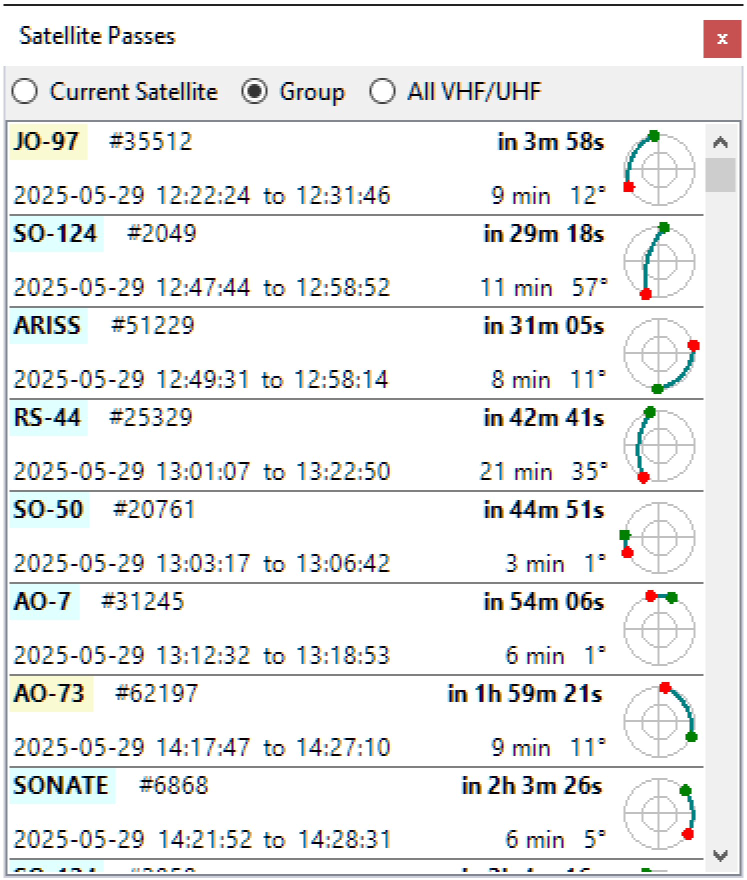
The Satellite Transmitters panel shows the list of transmitters carried by the currently selected satellite. It is similar to the bottom part of the [Satellite Details window]() but is dockable for permanent visibility:



Click on the transmitter in the **SatNOGS** section to select it. Move the mouse cursor over the transmitter name to see the details on the mouse tooltip.

Satellite Passes

The Satellite Passes panel shows the list of predicted satellite passes over your location:



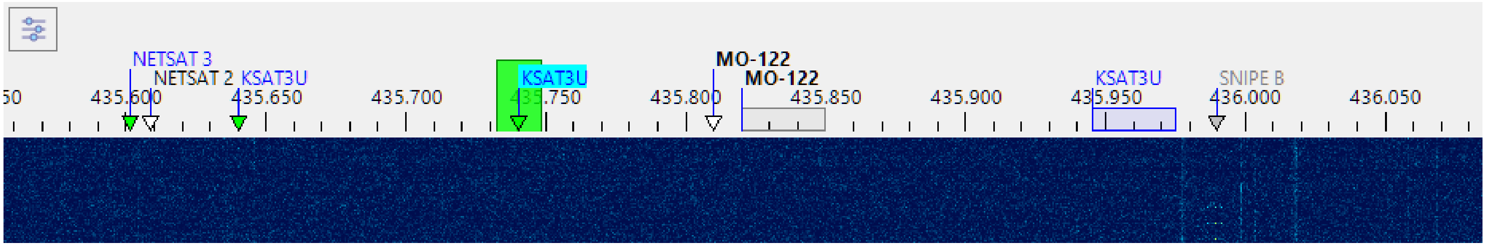
The radio buttons at the top allow you to view either the passes of the selected satellite, of all satellites in the currently selected group, or all satellites carrying a VHF or UHF downlink transmitter and thus visible on the Frequency Scale.

Click on a pass to make the satellite selected and to view its trajectory on the Sky View panel.

Move the cursor over the satellite name to view extra information on the mouse tooltip, or right-click to open the Satellite Details window.

Frequency Scale

The frequency scale appears on the [Waterfall Display](#) panel, above the waterfall:



Satellite Transmitters

The Doppler-corrected frequencies of the satellite transmitters are marked on the frequency scale with small triangles, labeled with the satellite names:

- **green triangles** - the satellites that are currently above the horizon;
- **white triangles** - the satellites that will rise in the next 5 minutes;
- **gray triangles** - the satellites that are already below the horizon, but whose signals are still may be visible on the waterfall;

- **blue rectangles** - the transponder segments of the satellites above the horizon;
- **gray rectangles** - the transponder segments of the satellites below the horizon.

The names of the satellites that belong to the current group are shown in bold;

The current transmitter has its satellite name on the light blue background.

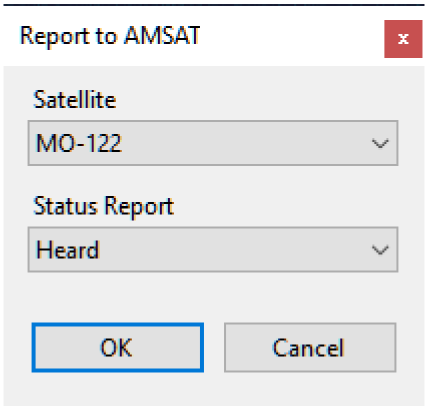
Move the mouse cursor over a satellite name to see the details on the mouse tooltip.

Right-click on a satellite name to open the popup menu with these commands:

- **Select Transmitter** - if the satellite has more than one transmitter on the same frequency, this command is enabled and allows you to set one of the transmitters as selected;

- **Add to Group** - add the satellite to one of the existing groups;

- **Report to AMSAT** - open the dialog to report your observation of this satellite to [AMSAT Live OSCAR Satellite Status Page](#)



- **Satellite Details** - opens the Satellite Details window

The frequency scale does not show all satellites at all times, only those that are currently available at your location. The satellite labels appear about 5 minutes before the AOS, and stay for a while after LOS, so that the existing signal traces could be identified, but then they disappear. This also applies to the linear satellites and their transponder segments. When the segment is not visible, you cannot tune within it using the mouse controls. The operator tunes in the transponder segment to tune to some station, or to find a clear space to send CQ. If the satellite is below the horizon, this, of course, cannot be done, and if you are still using the tuning commands, the program assumes that you are trying to do something else, e.g., tune to a terrestrial signal.

SDR Receiver

The passband of the SDR receiver is shown on the frequency scale as a green rectangle. To tune the receiver:

- click on one of the satellite labels to start tracking satellite's transmitter;
- click within a blue transponder segment to select the transmitter and set the transponder offset;
- click anywhere on the frequency scale to tune to a terrestrial signal;
- drag the green rectangle to another frequency with a mouse;
- spin the mouse wheel on or near the green rectangle. Hold the **Alt** key down to increase the tuning speed.

The effect of tuning depends on the transmitter selection:

- when tuned to a terrestrial signal, tuning the SDR receiver just changes the receiver frequency;
- when a satellite transmitter is tracked, tuning adjusts the **Manual Offset** of the satellite (see below);
- if a satellite transponder is selected, tuning changes the receiver offset within the transponder segment.

Another way to tune the SDR receiver is to use the [Frequency Control]() on the toolbar.

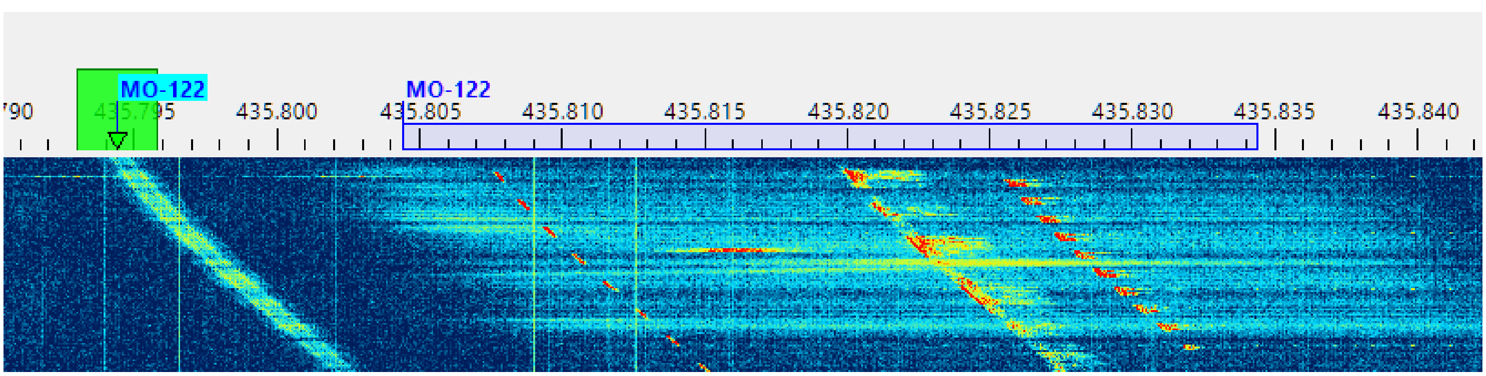
If [RX CAT]() is enabled, tuning the SDR receiver also tunes the external radio to the same frequency.

> **NOTE**
>
> Get a mouse with a free-spinning wheel, such as Logitech MX Master 3S, this makes tuning much easier.
>
> 

Manual Offset

The Manual Offset setting compensates for the transmitter frequency error, see [Frequency Control]() and [Doppler Tracking]() for details.

The offset value is usually the same for all transmitters of a satellite, so you can adjust it for some non-transponder transmitter before using the transponder. Most satellites with a transponder also carry a telemetry or beacon transmitter that you can use to set the manual offset. The screenshot below shows the telemetry signal of the MO-122 satellite, and the green rectangle perfectly aligned with its frequency by adjusting the Downlink Manual Offset:



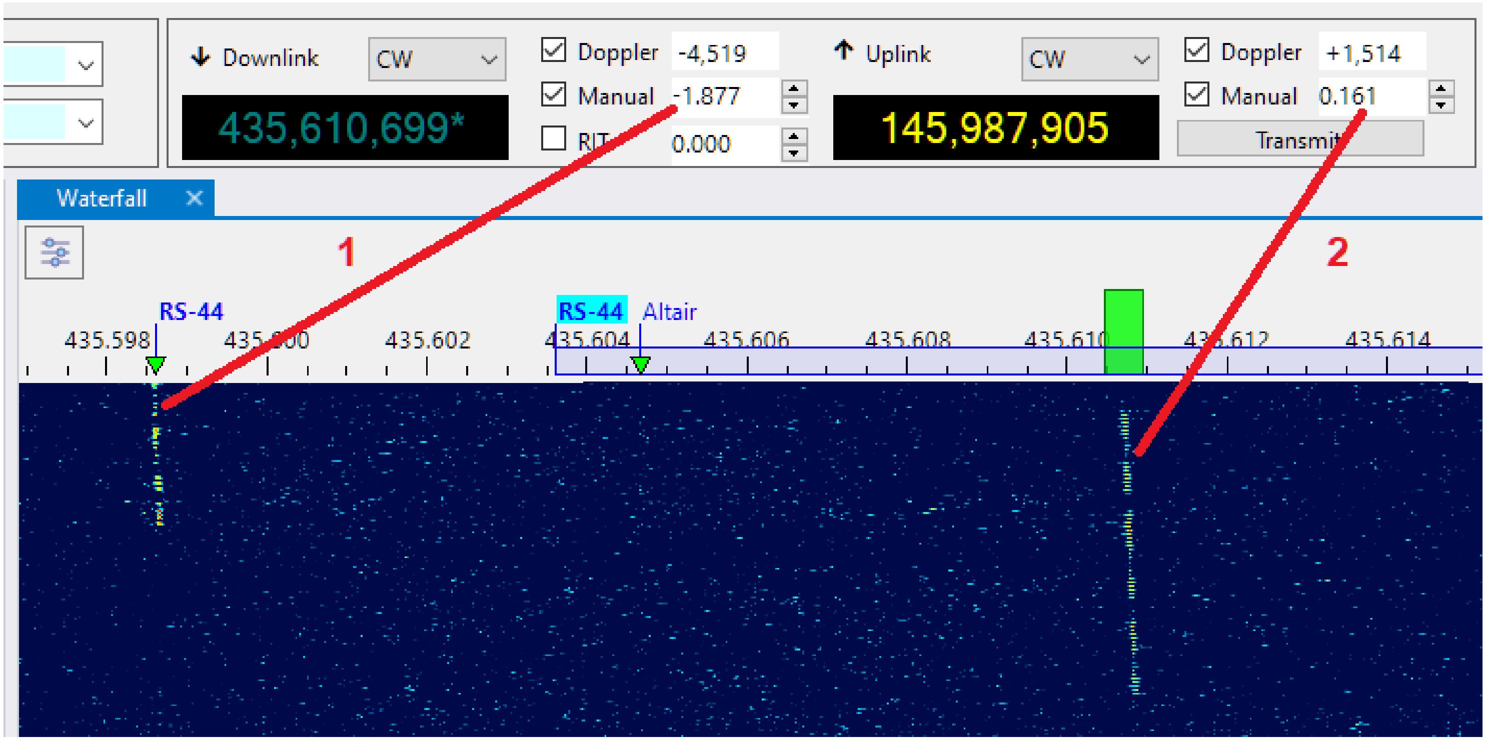
Aligning The Uplink and Downlink Frequencies

The screenshot below shows how to set Manual Correction for the uplink and downlink for a linear transponder, such as RS-44.

1. Select the beacon transmitter of the satellite and adjust the Downlink Manual Correction setting to align the transmitter label and its signal trace on the waterfall. For

RS-44 the required correction is about -1900 Hz. If your offset is significantly different from this value, this means that your SDR requires PPM calibration.

2. Select the transponder transmitter and find a clear frequency within the transponder segment. Send a sequence of dots and adjust the Uplink Manual Correction to align the center of the green rectangle with the trace of your signals coming from the satellite.

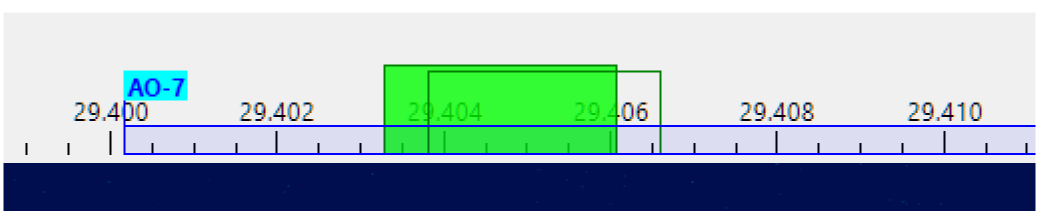


These adjustments need to be done only once. They stay the same, within a few tens of Hertz, between the satellite passes.

When TX CAT is enabled, RX CAT is disabled and the Ignore Dial Knob is set to false in the Settings, it is possible to adjust the Uplink Manual Offset using the dial knob on the radio.

RIT

While the RIT function could be controlled using the Frequency Control panel, it is more convenient to do this on the frequency scale:



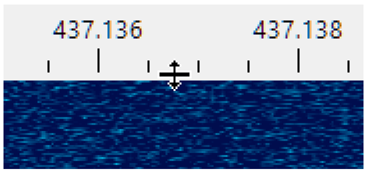
When RIT is enabled, the clear rectangle shows the current receiver passband, while the green rectangle stays on the main frequency.

- spin the mouse wheel on or near the green rectangle while holding the **Ctrl** key down: this enables RIT and tunes its offset;

- spin the mouse wheel on or near the green rectangle WITHOUT holding the **Ctrl** key down: this disables RIT and tunes the main frequency;
- right-click on or near the green rectangle to turn RIT on and off.

Resizing

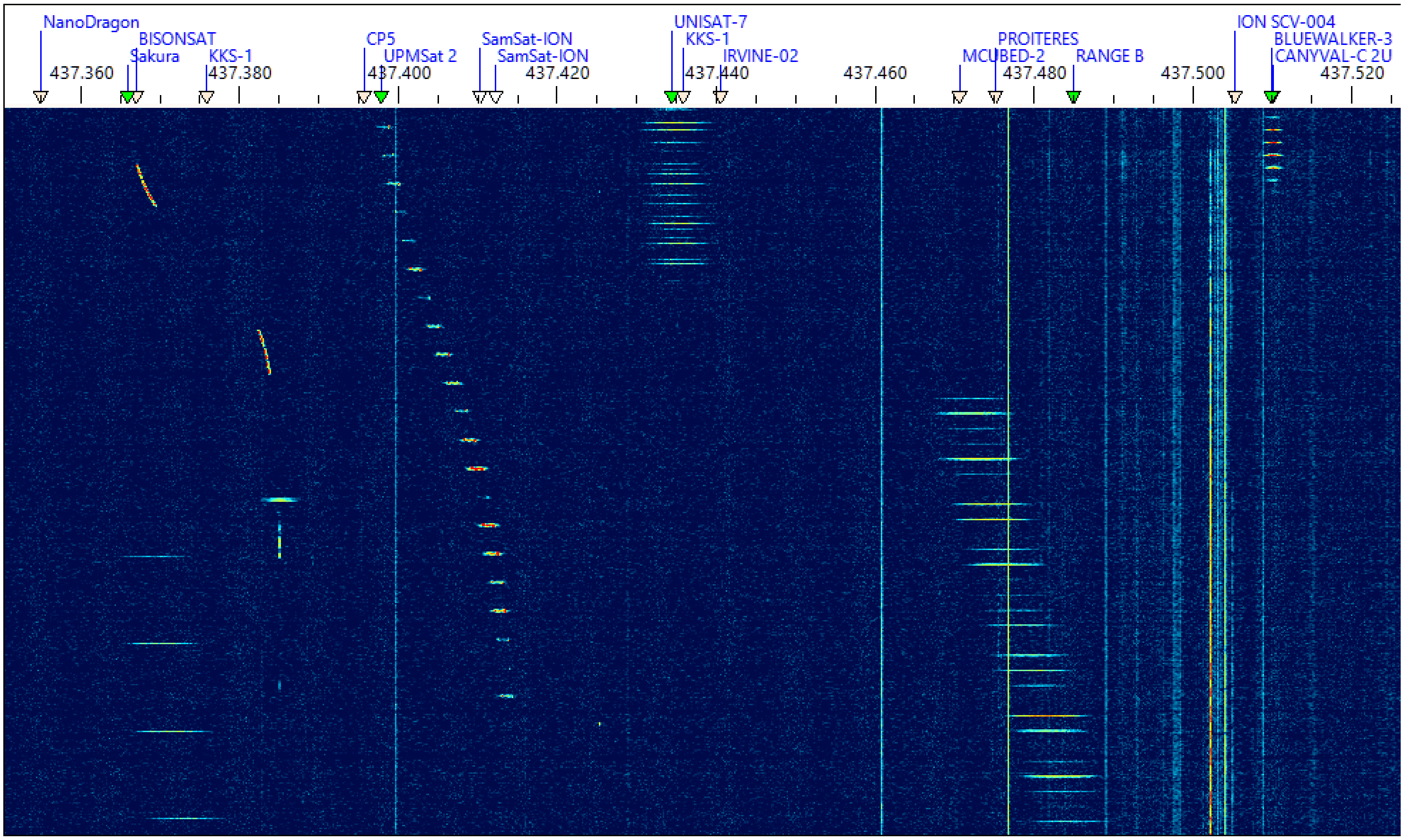
Adjust the height of the frequency scale by dragging the splitter between the scale and the waterfall:



Waterfall Display

Using Waterfall Display

The waterfall display and associated [Frequency Scale]() is the central piece of SkyRoof that integrates most of the functions available in the application:



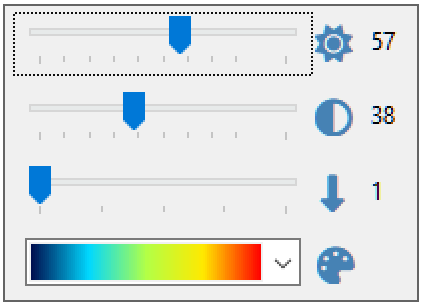
The waterfall spans over 3 MHz of spectrum (depending on the SDR model) and covers the whole satellite segment, 435-438 MHz, on the 70 cm band. On the 2 m band the satellite segment is only 200 kHz wide, 145.8-146 MHz, so it also fits completely in the waterfall.

- Zoom in and out using the mouse wheel
- Pan by dragging the waterfall horizontally with your mouse

A mouse-click on the waterfall display:

- tunes the SDR and external radio to a terrestrial signal
- or, if the frequency is within the transponder segment of a passing satellite, selects that satellite and sets the transponder offset to the clicked signal.

A click on the **Sliders** button in the top left corner of the panel opens the sliders that adjust brightness, contrast and scrolling speed of the waterfall, and select a color palette:



See Also

- Frequency Scale
- Doppler Tracking

Waterfall Display Characteristics

Finding and tracking satellite signals on the VHF and UHF bands is a difficult task. These signals are weak because the output power of most satellite transmitters is in the milliwatt range. The fact that an omnidirectional antenna needs to be used to receive all in-range satellites at the same time makes this task even more difficult. On top of that, we need to see the whole 3-MHz frequency segment where the satellite signals may appear, and at the same time we want a very high resolution to examine the structure of the signals, and to tune precisely to CW, SSB and digital transmissions.

The waterfall display in SkyRoof solves these problems by computing power spectra oversampled by a factor of about 100. This has three important consequences:

1. You can zoom in the waterfall display without changing its resolution in Settings, just by spinning the mouse wheel. When zoomed in to the maximum, you can see the signals with a **100x** magnification (20 Hz resolution).

2. When zoomed out to the maximum to see the whole 3-MHz segment, the sensitivity of the waterfall display to narrowband signals improves by about **15 dB** due to oversampling.

3. Spectrum oversampling, however, increases the requirements to the hardware:

    - more CPU power is needed to compute very large spectra;
    - more texture memory is needed in the video card to store the spectra;
    - support of OpenGL 3.3 or higher is required for the zoom function.

    Please see the System Requirements section for the exact requirements.

If your computer does not meet these requirements, you can still use the pass prediction, Doppler correction, radio control and rotator control functions in SkyRoof, just disable the SDR function and close the Waterfall Display panel.

Waterfall Quality

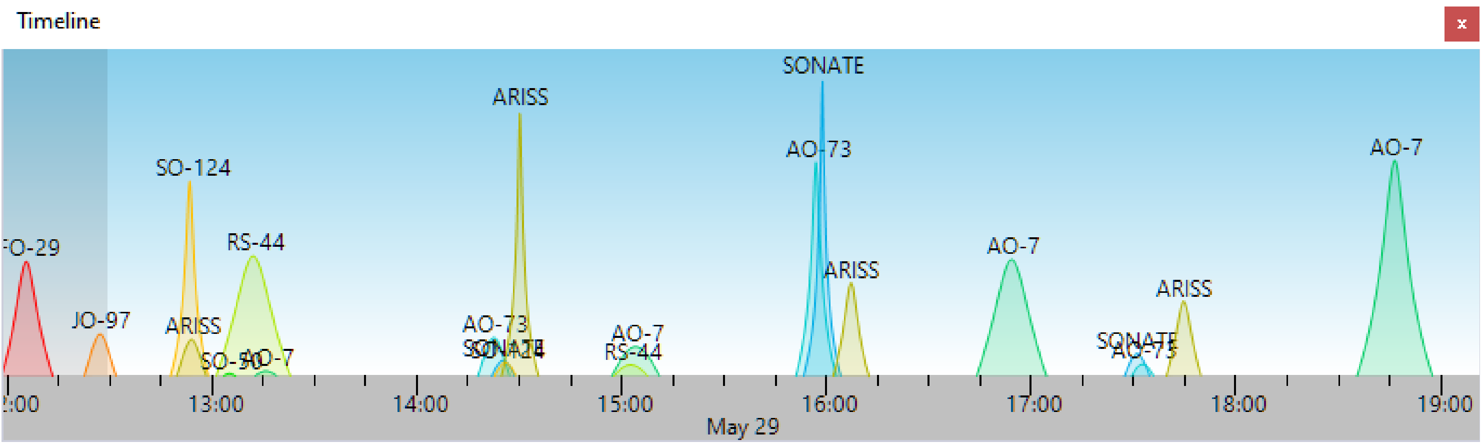
If the waterfall scrolls sluggishly or your video card struggles with the large oversampled spectra, you can lower the **Quality** setting. Open **Tools / Settings**, find the **Quality** field in the Waterfall section, and choose one of the following values:

- **Auto (maximum detail)** - the default. SkyRoof uses the largest spectrum and texture size your hardware can handle, giving the highest resolution and sensitivity.
- **High**, **Medium**, **Low (best for older GPUs)** - progressively reduce the internal spectrum and texture size. A lower quality requires less CPU power and less video card texture memory, which improves performance on older or low-end hardware.

The Quality setting only affects the internal resolution of the waterfall image. It does not change the SDR sampling rate, so all signals in the satellite segment remain visible at every quality level.

Time Line

The TimeLine panel shows the satellite elevation chart as a function of time for all satellites in the selected group:



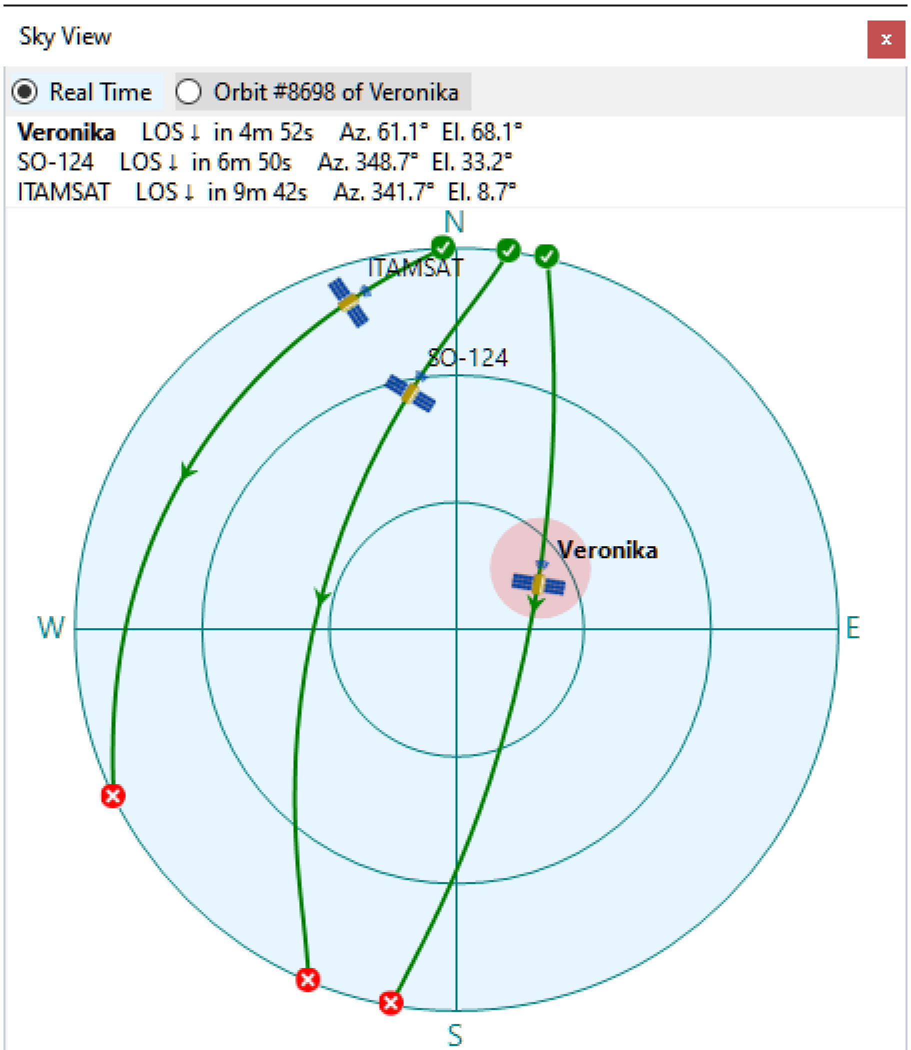
- Zoom in and out using the mouse wheel
- Pan by dragging the chart with your mouse
- Click on the satellite name to make it current, and to view the pass on the [Sky View panel](#).

The dark part of the chart represents the past time.

Sky View

The Sky View panel shows the trajectories of the satellites in the sky, as visible at your location:

The radio buttons at the top switch the chart between the real-time display showing all satellites in the selected group that are currently above the horizon, and a specific pass of a specific satellite.



The pink spot indicates the current antenna bearing if [Rotator Control] is enabled.

To select the pass to be displayed, click on it in the [Current Group panel], [Satellite Passes panel] or [Time Line panel].

Click on the satellite name next to the satellite icon to make it selected.

Earth View

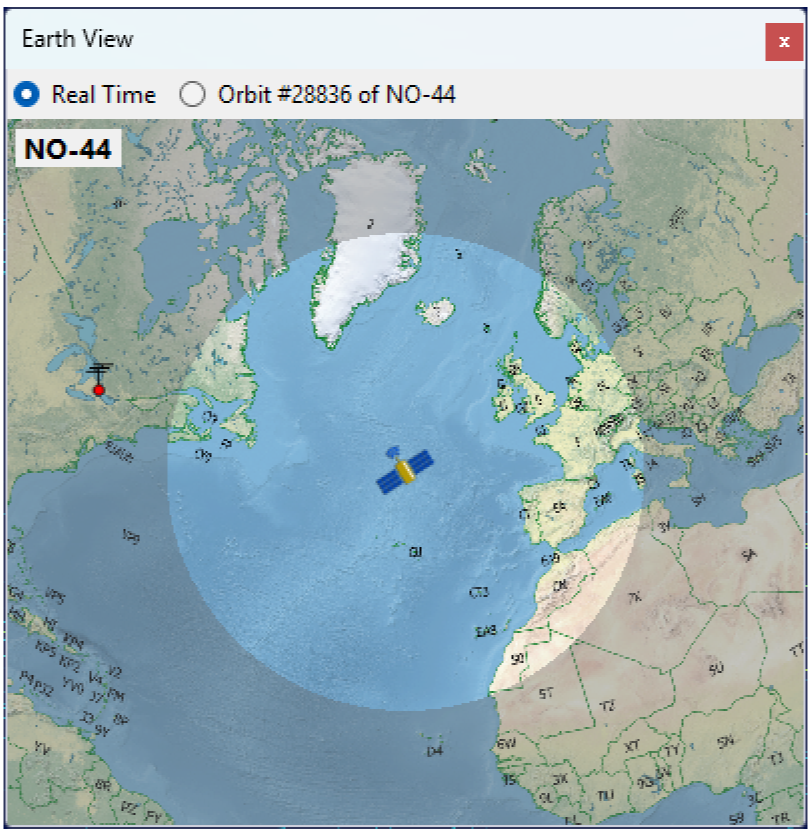
The Earth View panel shows the view of the Earth from the satellite. Use the radio buttons at the top of the panel to switch between two modes:

- **Real Time** centers the view on the satellite and highlights the area it can see at this moment.
- **Selected Pass** centers the view on your station and draws the coverage contour of the currently selected pass.

Use the mouse wheel to zoom the view in and out.

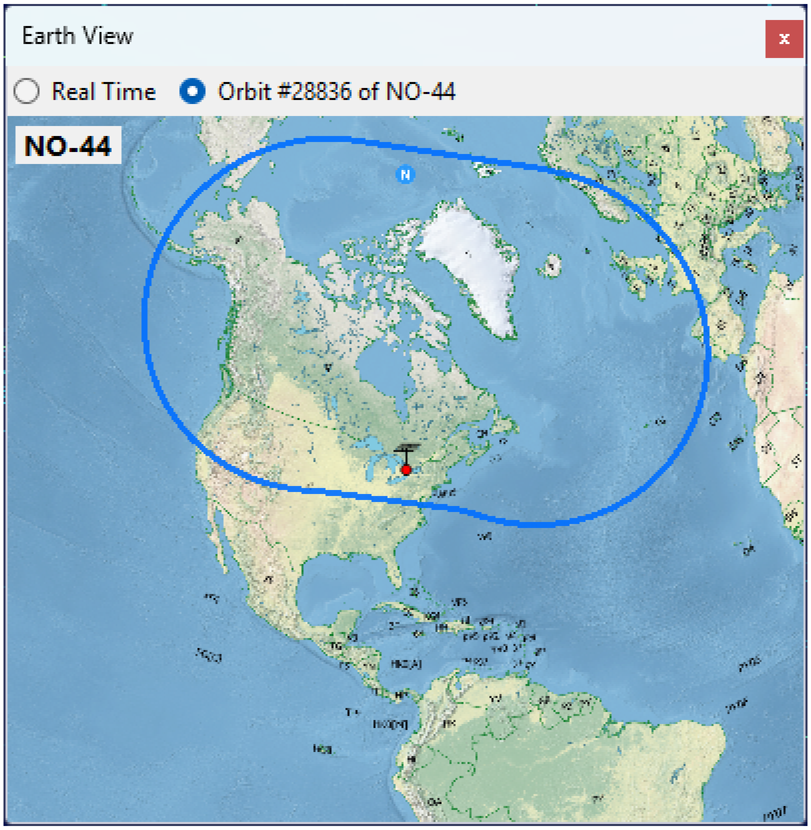
Real Time

The highlighted area is what the satellite can see from its current position.
The satellite is above the horizon for the observers located in this area.

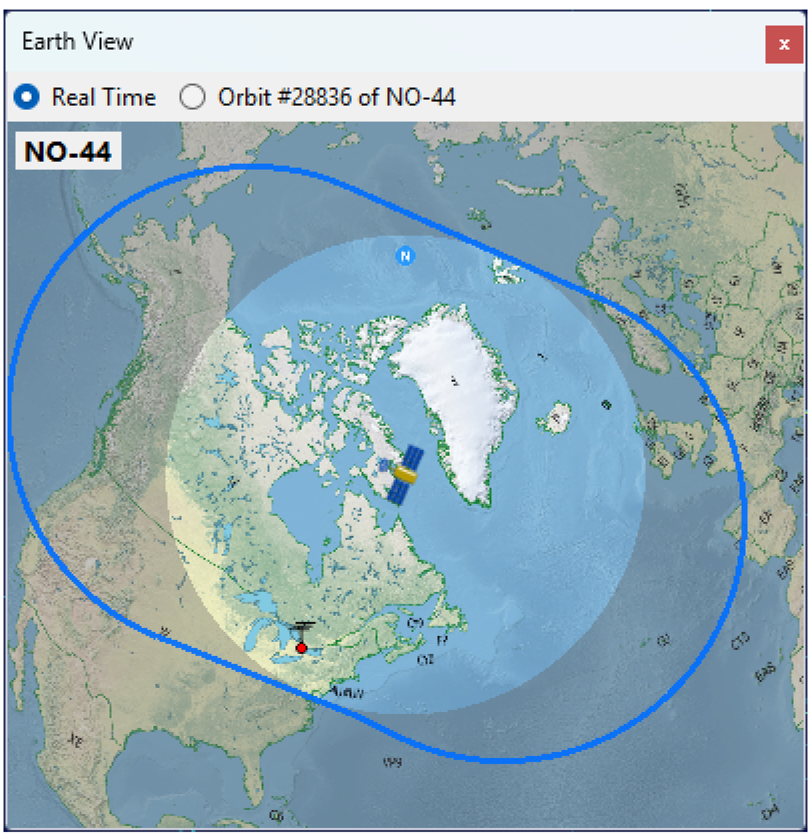


Selected Pass

The blue contour outlines the entire area swept by the satellite footprint during the pass. Any observer inside this contour will have the satellite above the horizon at some point during the pass.



While the pass is in progress, the panel switches back to Real Time mode so you can watch the satellite move across its coverage area:

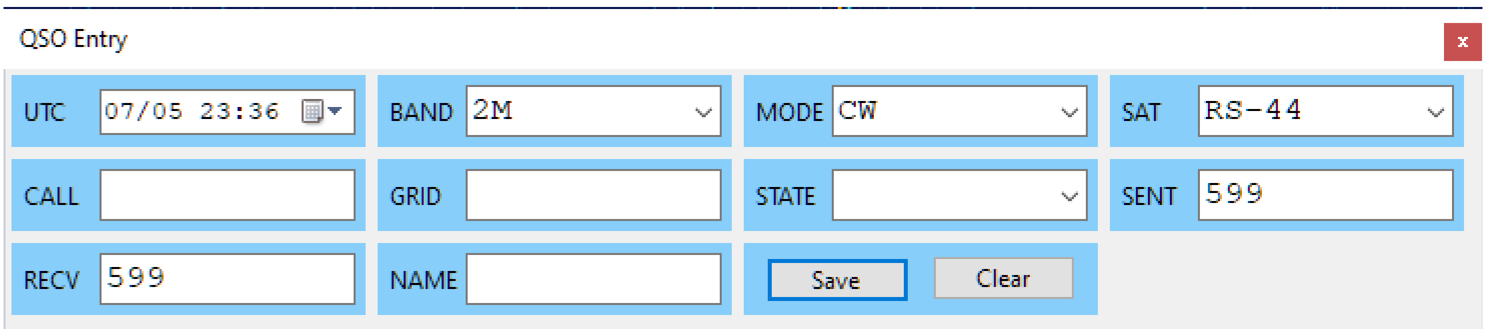


QSO Entry Panel

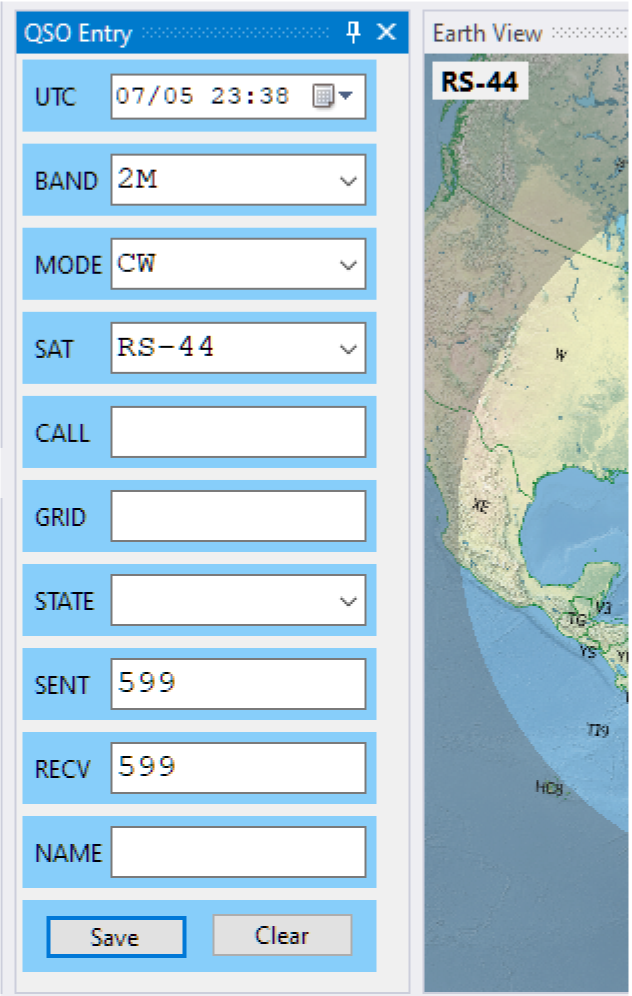
The QSO Entry panel allows the operator to enter the QSO details with minimum effort and save them to an ADIF file. This file may be later imported to the logging software for award tracking, QSL management, etc.

Configuring The Panel

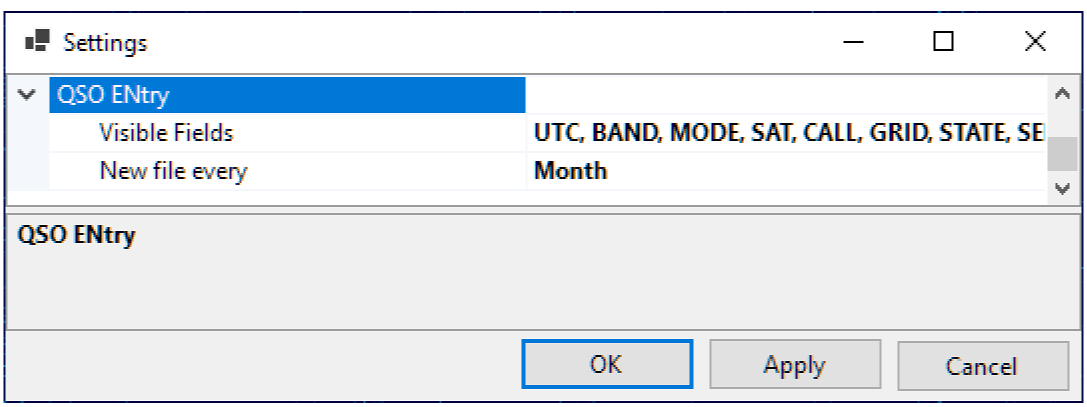
Click on **View / QSO Entry** to open the panel:



You can either dock the panel somewhere in the main window as described in [Configuring Window Layout](), or keep it floating. By resizing the panel you can arrange the input fields in a row, a column, or a matrix:



If you do not need some input fields, disable them in the Settings dialog:



Specify if a new ADIF file should be created every day, month or year, using the **New file every...** setting.

Entering The QSO Data

- **UTC** - this field shows the current UTC time. Click on it to freeze the clock or to make it run again, or enter the date and time manually if saving an old QSO;
- **Band, Mode, Sat** - These fields are populated automatically, based on the currently selected satellite transmitter. You can select different values from the drop-down lists,

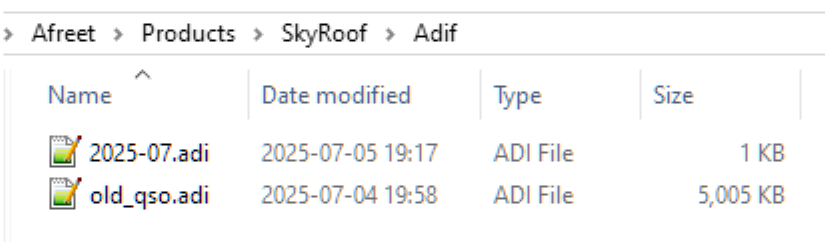
or type them in, if needed;
- **Call** - as you enter the callsign, the program tries to guess its grid square, US state and operator's name (see below).
- **Grid, State** - if the program fails to guess these values, or guesses them incorrectly, enter them manually;
- **Sent, Recv** - sent and received reports. Auto-populated with default values when the mode is set, change them manually as needed;
- **Name** - auto-populated if known to the program, otherwise enter manually.

The fields with manually entered values have a blue frame around them, these fields are not auto-populated.

The minimum QSO record consists of **UTC, Band, Mode** and **Call**, all other fields are optional.

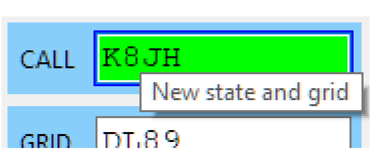
Information Lookup

When the QSO Entry panel opens, it loads all *.**adi** files in the **Adif** sub-folder of the **Data Folder** and builds the lookup lists of the grid squares, states and operator names associated with the callsigns. These lists are used to auto-populate the corresponding input fields. You can put an adif file with all your previous QSO in that folder to help the program build good lookup lists:



Callsign Status

When the program loads the adif files, it also builds the lists of worked callsigns, grid squares and states. Only the QSO with the `PROP_MODE` field set to `SAT` are counted. As you enter the callsign, the color of the input box changes to indicate the status:



Interfacing with Logging Software

The QSO Entry panel can interface with logging software, using a simple plugin system, to enter the QSO directly to the log and to get the callsign status and lookup values. The plugin DLL has to export 4 functions:

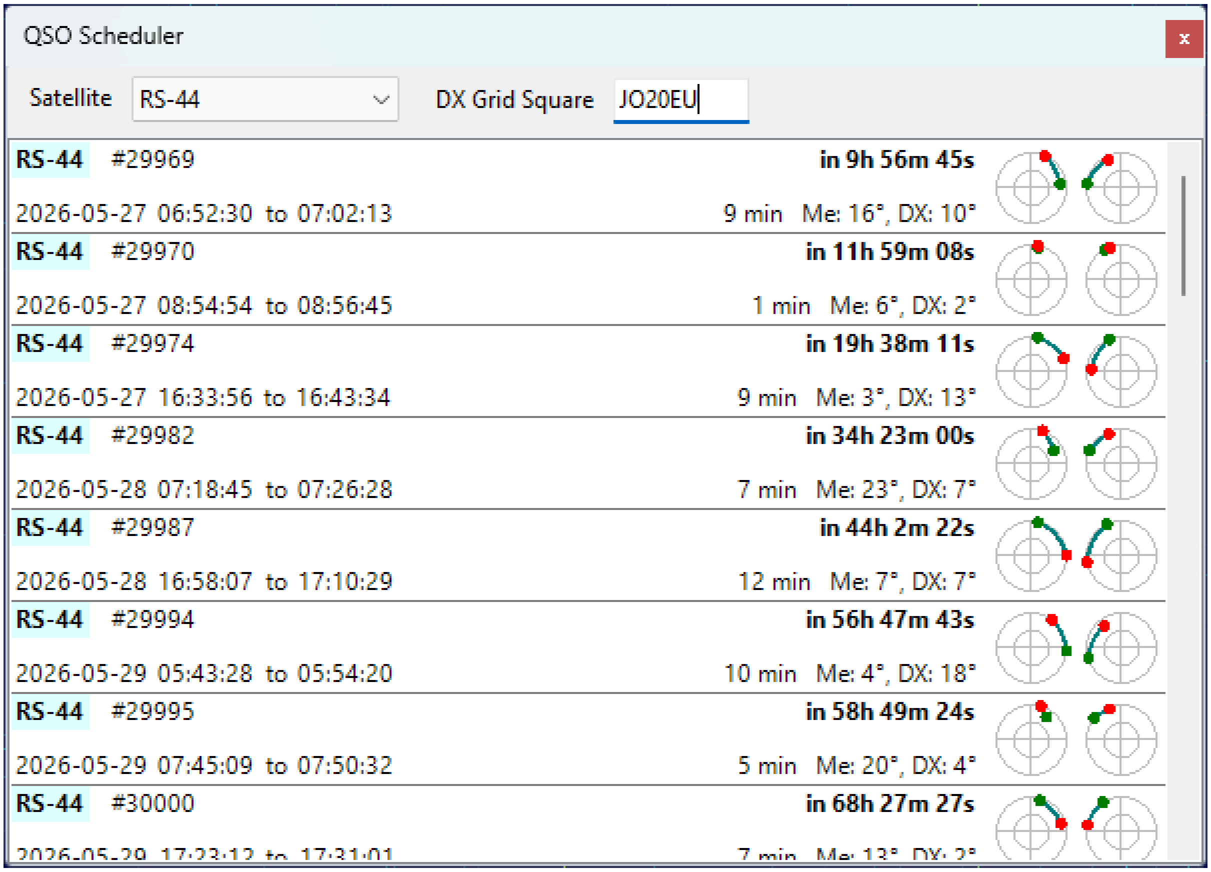
- Init();
- SaveQso(qso_data);
- GetStatus(qso_data);
- Augment(qso_data).

If you are the author of a logger, please contact me directly for details.

QSO Scheduler

The QSO Scheduler panel helps you plan a satellite QSO with a DX station by finding the time windows when the satellite is simultaneously visible from your location and from the DX location.

Click on **View / QSO Scheduler** to open the panel:



Selecting the Satellite and the DX Station

- **Satellite** - select the satellite for the QSO from the drop-down list. The list contains the satellites in the currently selected group, see Creating Satellite Groups;
- **DX Grid Square** - enter the 4- or 6-character Maidenhead grid square of the DX station. The input box turns pink while the value is incomplete or invalid.

Your own grid square and altitude are taken from the user details in the Settings Window. The DX station is assumed to be at sea level.

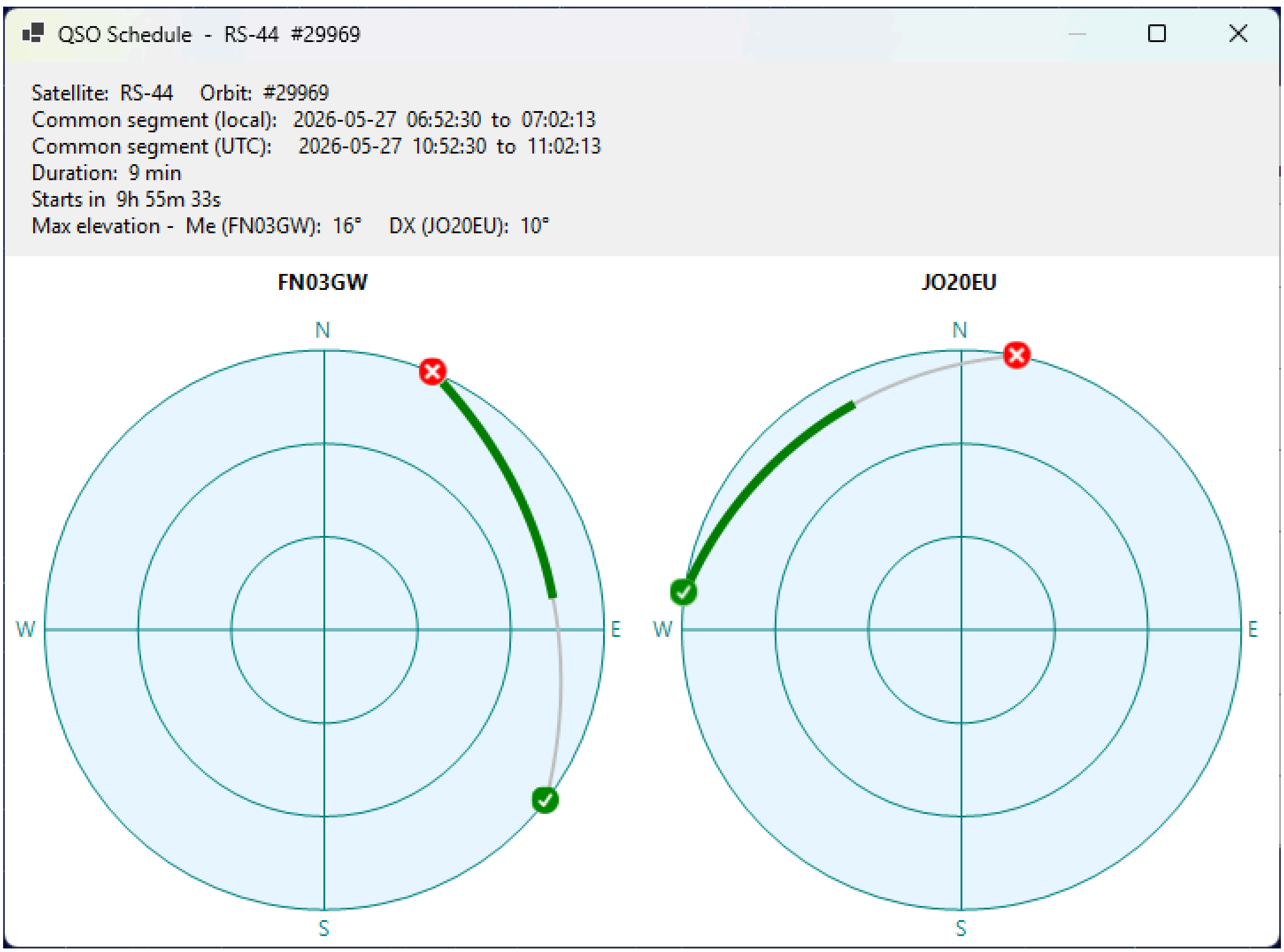
Reading the Prediction List

Each item in the list represents one common visibility window for the next 14 days, sorted by start time:

- **Satellite name** is highlighted in light yellow for VHF downlinks and light cyan for UHF downlinks;
- **Orbit number** follows the satellite name;
- **Wait time** on the upper right shows how long until the window starts, or **Now** in green if the window is already in progress;
- **Start and end times** of the common segment are displayed in local time (the label shows **Geostationary** instead for geostationary satellites);
- **Duration** of the common segment and the maximum elevation of the satellite from each station during that segment;
- **Two mini sky-views** on the right show the satellite path during the common segment, as seen from your station (left) and from the DX station (right). The green dot marks the start of the segment, the red dot marks the end.

Pass Details

Click on a list item to open the QSO Schedule window with more information about the pass:



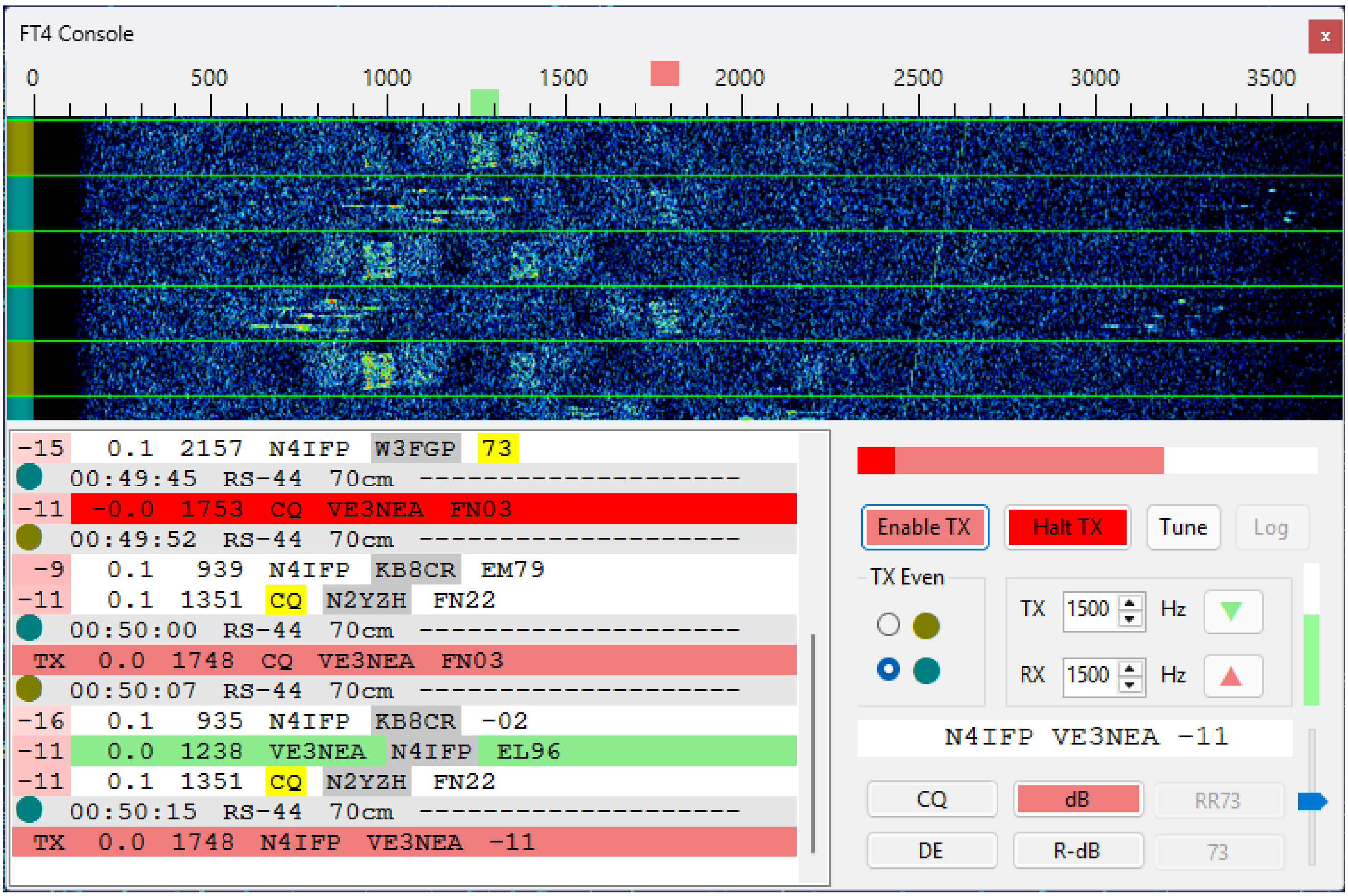
The window displays the full sky view for the pass from each station, with the entire pass arc drawn in silver and the common segment highlighted in green. The green check mark indicates the AOS point, the red X mark indicates the LOS point. The text above the charts lists the satellite, orbit number, common segment in local and UTC time, duration, time until the segment starts, and the maximum elevation of the satellite from each station during the common segment.

FT4 Console Panel

FT4 Console provides a streamlined, satellite-optimized interface for conducting FT4 QSOs via satellites with linear transponders (e.g., RS-44, AO-73, JO-97). While similar results can be obtained by running two instances of WSJT-X side by side with SkyRoof, the Console removes the need for external programs and simplifies the workflow. It offers satellite-specific features not available in general-purpose FT4 software, and an intuitive user interface that improves efficiency and usability.

Be sure to configure the [FT4 Console settings](#) before you start using the FT4 Console panel.

User Interface



Audio Waterfall Diaplay

Click with the left mouse button on the waterfall or on the frequency scale to set RX and TX audio tones:

- **click** - set the RX tone;
- **Shift-click** - set the TX tone;
- **Ctrl-click** - set the RX and TX tones.

Move the mouse cursor over an FT4 signal trace on the waterfall to see sender's callsign.

Message List

The message list shows transmitted and received messages, including your own messages coming back from the satellite.

The list auto scrolls as the new messages are added. To disable auto-scrolling, for example, when looking at the old messages, just scroll the list up manually. To resume auto-scrolling, scroll it down to the last entry.

When the mouse cursor is moved over the list, auto-scrolling is temporarily disabled to prevent clicks on a wrong entry. The frozen status is indicated by a blue frame around the list panel. The list un-freezes when you click on an entry or move the cursor out of the list, and when the cursor has not been moving for 2 seconds.

When the mouse cursor is over a received message, a label on the waterfall display indicates the frequency of the decoded signal and sender's callsign.

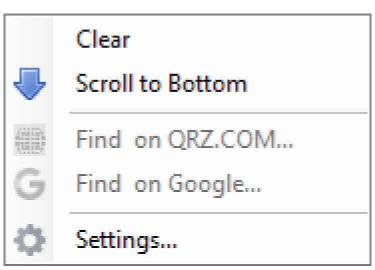
Right-click on the message list to open the popup menu. This menu has commands to clear the list and to scroll down to the bottom.

The first column in the received messages is the SNR of the received signal. It is color-coded, with weak signals represented with pale colors and strong ones with bright colors.

Sender's callsigns are color-highlighted based on their status from the logging interface. See QSO Entry for information about callsign colors.

The popup menu has commands to

- clear the message list;
- scroll to the bottom and enable auto-scrolling;
- find the callsign on QRZ.COM or on Google;
- open the Settings dialog and expand the FT4 settings.



Control Area

- **Enable TX button** - enable transmissions;

- **Halt TX button** - abort current transmission;
- **Tune button** - transmit a tone for tuning. This button works only when Enable TX is turned off. Similarly, the Enable TX works only when Tune is off;
- **Log button** - log the current QSO. Disabled until the first QSO is made;
- **TX Odd/Even** - select odd or even time slots for transmission;
- **TX #### Hz** - transmitted audio tone;
- **RX #### Hz** - primary received audio tone;
- **up and down arrow buttons** - set RX tone equal to TX tone or vice versa. The buttons are color-coded, e.g., if you click on a green arrow, the green maker on the frequency scale moves;
- **amplitude bar** - shows the peak amplitude of the input audio, see the **FT4 Console Settings** section for details;
- **TX Gain slider** - allows you to adjust the amplitude of FT4 audio signals sent to the radio;
- **message text** - the current message to be transmitted;
- **message buttons** - allow you to select the message to transmit. A single click on a button changes the message immediately, even during the transmission. Only the messages applicable in the current stage of the QSO are enabled. The currently selected message is shwon in pink.

Here are the examples of the messages:

Run Sequence

Button	Example
CQ	CQ VE3NEA FN03
dB	AA0AA VE3NEA -10
RR73	AA0AA VE3NEA RR73

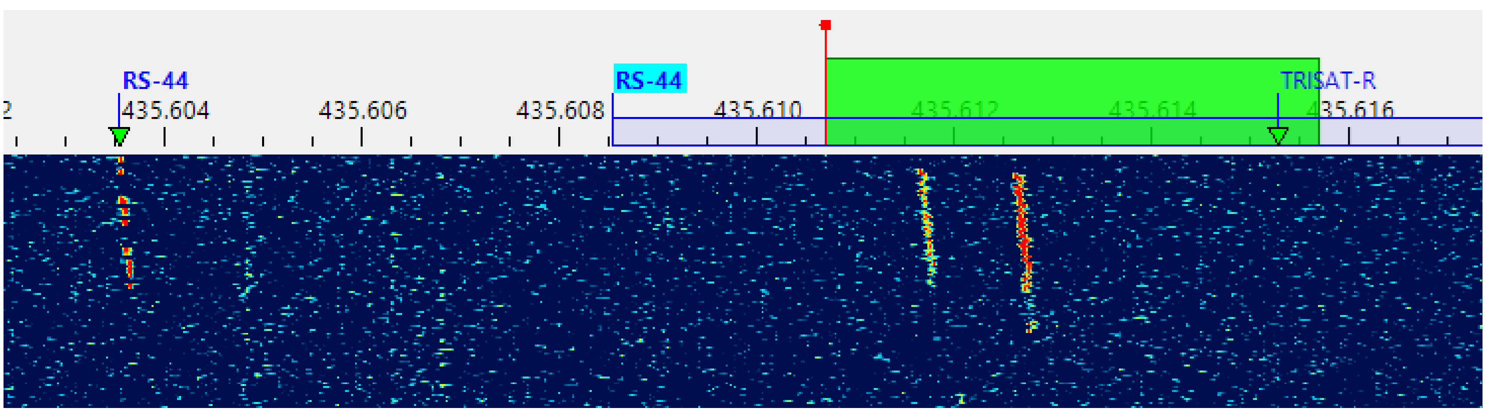
S&P Sequence

Button	Example
DE	AA0AA VE3NEA FN03
R-dB	AA0AA VE3NEA R-10
73	AA0AA VE3NEA 73

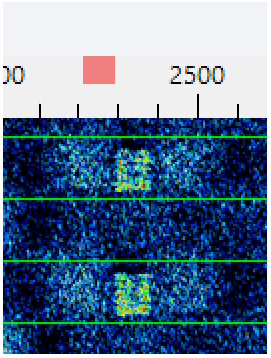
Making Contacts

At the time of this writing RS-44 was the best choice for FT4 contacts, so we will use this satellite in the examples below. When the satellite rises above the horizon and you start receiving its signals, do a quick check:

- verify your Downlink Manual Correction. This is easy to do with an SDR receiver: the label of the RS-44 CW beacon, shown on the left hand side in the screenshot below, should be aligned with its signal trace on the wideband waterfall display. If there is a mismatch, you can correct it as described in the Frequency Scale and Frequency Control sections;
- check that the receiver (SDR or external) is tuned to the frequencies where the FT4 signals are present. Usually FT4 stations operate a few kHz above the lower end of the transponder passband. You can just drag your receiver passband (green rectangle) to the required frequency.

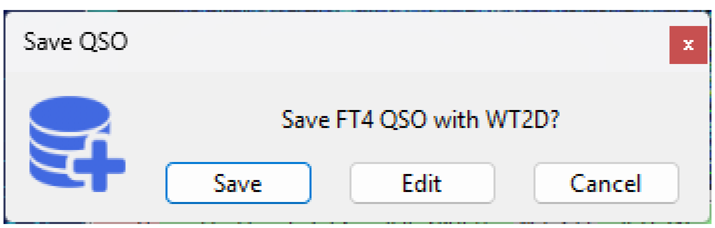


Now you can click on Tune or Enable TX to start transmitting. If all is good, you will receive your own signals back from the satellite:



The received signals are often slightly offset from the intended frequency (the pink mark on the scale), as shown above. This should be corrected by adjusting Uplink Manual Correction. You can do this manually in the Frequency Control widget, or automatically, by **Ctrl-click**'ing on your own received message (with red background by default) in the message list. Automatic correction works only if the error is less than 1 kHz. If it is greater than that, you have to find out the cause. Large frequency errors could be caused by bad SDR calibration, inaccurate system clock, old TLE data, wrong downlink offset correction, etc.

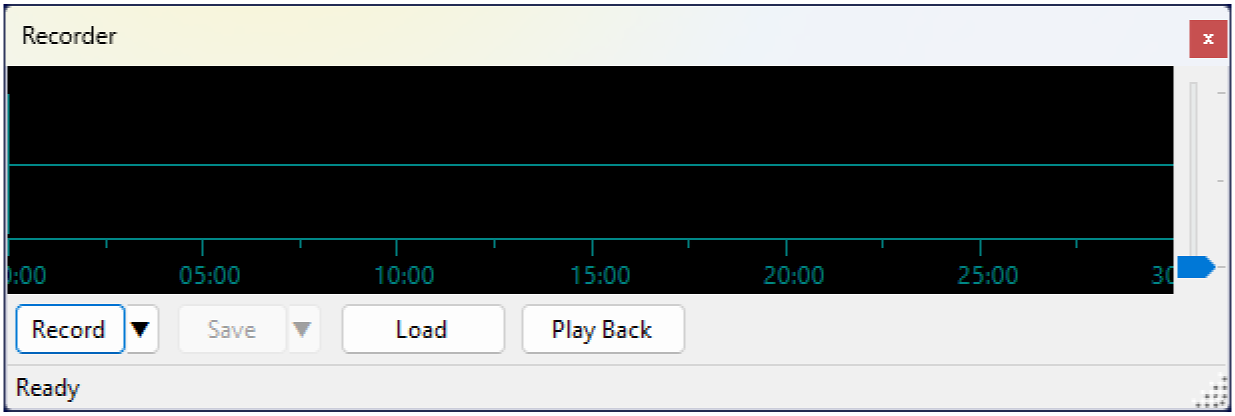
Wait until someone answers your CQ, or call another station by clicking on their message. A single click is enough, no need to double-click. The QSO sequence is followed automatically, and the Save QSO dialog pops up when the QSO is completed:



Click on **Save** to save the QSO , or click on **Edit** to load the QSO details in the QSO Entry form. See QSO Entry for information about editing and saving QSO.

Recorder Panel

The Recorder panel records the received audio or I/Q signal and plays it back later. Up to 30 minutes of data can be stored in the recording buffer.

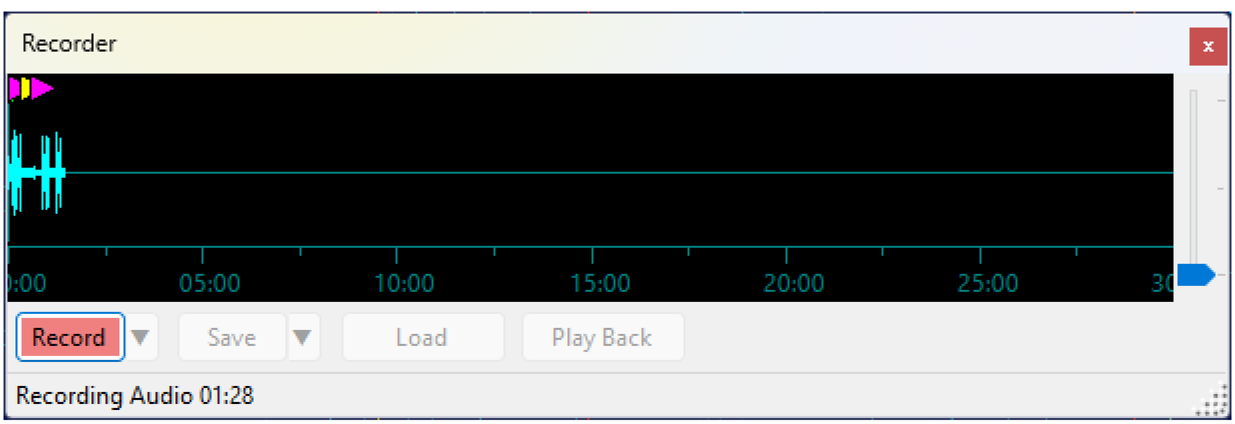


Recording

Click **Record** to start recording audio. To choose what to record, click the ▼ button next to Record:

- **Record Audio** — records the demodulated audio signal (mono);
- **Record I/Q** — records the complex baseband I/Q signal (stereo).

The button turns red and the status bar shows the recording type and elapsed time while recording is active. Click **Record** again to stop.



The waveform display shows the recorded signal growing from the left. The full 30-minute buffer width is always shown during recording, so the waveform expands to fill it as the recording progresses.

Event Markers

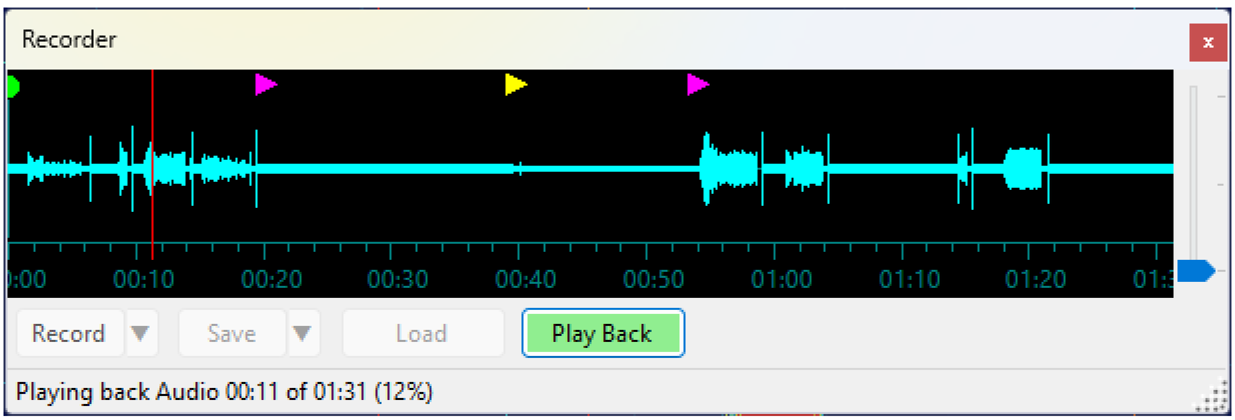
While recording is active, the panel automatically tracks changes in the selected satellite, transmitter, and mode, as well as QSOs saved in the QSO Entry panel. These events appear as colored markers along the top edge of the waveform:

- **green circle** — satellite selection changed;
- **magenta triangle** — transmitter selection changed;
- **yellow triangle** — demodulation mode changed;
- **cyan square** — QSO saved.

Move the mouse cursor over a marker to see a tooltip with the details of that event.

Playback

Click **Play Back** to play back the recording. The button turns green and a red vertical line moves across the waveform to show the current position. The status bar shows the elapsed time, total duration, and progress percentage.

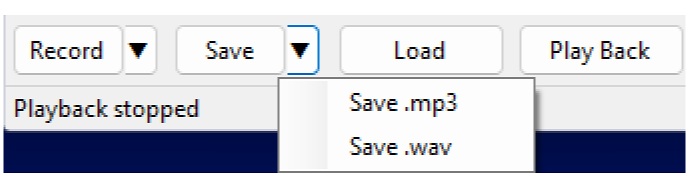


Click anywhere on the waveform during playback to seek to that position.

Click **Play Back** again to stop playback.

Saving and Loading

Click **Save** to save the recording. To choose the file format, click the ▼ button next to Save:



- **Save .mp3** — available for audio recordings only; saves a compressed MP3 file resampled to 16 kHz;

- **Save .wav** — saves a 48 kHz 32-bit float WAV file. Audio recordings are saved as mono; I/Q recordings are saved as stereo (left channel = I, right channel = Q).

Click **Load** to load a previously saved .wav or .mp3 file into the buffer for playback.

The Save and Load dialogs open in the **Recordings** subfolder of the data folder by default.

Playback Routing

During playback, audio is sent to the same soundcard that is selected in Settings for SDR audio output. Both audio and I/Q are also routed through the same output channel (VAC or UDP) that is configured for the Output Stream function in the Settings, so external applications can receive the played-back signal just as they would a live signal.

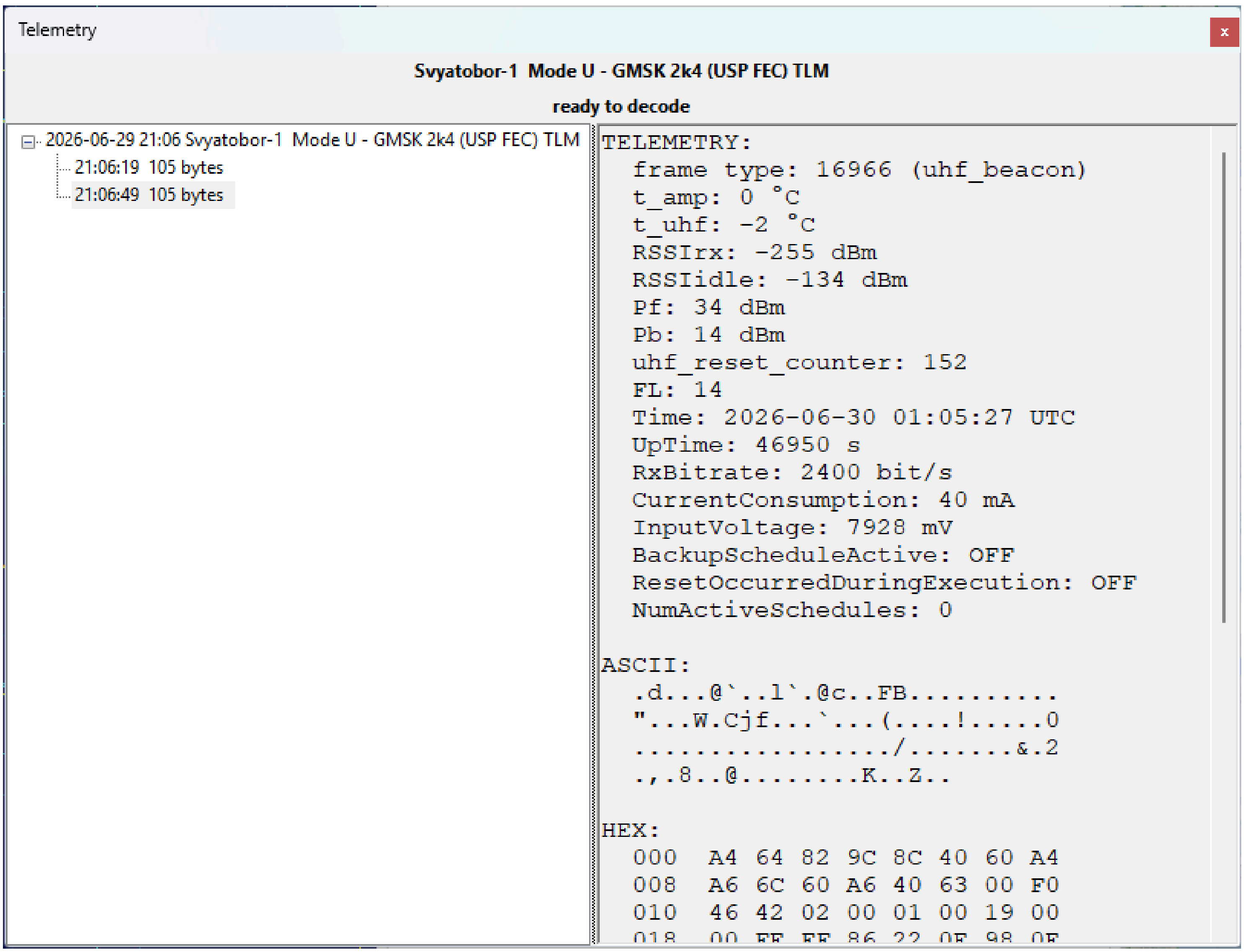
Waveform Display

The vertical gain slider on the right edge of the waveform adjusts the display amplitude in dB, making quiet signals easier to see. It does not affect the recording or playback level.

Telemetry Panel

The Telemetry panel shows the telemetry frames that SkyRoof decodes from the satellite signal. SkyRoof can decode the telemetry transmitted by many satellites without any external software: the built-in decoder demodulates and deframes the signal directly from the SDR, right inside the program. Unlike the external decoder and sound modem workflows, it needs no separate program, no Virtual Audio Cable, and no output stream — SkyRoof feeds the Doppler-corrected receiver passband straight to the decoder.

The panel is opened from the **View / Telemetry** menu. Its optional outputs — saving to a file, sharing over a KISS server, and uploading to SatNOGS — are configured as described in Setting Up Telemetry Decoding.



Supported Signals

The decoder supports the **FSK**, **GFSK**, **MSK**, **GMSK**, **AFSK**, **BPSK** and **DBPSK** modulations with the framing formats used by the supported satellites. The modulation, baud rate, and framing are looked up automatically from the transmitter description in the satellite database, so there is nothing to configure for the signal itself — you only have to select the right transmitter.

The panel also decodes **SSTV** images from satellites that transmit them over an FM downlink. This is covered separately in Receive SSTV Images.

Some satellites send more than telemetry in their frames, and the panel reconstructs that too, with nothing extra to turn on:

- **SSDV images** — still pictures sent as data, either as SSDV packets (HADES-SA, JY1SAT) or as raw JPEG fragments (the Geoscan fleet, Lobachevsky, and the Sputnix satellites). See Receive SSDV Images;
- **Codec2 voice messages** — short recorded speech compressed to a few hundred bits per second and carried in the telemetry frames of HADES-SA. See Receive CODEC2 Voice Messages.

The rest of this page describes telemetry frame decoding.

If the selected transmitter uses an unsupported modulation, or its parameters are unknown, the panel names what it cannot decode — `telemetry format not supported`, `CW decoding not supported`, or `FM decoding not supported` — and no telemetry frames are decoded. Other transmissions on the same frequency may still be decoded — an SSTV image, or an FM voice transcript.

Decoding a Pass

1. Open the Telemetry panel from the **View / Telemetry** menu.

2. Select the satellite in the Satellite Selector on the toolbar.

3. Select a data transmitter for that satellite. The decoder follows the transmitter selection, so pick the transmitter whose telemetry you want to decode. You can select it in the Satellite Transmitters panel, or by clicking on its label on the frequency scale. The panel header shows the satellite name and the selected transmitter; hover over it to see the resolved signal parameters.

   Many satellites have several transmitters on the same frequency, and only some of them may be supported by the decoder - for exmple, a CW beacon and a GMSK telemetry. The label you click on the frequency scale is not necessarily the one that gets selected, so after clicking check that the right transmitter is highlighted in the Satellite Transmitters panel.

Some satellites can transmit their telemetry at different Baud rates, selected by the ground station. Such satellites have multiple transmitters in the database, one per Baud rate. To switch to a different rate, just select the right transmitter.

4. Make sure the SDR is running and tuned to the satellite. The decoder uses the same Doppler-corrected passband as the receiver, so the satellite's signal must be visible on the waterfall at the tuned frequency.

That is all that is needed to start decoding. When the satellite is above the horizon and a supported transmitter is selected, frames are decoded automatically and appear in the panel.

Layout

- The **header** at the top shows the selected satellite and transmitter. Hover over it to see the resolved signal parameters (modulation, baud rate, framing). The **gear button** to its right opens the Signal Details dialog.

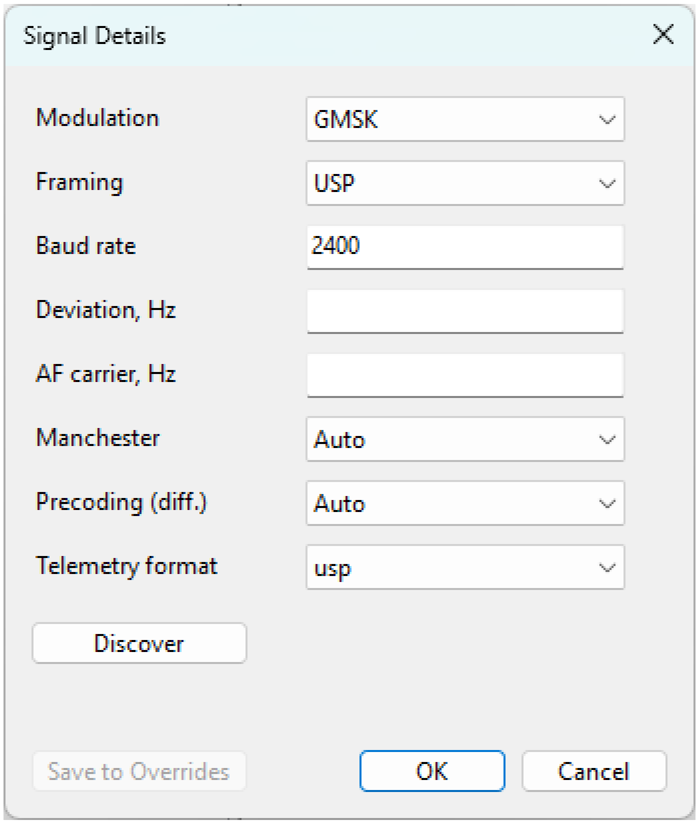
- The **status line** below the header shows the current state of the decoder, in color:

  - **satellite below horizon** — the format is supported and the decoder is waiting for AOS; decoding starts when the satellite rises above the horizon;
  - **ready to decode** — the satellite is up and the decoder is listening;
  - **DECODING...** in green — a burst is being processed;
  - **telemetry format not supported** in red — the transmitter's modulation or framing is not supported for telemetry; SSTV or FM voice on the same frequency may still be decoding;
  - **CW decoding not supported** in red — a CW transmitter is selected; SSTV on the same frequency may still be decoding;
  - **FM decoding not supported** in red — an FM transmitter is selected but the speech model is not installed;
  - **terrestrial, signal parameters not set** in red — the radio is tuned to a terrestrial signal and no parameters have been entered for it yet. Enter them in the Signal Details dialog and decoding starts; see Setting Up Telemetry Decoding.
- The **tree** on the left lists the decoded passes and frames.

- The **detail pane** on the right shows the contents of the pass or frame selected in the tree.

Signal Details

For most satellites the signal parameters resolved from the database are correct and there is nothing to do here. Occasionally a transmitter is described incorrectly in the database, or its description is incomplete, and the decoder needs a hand. The **gear button** in the header opens the **Signal Details** dialog, which shows the parameters actually in use for the selected transmitter and lets you override any of them:



- **Modulation** and **Framing** — the modulation and framing format of the signal;
- **Baud rate** — the symbol rate;
- **Deviation, Hz** — the FSK deviation;
- **AF carrier, Hz** — the audio carrier frequency for AFSK downlinks;
- **Manchester** and **Precoding (diff.)** — the line coding and the differential precoding. These are tri-state: **Auto** leaves the decision to the decoder, **On** and **Off** force it;
- **Telemetry format** — the definition used to parse the decoded frames into named telemetry values. It is normally chosen from the satellite's NORAD number; select a different one here when the frames decode but their values do not. The field is empty, as in the picture above, when SkyRoof has no telemetry definition for the satellite — its frames are then shown without telemetry values in the PAYLOAD section.

Each change takes effect as soon as you finish making it — when you pick a value from a list, press Enter in a text box, or move on to another field. A change to any of the demodulator fields rebuilds the decoder there and then, so it applies to the next burst; a change to the telemetry format applies to frames decoded from that point on and does not re-parse the frames already in the tree.

To undo a single override, right-click the field and choose **Reset to database value** — the field goes back to the value the database gave it, leaving your other overrides in place. **Cancel** puts back the last set of parameters that was written down: the ones the dialog opened with, or the ones you last saved. **OK** simply closes the dialog and keeps what is in use.

Discovering Signal Parameters

When the parameters are wrong and you do not know what the right ones are, the **Discover** button works them out from the signal itself. Press it during a pass: SkyRoof analyses the bursts that arrive from the press onward, trying candidate combinations of modulation, framing, baud rate, and deviation against each one, while normal decoding continues undisturbed. The status line beside the button reports what the search is doing — **waiting** between bursts, with the number of bursts analysed and skipped so far, and **analyzing** while it works on one that has just arrived.

While the search runs, the parameter fields are emptied and locked. The search is about to answer for them, and leaving values on screen that it is about to replace would only mislead. If it ends without an answer, they come back exactly as they were.

The search ends as soon as a candidate decodes a frame with a valid checksum. That is a strong result: a wrong set of parameters does not produce a correct checksum by accident. The parameters found are applied to the current pass immediately, so decoding continues with them, and they appear in the dialog with green dots.

Two other outcomes are possible:

- **the parameters belong to another transmitter** of the same satellite on the same downlink frequency — a satellite often has several. Nothing is overridden in that case: SkyRoof selects that transmitter instead and says so, because the database was right all along and the signal was simply not the one you had selected;
- **the pass ends with nothing found**, and the line says *no parameters found*. That is a useful answer too: it says the trouble lies somewhere other than the parameters — too weak a signal, the wrong frequency, or a modulation SkyRoof does not support.

Press **Discover** a second time to stop the search early. It cannot be started while the satellite is below the horizon, because no burst can arrive then, and the status line says so if you try.

Where a Value Came From

The dot to the right of each field shows where its value came from:

- **no dot** — the value from the satellite database, unchanged;
- **orange** — a value you edited that has not yet produced a frame;
- **green** — an override that has produced a valid frame, or a value the decoder itself discovered at run time (it locks the baud rate, deviation, and precoding as it decodes).

An orange dot that turns green is the confirmation that your override was right. One that stays orange means no frame has decoded with it yet.

The gear button's own color mirrors the dots, so you can see the state of the overrides without opening the dialog: gray when there is nothing to show, orange while an override is waiting for a confirming frame, and green once every override has produced one.

A green dot says the parameters are working now. It does not by itself unlock the **Save to Overrides** button, which asks for more than that — see below, where the status line counts down the frames the save decision is still waiting for.

Saving the Parameters

Overrides are remembered per transmitter for the current session only. They are discarded when you select a different transmitter, and they are not saved between runs — unless you save them with the **Save to Overrides** button. Saving writes them to `transmitters-override.json` in the data folder, where they are read on every later run and survive updates of the satellite database.

Because that file is long-lived, the button asks for more evidence than a dot does. A green dot means one frame decoded with the value. Save unlocks only after **2 more frames** have decoded with the parameters on screen, and the status line counts them down: *2 more frames to save*, then *1 more frame to save*, then *3 frames decoded — parameters can be saved*. The count keeps running while the dialog is closed. Editing a field afterwards greys the button out again: the frames proved the values they were decoded with, not the new one. The count also starts again at the end of the pass, so a Save left unclicked at LOS needs two fresh frames on the next pass — an override already saved is not affected.

The same gate decides when frames go to the SatNOGS DB. While a transmitter's parameters carry unsaved edits, its decoded frames still appear in the tree, are still written to the log file, and still reach any KISS client, but they are not uploaded. Saving is what tells SkyRoof the frames are attributed to the right transmitter, and uploading runs from that point on; frames decoded before the save are not sent retroactively. Transmitters you have not edited are unaffected.

Passes and Frames

The frames are grouped by pass. Each top-level node in the tree is one pass of one transmitter, labeled with the start time, satellite name, and transmitter description. A pass node is created as soon as the first burst is detected and stays grayed out until the first valid frame is decoded.

Each child node is a single decoded frame, labeled with the time of arrival, the frame length in bytes, and, for AX.25 frames, the source and destination addresses. The newest pass is expanded automatically, and the view scrolls to follow new frames as long as the latest frame is selected.

Frames are not the only children a pass can have. An SSTV or SSDV image and a voice message each get a node of their own, which is updated in place as the picture or the recording is built up, and which shows the picture or plays the audio when it is selected.

Select a **pass** node to see a summary in the detail pane: the start time, satellite, transmitter, orbit number, the number of bursts, frames, and images decoded, and the signal parameters.

Select a **frame** node to see its full contents:

- **PAYLOAD** — everything SkyRoof can name in the frame: the AX.25 source and destination addresses where the frame has them, the named telemetry values decoded from it (battery voltage, temperatures, and so on) for satellites that SkyRoof has a telemetry definition for, and, on the Geoscan fleet, the sending satellite and the message type, including where an image frame's bytes belong in the picture. The section is omitted when none of these apply;
- **ASCII** — the frame bytes rendered as text;
- **HEX** — a hex dump of the frame bytes;
- **META** — the carrier frequency offset (CFO), signal-to-noise ratio (SNR), CRC check result, and the number of corrected bits and erased bytes from forward error correction.

Sharing the Frames

As each frame is decoded it is also, depending on the decoder settings:

- appended to a log file in the `TelemetryDecodes` subfolder of the data folder;
- sent to any client connected to the KISS-over-TCP server;
- uploaded to the SatNOGS DB.

Clearing the List

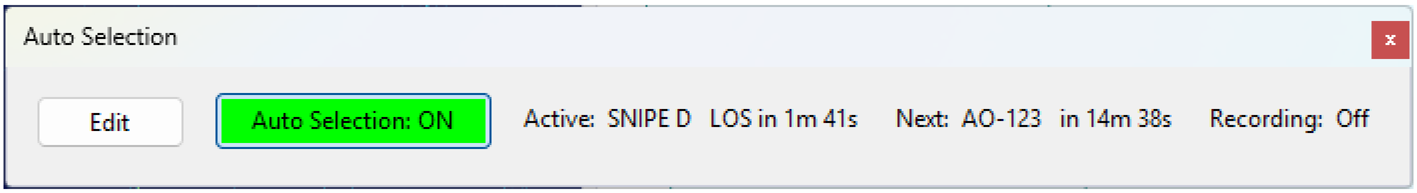
Right-click anywhere in the tree and choose **Clear All** to remove all passes and frames from the panel. This clears only the display; frames already saved to file or uploaded are not affected.

Auto Selection Panel

The Auto Selection panel lets SkyRoof automatically tune to each satellite in a schedule at the start of its pass (AOS) and hold it until the pass ends (LOS), cycling through a set of passes you choose. The selected passes can optionally be recorded, each to its own file, or left unrecorded so that only their live telemetry is decoded as they pass over.

Auto selection works on the **current satellite group**. Each group keeps its own schedule, so switching groups switches schedules.

Status Panel



The panel shows the live status of auto-selection for the current group:

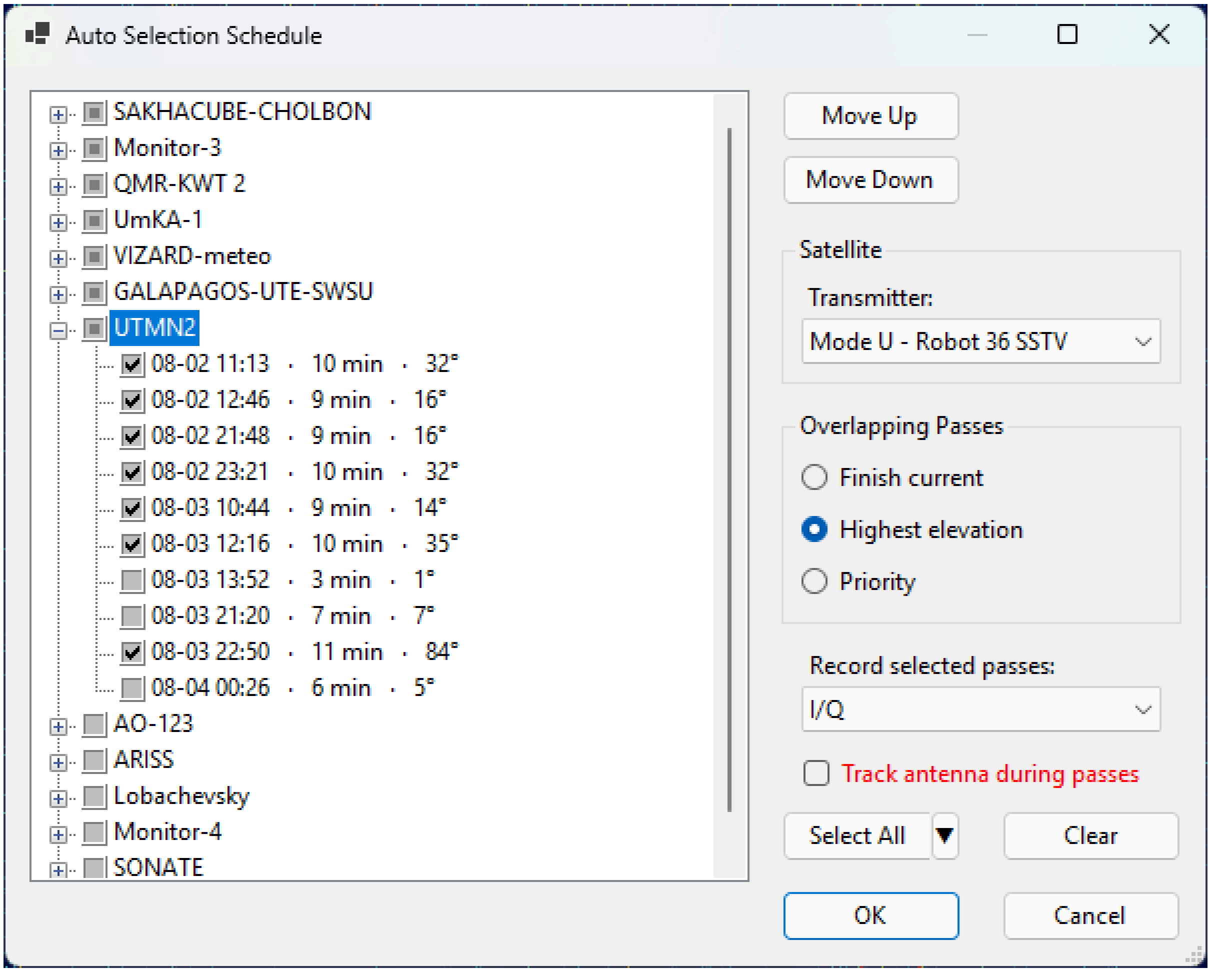
- **Edit** — opens the schedule dialog for the current group.
- **Auto Selection: ON / OFF** — the master toggle. It is disabled until the current group has a schedule, and lights up in green while auto selection is running. Auto selection always starts **OFF** when the program launches; you turn it on deliberately each session.
- **Active** — the pass currently tuned, with the time remaining until its LOS, or **none** when no pass is active.
- **Next** — the next scheduled pass and the time until its AOS, or **none** when nothing is coming up.
- **Recording** — the recording type and elapsed time while a pass is being recorded, or **Off** when nothing is being recorded.

While auto selection is on, changing the satellite, transmitter, or group **by hand** stops auto selection and shows a message. Your manual change is applied normally; re-enable auto selection when you are ready.

Closing the Auto Selection panel also stops auto selection, since there is no longer anywhere to monitor or control it.

The Schedule Dialog

Click **Edit** to open the schedule dialog for the current group.



Rotation Tree

The tree lists every satellite in the group (except geostationary satellites, which have no passes) as a parent node, with its upcoming passes over the next 48 hours as child nodes. Each pass leaf shows its AOS local time, duration, and maximum elevation.

Use the checkboxes to choose which passes take part in the rotation:

- clicking a **satellite** checkbox selects or clears all of its passes;
- clicking a **pass** checkbox selects that single pass;
- a satellite checkbox is shown in an **indeterminate** state when only some of its passes are selected.

The order of the satellite nodes is the **priority** order (top = highest). Select a satellite and use **Move Up** / **Move Down** to change its priority.

Use **Select All** to check every pass of the group at once, or **Clear** to uncheck them all. Clicking **OK** with no passes checked removes the current group's schedule.

To skip the low passes, click the small arrow next to **Select All** and pick a maximum-elevation threshold from the drop-down menu: **Select All**, **Select > 5º**, **Select > 10º**, **Select > 15º**, or **Select > 20º**. Only the passes that rise at least that high are checked.

The menu adds to your current selection and never unchecks anything, so click **Clear** first if you want the threshold to be the only thing selected.

Satellite Settings

Select a satellite node to edit its settings:

- **Transmitter** — the transmitter to tune when a pass of this satellite is selected.

Overlap Mode

When two selected passes are up at the same time, the **overlap mode** decides which one is tuned:

- **Finish current** — once a pass is entered at its AOS, it is held until its LOS, then the next selected pass that is still up is entered.
- **Highest elevation** — the currently-up selected pass with the greatest elevation is tuned, switching only when another pass is clearly higher (to avoid rapid switching back and forth).
- **Priority** — the currently-up selected pass whose satellite has the highest priority is tuned; a higher-priority pass rising over the horizon takes over, and when it sets the next-highest still-up pass is entered.

Record Selected Passes

Record selected passes sets how the passes in the rotation are recorded. Like the overlap mode, this option belongs to the schedule rather than to a satellite: it applies to **all satellites** in the group, and thus to every selected pass. Each satellite group has its own setting:

- **Off** — do not record. The satellites are still tuned, so live telemetry is decoded during each pass.
- **Audio** — record the demodulated audio as an `.mp3` file.
- **I/Q** — record the complex baseband signal as an `.iq.wav` file.

Track Antenna During Passes

Tick **Track antenna during passes** to let auto selection also point your antenna at each pass. Like the overlap mode, this option belongs to the schedule, so each satellite group has its own setting.

- When a pass is entered, the antenna starts tracking it and follows it until LOS.
- One minute before the AOS of the next selected pass, the antenna is sent to the pass's rise point ahead of time, so it is already in position when the pass starts. Only the antenna moves at this point; the satellite is not selected or tuned until AOS.

- Outside that one-minute window, auto selection does not touch the rotator between passes, so you are free to move or track the antenna by hand in the gaps. If you click **Stop** during the pre-roll, the antenna stays where it is until the pass begins.
- Turning **Auto Selection** off stops the antenna if auto selection was tracking it.

The option takes effect as soon as you click **OK**: ticking it starts tracking the pass that is currently active, and clearing it stops the rotator.

When the box is not ticked, auto selection never moves the antenna, and you can track passes manually with the Rotator Control panel as usual.

The option requires rotator control to be enabled in the rotator settings. If it is disabled, the checkbox caption is shown in red to warn you that antenna tracking will have no effect until you enable it.

Click **OK** to save the schedule, or **Cancel** to discard your changes. Reopening the dialog extends the list to the next 48 hours from that moment, keeping your previously selected passes checked. The newly added passes are unchecked, so they do not join the rotation until you check them yourself.

Recordings

Auto-selection recordings are saved into the **Auto** subfolder of the **Recordings** folder inside the data folder. Each contiguous segment becomes one file, named with its start time and the satellite name. Audio is saved as `.mp3` and I/Q as `.iq.wav`, the same formats as the Recorder panel.

How to Decode Telemetry from PEARL-1C

> **NOTE**
>
> This approach is **deprecated**. SkyRoof now has a built-in telemetry decoder that needs no external program, Virtual Audio Cable, or output stream — use it whenever the satellite's modulation is supported. The instructions below are kept for satellites and formats that the built-in decoder does not yet handle, where an external decoder such as gr_satellites is still useful.
>
> Note also that **PEARL-1C is no longer active**; the NORAD ID and parameters in the commands below are only an example — substitute the values of a satellite that is currently transmitting.

I/Q or Audio data, streamed via VAC or UDP, may be used to decode telemetry transmitted by the satellites. There is a number of telemetry decoders to choose from. One such decoder is gr_satellites.exe command line tool, its installation instructions are here.

This command runs gr_satellites.exe to decode telemetry of the PEARL-1C satellite using the I/Q UDP stream from SkyRoof:

```
(base) C:\Ham>gr_satellites 58342 --udp --udp_port 7355 --udp_raw --iq --samp_rate 48e3 --hexdump
```

This command decodes the same satellite via I/Q output to VAC :

```
(base) C:\Ham>gr_satellites 58342 --audio "CABLE Output (VB-Audio Virtual Cable)" --samp_rate 48000 --iq
```

To use the second command, you will need a virtual audio cable, such as VB-Audio.

Run one of these commands, then in SkyRoof settings:

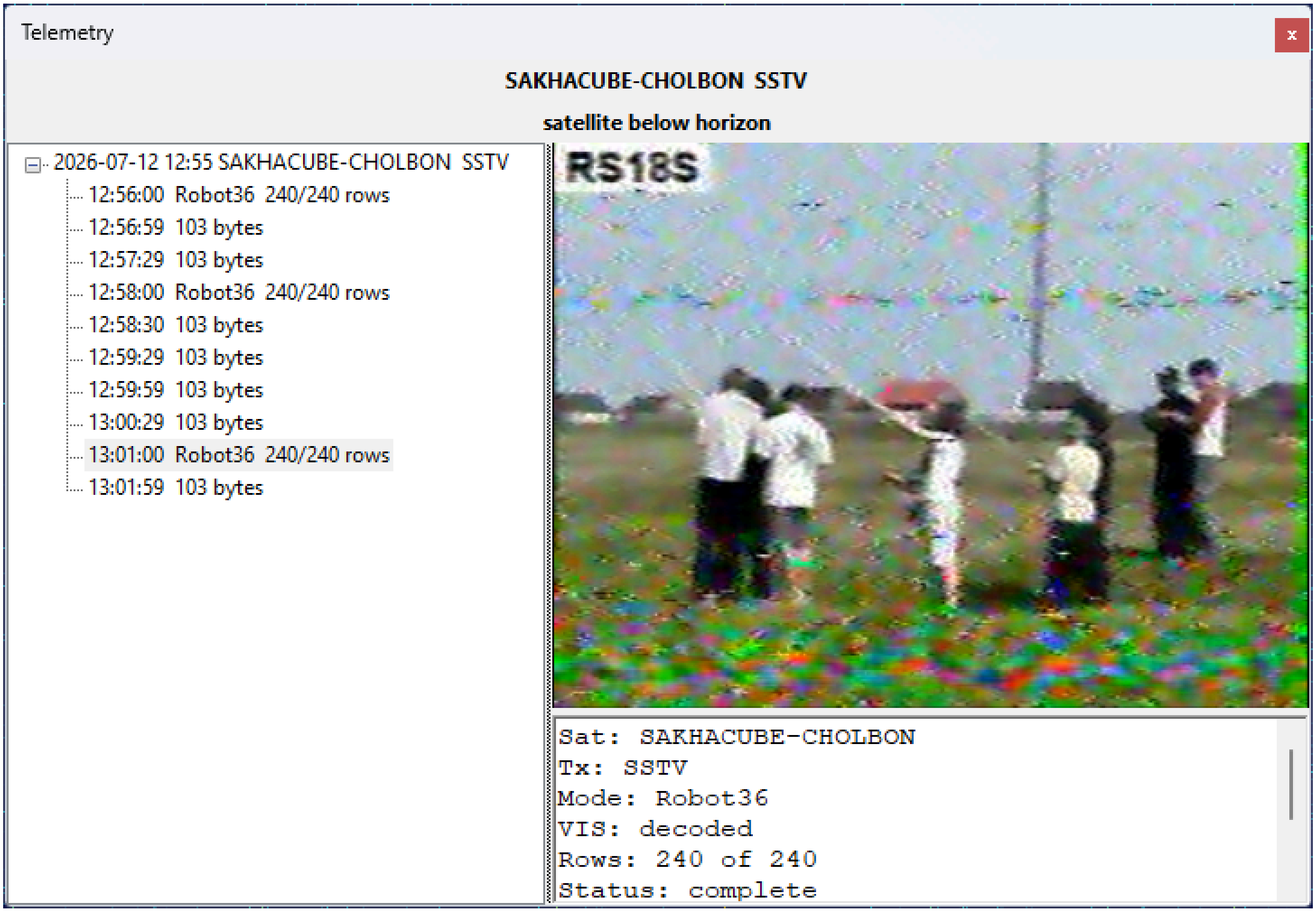
- select either I/Q to VAC or I/Q to UDP, depending on the command you use;
- set Gain, dB to 0;
- select the VAC in the list of audio devices;
- click on the Output Stream label on the status bar to enable the output.

Example output from gr_satellites.exe v.5.7.0:

```
pagesize :debug: Setting pagesize to 4096 B
top_block_impl :debug: Using default scheduler "TPB"
udp_source :info: Listening for data on UDP port 7355.
***** VERBOSE PDU DEBUG PRINT ******
((transmitter . 9k6 FSK downlink))
pdu length =         64 bytes
pdu vector contents =
0000: 9c 86 aa 8e a6 62 e0 a0 8a 82 a4 98 86 e1 03 f0
0010: f9 11 01 83 43 33 a9 e7 a4 10 00 00 11 00 00 00
0020: 00 00 00 00 7a 0f 01 00 00 00 00 00 00 00 00 00
0030: 05 00 00 00 77 fb 01 00 67 aa 00 00 00 00 00 00
************************************
***** VERBOSE PDU DEBUG PRINT ******
((transmitter . 9k6 FSK downlink))
pdu length =         88 bytes
pdu vector contents =
0000: 9c 86 aa 8e a6 62 e0 a0 8a 82 a4 98 86 e1 03 f0
0010: fb 11 01 81 43 33 00 00 00 00 00 00 00 00 00 00
0020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0030: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0040: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0050: 00 00 00 00 00 00 00 00
************************************
***** VERBOSE PDU DEBUG PRINT ******
((transmitter . 9k6 FSK downlink))
pdu length =         56 bytes
pdu vector contents =
0000: 9c 86 aa 8e a6 62 e0 a0 8a 82 a4 98 86 e1 03 f0
0010: 06 11 01 82 43 33 83 8a 01 00 9e 00 00 00 04 00
0020: 00 00 88 ff f8 ae 00 00 80 01 00 00 0f 02 00 00
0030: 00 00 00 00 00 00 00 00
```

Receive SSTV Images

Some satellites transmit **SSTV** (Slow-Scan Television) images over an FM downlink. SkyRoof decodes these images with its built-in decoder, right inside the program — there is no need for an external SSTV program, a Virtual Audio Cable, or an output stream. The decoded image is built up line by line in the [Telemetry](#) panel as the satellite passes overhead.



Supported Modes

The decoder supports the YCrCb mode family used by satellites:

- **Robot 36** and **Robot 72**;
- **PD 50**, **PD 90**, **PD 120**, **PD 160**, **PD 180**, **PD 240**, and **PD 290**.

The mode is detected automatically from the **VIS** header and the sync cadence of the received signal, so you do not have to select it by hand. The RGB Martin and Scottie modes, which are rarely used by satellites, are currently not supported.

Decoding an Image

1. Open the [Telemetry](#) panel from the **View / Telemetry** menu.

2. Select the satellite in the Satellite Selector on the toolbar. If it is not in the current group, add it using the Satellites and Groups dialog.

3. Select the satellite's **SSTV transmitter**. You can select it in the Satellite Transmitters panel, or by clicking on its label on the frequency scale. The decoder follows the transmitter selection, so make sure the SSTV transmitter is the one highlighted in the Satellite Transmitters panel.

4. Make sure the SDR is running and tuned to the satellite. The decoder uses the same Doppler-corrected passband as the receiver, so the satellite's signal must be visible on the waterfall at the tuned frequency.

When the satellite is above the horizon and its signal starts to appear, the decoder detects the SSTV transmission and begins building the image. Each image appears as a node in the tree, under the pass node, and updates in place as new scan lines arrive. Select an image node to watch it build up in the detail pane on the right, together with a **META** section that shows the satellite, transmitter, mode, whether the VIS header was decoded, the number of rows received, and the decoding status.

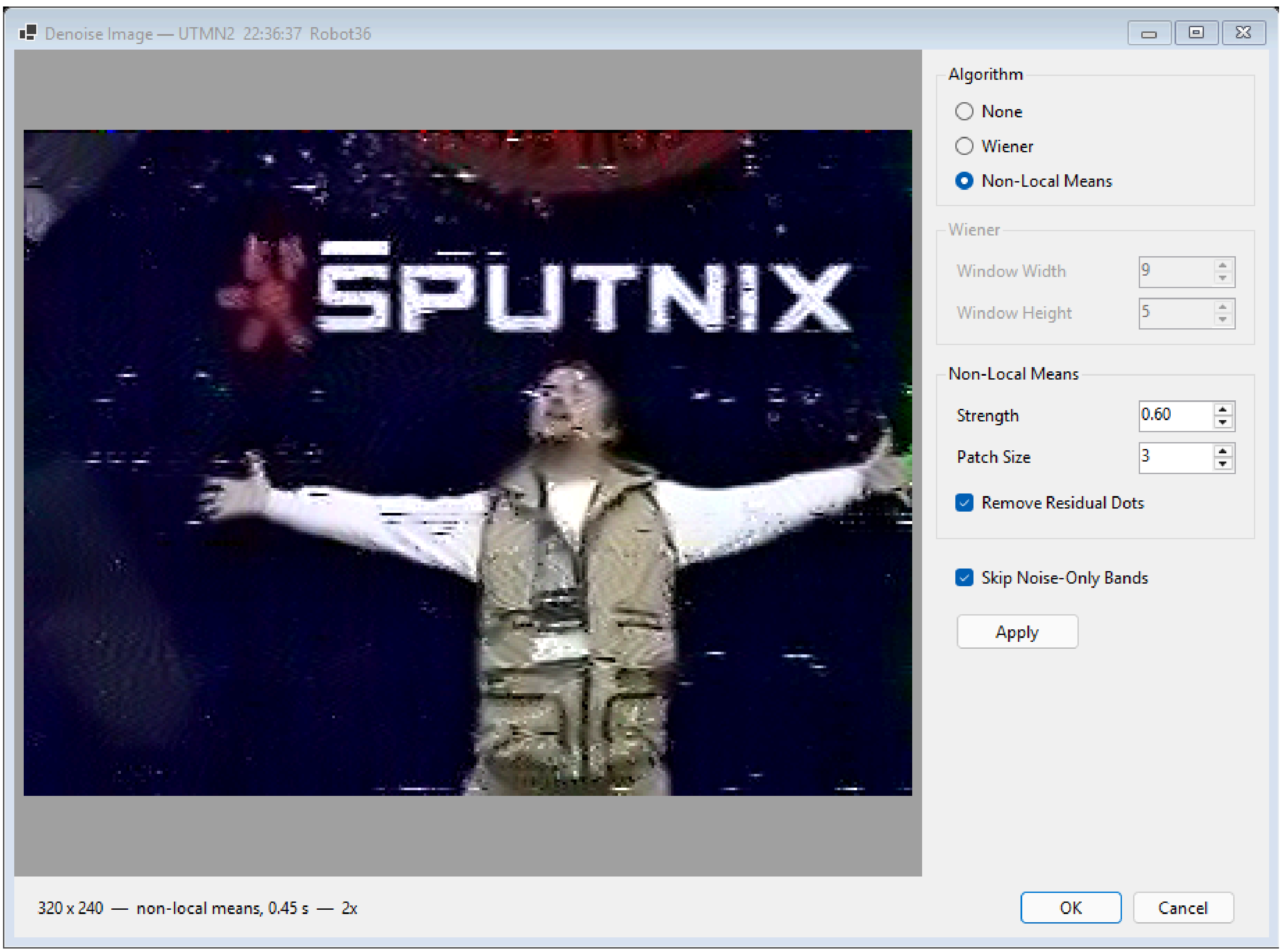
There is nothing to start or stop manually: the decoder rides through short signal fades and finalizes each image when it is complete or the signal is lost. A satellite whose transmitter alternates between telemetry and SSTV (such as UmKA-1) is handled automatically — the telemetry frames and the SSTV images both appear in the same panel.

Denoising an Image

A satellite SSTV picture arrives over an FM link, and once the signal drops toward the FM threshold the picture fills with coloured speckle. SkyRoof applies a mild **Wiener** filter to every decoded image automatically, which is what you see while the image is building. When a pass was marginal you can do considerably better afterwards: right-click a finished image and choose **Denoise Image...**.

As received	After non-local means
	

The command becomes available once an image is complete, because the filter works on the raw reconstruction that is stored with the finished image.



The dialog shows the picture at double size — the speckle it removes is one to three pixels across and is hard to judge at 1:1. Roll the **mouse wheel** over the picture to zoom, from 1:1 up to the point where it fills the pane; enlarging or maximizing the window raises that limit, so maximizing gives you a close look at fine detail.

Choose the algorithm:

- **None** — the raw reconstruction, with no filtering at all. Note that this is *not* the image you started with: the automatic Wiener filter is switched off too, so this is the noisiest of the three.
- **Wiener** — the same filter the decoder applies automatically. Cheap, and gentle on smooth areas, but it judges each pixel from its immediate neighbourhood alone, so it thins fine detail such as text and thin lines.
- **Non-local means** — much slower, but far better on a noisy pass. Instead of averaging a pixel with its neighbours, it searches the surrounding area for patches that look like the one being cleaned and averages those. Repeated fine structure — lettering, edges, texture — therefore reinforces itself instead of being smoothed away. This is the one to reach for.

Only the settings of the selected algorithm are enabled. For non-local means:

- **Strength** — how much noise the filter assumes is present. Higher removes more speckle but starts to flatten real detail; 0.40 to 0.80 suits most pictures, and 1.60 and above is usually too much. The best value depends on the pass, which is why the filter is manual rather than automatic.
- **Patch Size** — the size of the patches compared against one another. 3 is a good default.
- **Remove Residual Dots** — a second pass that clears isolated specks the first pass leaves behind. An isolated speck resembles nothing else in the picture, so nothing matches it and the plain filter preserves it faithfully.

For the Wiener filter, **Window Width** and **Window Height** set the size of the neighbourhood it averages over. A larger window removes more noise and loses more detail.

Skip Noise-Only Bands applies to both algorithms and is on by default. Where the signal dropped below the FM threshold the picture carries bands of pure noise with no image in them at all. Averaging those bands turns them into a soft grey wash that reads as a blurred picture rather than as absent signal, so by default they are left exactly as received and the filtering is concentrated where there is a picture to improve. Turn this off for a very weak pass in which you can see there is a faint picture buried in the noise — the program cannot always tell the two cases apart, but you can.

Press **Apply** to filter the picture, and try as many settings as you like: every Apply starts again from the original, so filters never pile up on one another and **None** always brings back exactly what was received. **OK** keeps what is on screen and replaces the automatically saved PNG file; **Cancel** leaves the image as it was.

Saved Images

Each finalized image is saved automatically as a PNG file, with a JSON sidecar holding its metadata, in the **SstvImages** subfolder of the data folder. You can also right-click an image in the detail pane and choose:

- **Save As...** — save the image to a location of your choice. This writes the image as it is currently shown, so a denoised picture is saved denoised;
- **Copy** — copy the image to the clipboard;
- **Denoise Image...** — clean up a noisy picture, as described above;
- **Open in Viewer** — open the automatically saved PNG file in the program that Windows associates with PNG images, where you can see it at full size. This command is available only after the image has been finalized and saved, so it stays grayed out while an image is still being received.

Receive SSDV Images

Some satellites send still pictures down as data, packed into the same telemetry stream that carries their housekeeping frames. SkyRoof reconstructs these pictures with its built-in decoder, right inside the program — there is no need for an external decoder, a Virtual Audio Cable, or an output stream. The picture is built up fragment by fragment in the Telemetry panel as the satellite passes overhead.

This is a different thing from SSTV, which sends an analog picture over an FM downlink, line by line. Here the picture is a JPEG file, cut into numbered packets that ride the telemetry downlink interleaved with ordinary telemetry frames. Nothing extra has to be turned on: if you are decoding telemetry from a satellite that sends images, the images are decoded too.

Supported Satellites

Two families of image transport are supported:

- **SSDV packets** — the amateur-satellite digital SSTV format, where each packet carries the image ID, its own position in the picture, and a checksum. Sent by **HADES-SA** and by **JY1SAT (JO-97)**.

- **Raw JPEG fragments** — the picture is sent as byte ranges of the JPEG file itself. Sent by the **Geoscan** fleet and **Lobachevsky**, and by the **Sputnix** satellites — **Luca**, **239Alferov** and **HyperView-1G** — as file transfers over their USP telemetry protocol.

The difference matters when the reception is imperfect. An SSDV packet that is lost costs only its own block of the picture, and the rest of the image is unharmed. In a raw JPEG a lost fragment desynchronizes everything after it, so the picture is good down to the first gap and noise below it — which is why the detail pane reports **Intact to** for these satellites and not for SSDV.

Receiving an Image

The steps are the same as for decoding telemetry:

1. Open the Telemetry panel from the **View / Telemetry** menu.

2. Select the satellite in the Satellite Selector on the toolbar. If it is not in the current group, add it using the Satellites and Groups dialog.

3. Select the satellite's **telemetry transmitter** — the one whose frames carry the pictures. Some satellites also have a separate **SSDV** row in the database. That row is not a transmitter of its own: it is the same signal on the same frequency, so selecting it

works as well, and SkyRoof quietly pairs it with the co-channel telemetry transmitter that actually carries the packets.

4. Make sure the SDR is running and tuned to the satellite. The decoder uses the same Doppler-corrected passband as the receiver, so the satellite's signal must be visible on the waterfall at the tuned frequency.

Each picture appears as a node in the tree, under the pass node, labeled with the time it was first seen, the image number, and how many fragments have arrived:

```
04:12:37  Image 238  9/15 fragments
```

Select the node to watch the picture fill in as the pass goes on. The detail pane below the picture shows:

- **Sat** and **Tx** — the satellite and the transmitter the picture came from;
- **Image** — the image number the satellite gave it;
- **Source** — the sender, where the protocol names one: the Geoscan satellite that took the picture, or the file name for a USP file transfer;
- **Size** — the picture's dimensions in pixels;
- **Fragments** — how many fragments were received, of how many the picture spans;
- **Intact to** — for raw-JPEG satellites, how many bytes of the file are good before the first gap;
- **Status** — `receiving...`, `complete`, or `incomplete` for a picture the pass ended in the middle of;
- **Saved** — the file it was written to, once it has been saved.

An incomplete picture is normal rather than exceptional: a pass ends when the satellite sets, whatever the satellite was in the middle of sending. Missing parts are left gray.

Saved Images

Each finalized image is saved automatically as a JPEG file, with a JSON sidecar holding its metadata, in the **SsdvImages** subfolder of the data folder. The bytes are written exactly as received — gaps filled in and the file properly terminated — so whatever fraction of the picture arrived is a valid JPEG that any program can open.

A picture built from a single fragment is not saved automatically, because one misread frame can look like one image fragment; neither is one with no recognizable geometry. Such an image is still shown in the tree and can still be saved by hand.

Right-click a picture in the detail pane to:

- **Save As...** — save the picture to a location of your choice;

- **Copy** — copy the picture to the clipboard;
- **Open in Viewer** — open the automatically saved JPEG in the program that Windows associates with JPEG files, where you can see it at full size. This command is available only after the image has been finalized and saved, so it stays grayed out while an image is still being received;
- **Combine with Previous Passes** — see below.

Combining Passes

Satellites usually send the same picture several times, over consecutive passes, so that stations that missed a piece of it can pick that piece up later. SkyRoof can put those receptions together.

Right-click a picture and choose **Combine with Previous Passes**. SkyRoof searches the **SsdvImages** folder for earlier receptions of the same picture by the same satellite, merges their fragments with the ones just received, and shows the result. The menu item is grayed out when there is nothing to combine with, and shows the number of earlier receptions found when there is: **Combine with Previous Passes (3)**.

The item is a toggle — choose it again to go back to what this pass alone heard. The detail pane reports both, on separate lines, so you can tell what the current pass contributed:

```
Fragments: 9 of 15
Combined: 14 of 15 (with 3 earlier passes)
```

Combining stays live while the pass runs: each new fragment is merged in as it arrives, and the combined picture keeps filling in on screen.

A few limits are worth knowing:

- It works only where each fragment carries its own checksum, so that two copies of one fragment can be judged against each other. That is SSDV as **HADES-SA** sends it. The raw-JPEG satellites and JY1SAT have no per-fragment check, and the menu item stays grayed out for their pictures.

- Only the last **30 days** of the archive are searched. Image numbers are 8-bit and satellites reuse them freely — HADES-SA sent images 237, 238 and 239 twice within 92 minutes — so the window is what keeps a merge from reaching back to a different picture that happens to share a number. If two unrelated pictures do get merged, the result looks obviously wrong: toggle the combination off to get the pass's own reception back, exactly as it was.

- The sidecar of each reception records the fragments of **that reception only**, never of a combination, so the archive stays a record of what was actually heard.

See Also

- Telemetry Panel
- Receive SSTV Images
- Receive CODEC2 Voice Messages

Receive CODEC2 Voice Messages

Some satellites send short recorded voice messages down as data rather than as FM audio: the speech is compressed with the **Codec2** vocoder to a few hundred bits per second and packed into ordinary telemetry frames. SkyRoof decodes these messages back to audio with its built-in decoder, and each message appears in the Telemetry panel, ready to play.

At the moment **HADES-SA** is the satellite that sends them, in Codec2 **700C** mode, interleaved with its telemetry frames and SSDV image packets on the same downlink. Nothing extra has to be turned on: if you are decoding HADES-SA telemetry, the voice messages are decoded too.

Receiving a Message

The steps are the same as for decoding telemetry — open the Telemetry panel, select the satellite and its telemetry transmitter, and make sure the SDR is running and tuned to it.

Each message appears as a node in the tree, under the pass node, labeled with the time it was first heard, how many sub-frames arrived, and how long the recording plays:

```
04:15:02  Voice  27 sub-frames, 10.8 s
```

The node updates in place as the message comes in. A gap of a few seconds with no sub-frames ends the message; the next sub-frame after that starts a new one.

Select the node to see the details, and **click anywhere in the detail pane to play the message**. You can play it while it is still arriving — what plays is always what has been received so far.

The detail pane shows:

- **Sat** and **Tx** — the satellite and the transmitter the message came from;
- **Duration** — the length of the reconstructed recording;
- **Sub-frames** — how many were received, and how many the message spans. There is no "of N": nothing in the transmission says how long a message is, so the second number is the span of what arrived, not the length of what was sent;
- **Numbered** — the range of sub-frame numbers received;
- **Gaps** — `none`, or how many sub-frames are missing. Missing sub-frames are played as silence, so a message with holes still plays at its right length and the words around the holes stay in place;
- **Status** — `receiving...` or `complete`;
- **Saved** — the file it was written to, once it has been saved.

A message the pass ended in the middle of is normal rather than exceptional, and there is no way to tell how much of it was missed — the downlink does not say.

About the Audio Quality

Codec2 700C compresses speech to 700 bits per second, which is roughly a hundredth of the rate of a telephone call. Even a message received without a single lost sub-frame sounds robotic and is often hard to make out. That is the vocoder's own limit at this bit rate, not a fault in the reception or in SkyRoof — a perfect decode still sounds like this. Comparing several receptions of the same message often helps more than trying to clean up one of them.

Saved Messages

Each finalized message is saved automatically as a WAV file, with a JSON sidecar holding its metadata, in the **Codec2Voice** subfolder of the data folder. Voice sub-frames carry no checksum of any kind, so a message of fewer than three sub-frames is not written to disk — too little to tell a real transmission from a misread frame. Such a message is still shown in the tree and still plays.

Right-click a message in the tree to:

- **Play** — play the message, the same as clicking in the detail pane;
- **Save As...** — save the WAV file to a location of your choice;
- **Open in Player** — open the automatically saved WAV file in the program that Windows associates with WAV files. This command is available only after the message has been finalized and saved, so it stays grayed out while a message is still being received.

See Also

- Telemetry Panel
- Receive SSDV Images

How to Receive FSK & AFSK Telemetry and other info using UZ7HO packet radio TNCs

by Marcus PY2PLL

> **NOTE**
>
> This approach is **deprecated**. SkyRoof now has a built-in telemetry decoder that decodes FSK, AFSK, GFSK, GMSK and BPSK telemetry directly, with no external sound modem or Virtual Audio Cable required — use it whenever the satellite's modulation is supported. The instructions below are kept for cases where an external sound modem is still needed, for example the UZ7HO modems support some packet/deframer formats that the built-in decoder does not yet handle.

An external program is needed to decode and display common information usually disseminated using FSK or AFSK. In this tutorial the UZ7HO sound modems will be used. More information about these modems can be found at the UZ7HO Personal page under **Packet-Radio** - English version.

Installing VAC

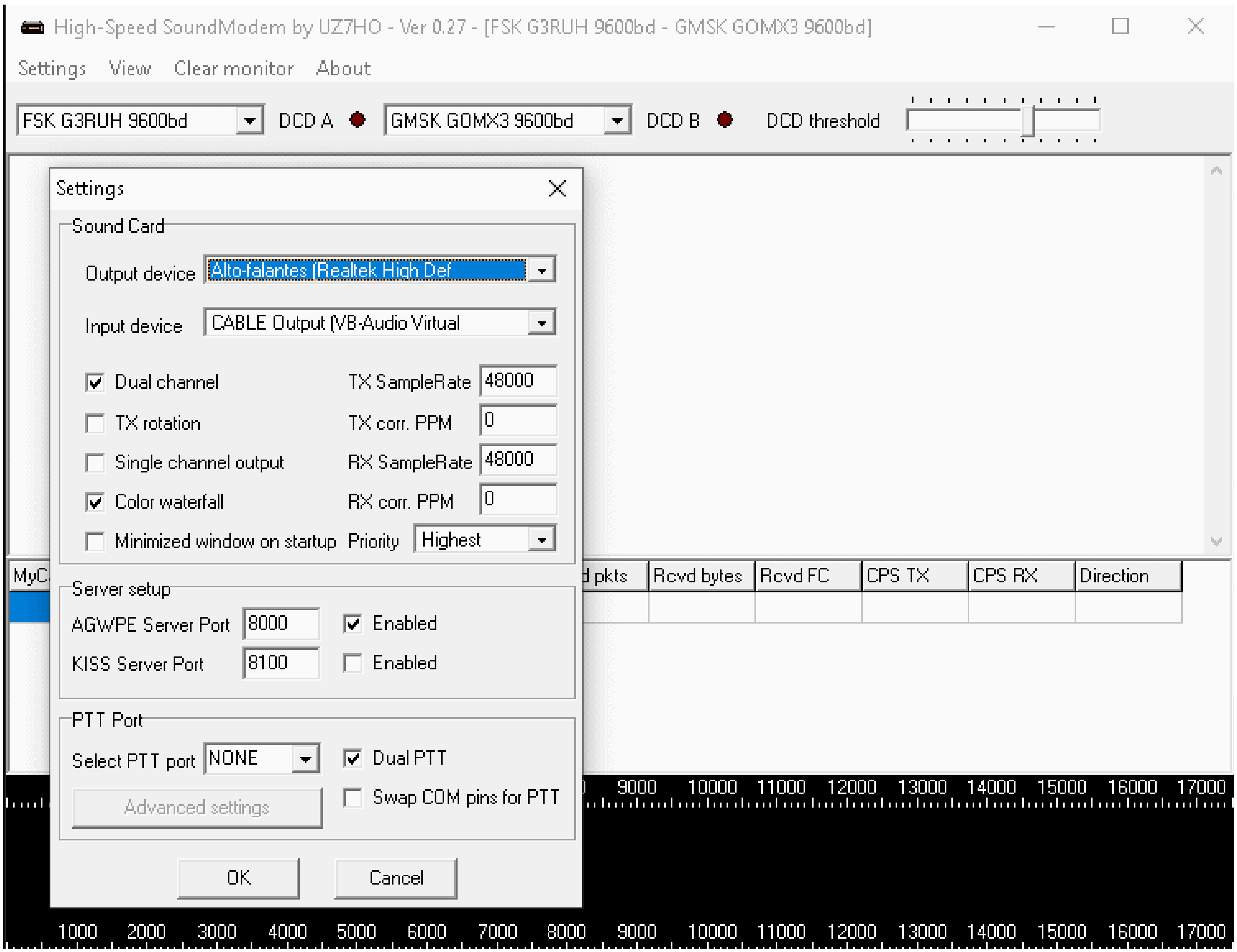
A virtual audio cable, VAC, is required to pass the satellite signals demodulated in SkyRoof to the decoding program. Download and install VB-Audio if you do not have it yet, and reboot your system. Be sure to get the latest version (2024) of VB-Audio, the old version may not work correctly.

Setting Up The Sound Modems

- Download and extract zipped files content to new folders -- one folder for each executable -- somewhere on your PC storage. For AFSK, e.g. the common modes for APRS, AX-25, 1200bps packet and other similar modes, use this low speed modem; for the higher speeds and formats such as 9600/19200 G3RUH FSK, etc., use this high speed modem Sometimes there is false virus detection or unsafe download alert from some web browsers. As already recommended in the SkyRoof FAQ, scan both links using one of the several online virus scanning services that you can use to check the link.

  P.S.: there are a few extra modems that can be tested for other modulation formats such as BPSK, GMSK. etc. in this archieve. But this How-To only covers AFSK and FSK associated to **FM_D** SkyRoof mode. For the UZ7HO itself, there is a more detailed user guide at Soundmodem User Guide v114

- Once downloaded and extracted, create corresponding desktop shortcuts to easy programs launch;
- The following examples will cover **Veronika** or **Tigrisat** satellites FSK data;
- Configure the sound modems audio setup using the already installed VACs. This setup is under Settings => Devices tab. Despite the fact that those modems can be used to TX and RX, the sample picture below assumes that the use is RX only;
- Select **FSK G3RUH 9600bd** in one of the drop down menus and adjust **DCD slider** up to the point that the "led" blinks a little bit.



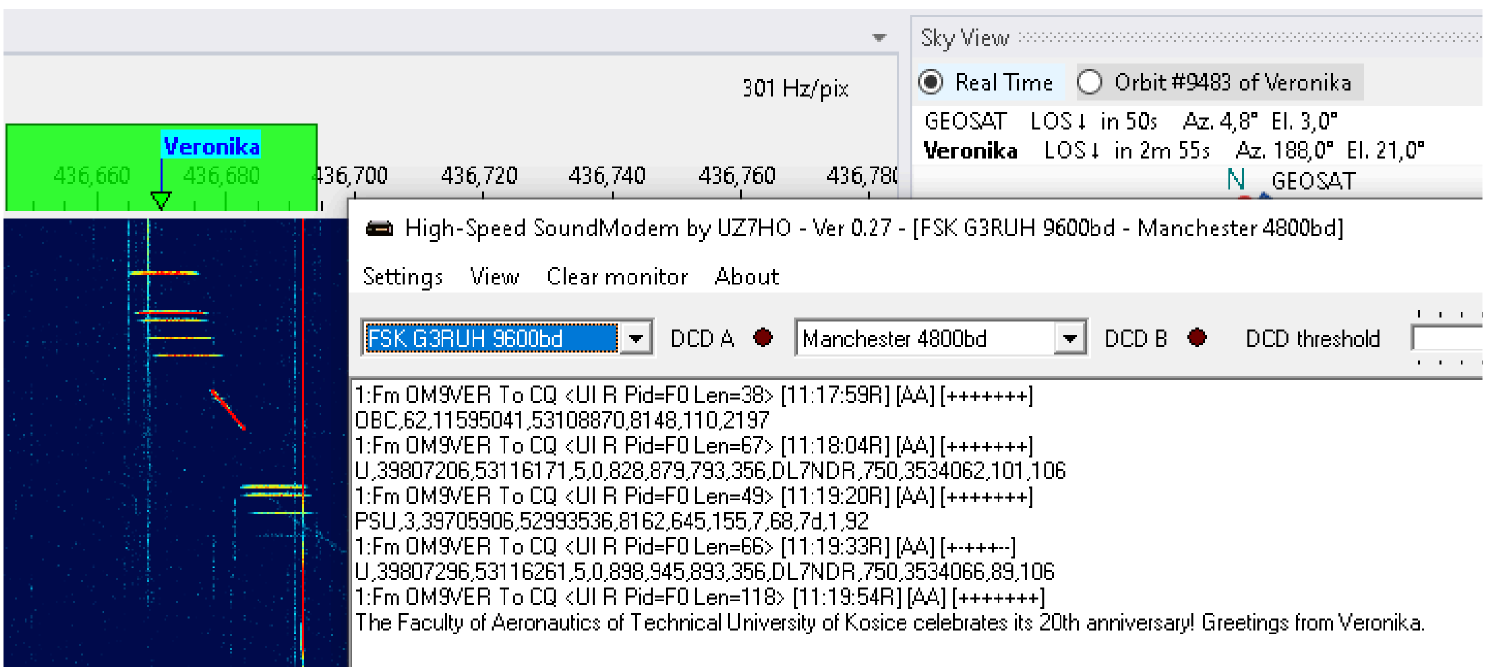
Configuring SkyRoof

In SkyRoof:

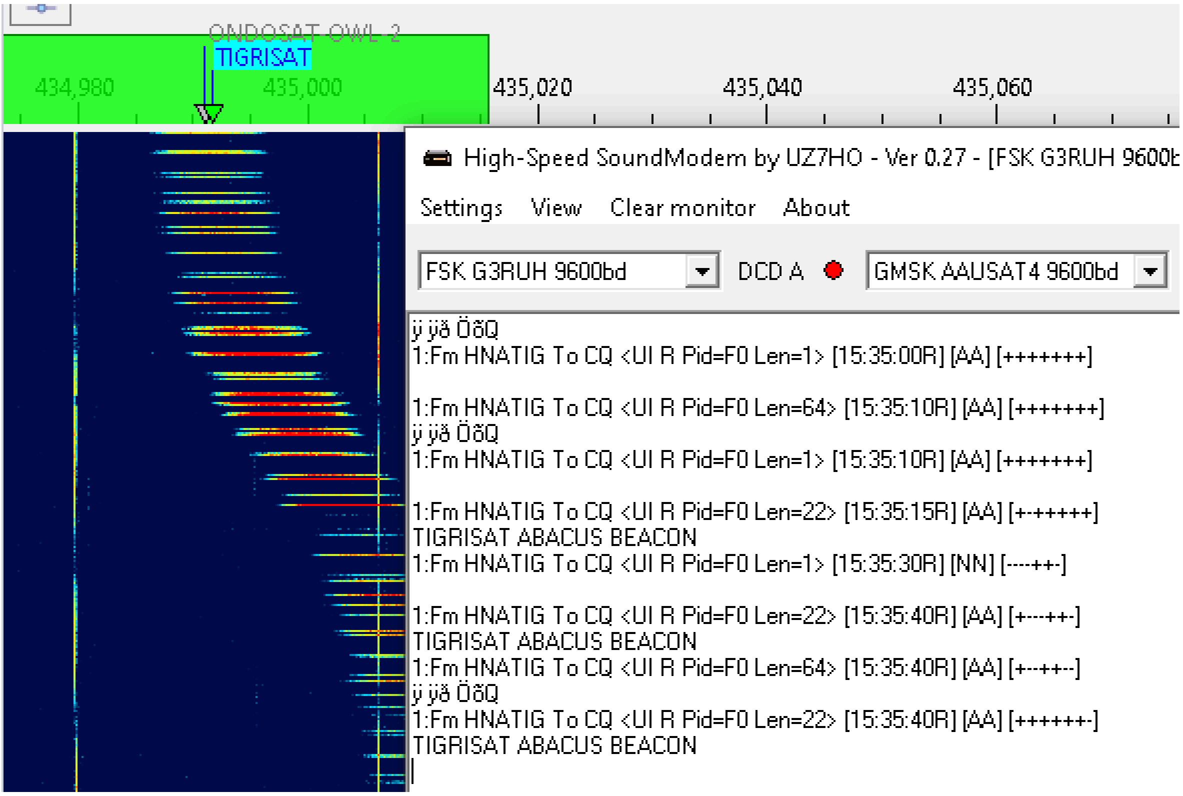
- select the Veronika satellite. If it is not in the current group, add it using the Satellites and Groups dialog;
- Select **GFSK9K6 AX25** transmitter on the UHF band from the list of transmitters;
- Select **FM_D** downlink mode in the drop-down list on the toolbar;
- in the **Output Stream** section of the Settings window:
- select **Audio to VAC**;
- set Gain, dB to 0;
- select the VAC in the list of audio devices;
- click on the Output Stream label on the status bar to enable the output.

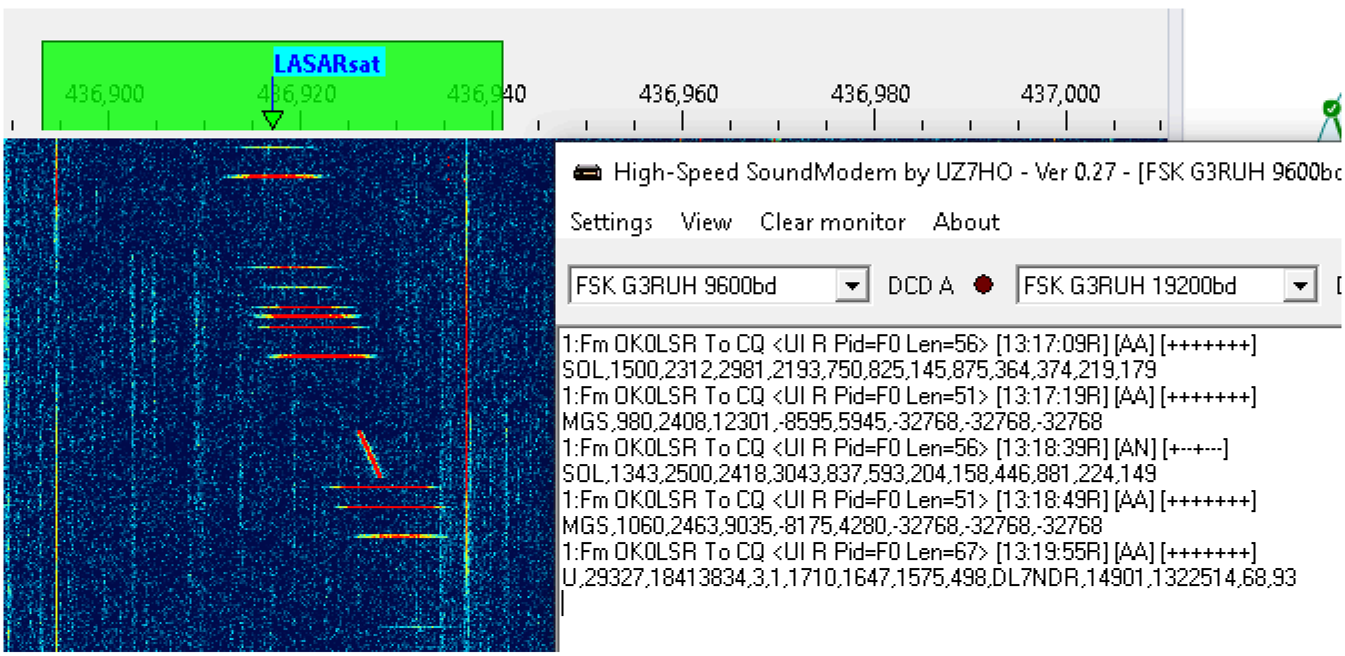
Receiving Data

When the selected satellite raises above the horizon, the UZ7HO FSK Modem will start decoding and produce output like this:



For Tigrisat, the decoded telemetry should be like the picture below:


ONDOSAT OWL-2
TIGRISAT
434,980
435,000
435,020
435,040
435,060
High-Speed SoundModem by UZ7HO - Ver 0.27 - [FSK G3RUH 9600b
Settings View Clear monitor About
FSK G3RUH 9600bd
DCD A
GMSK AAUSAT4 9600bd
ÿ ÿä ÖõQ
1:Fm HNATIG To CQ <UI R Pid=F0 Len=1> [15:35:00R] [AA] [+++++++]
1:Fm HNATIG To CQ <UI R Pid=F0 Len=64> [15:35:10R] [AA] [+++++++]
ÿ ÿä ÖõQ
1:Fm HNATIG To CQ <UI R Pid=F0 Len=1> [15:35:10R] [AA] [+++++++]
1:Fm HNATIG To CQ <UI R Pid=F0 Len=22> [15:35:15R] [AA] [+-+++++]
TIGRISAT ABACUS BEACON
1:Fm HNATIG To CQ <UI R Pid=F0 Len=1> [15:35:30R] [NN] [----++-]
1:Fm HNATIG To CQ <UI R Pid=F0 Len=22> [15:35:40R] [AA] [+---++-]
TIGRISAT ABACUS BEACON
1:Fm HNATIG To CQ <UI R Pid=F0 Len=64> [15:35:40R] [AA] [+--++--]
ÿ ÿä ÖõQ
1:Fm HNATIG To CQ <UI R Pid=F0 Len=22> [15:35:40R] [AA] [++++++-]
TIGRISAT ABACUS BEACON


LASARsat
436,900
436,920
436,940
436,960
436,980
437,000
High-Speed SoundModem by UZ7HO - Ver 0.27 - [FSK G3RUH 9600bd
Settings View Clear monitor About
FSK G3RUH 9600bd
DCD A
FSK G3RUH 19200bd
1:Fm OK0LSR To CQ <UI R Pid=F0 Len=56> [13:17:09R] [AA] [+++++++]
SOL,1500,2312,2981,2193,750,825,145,875,364,374,219,179
1:Fm OK0LSR To CQ <UI R Pid=F0 Len=51> [13:17:19R] [AA] [+++++++]
MGS,980,2408,12301,-8595,5945,-32768,-32768,-32768
1:Fm OK0LSR To CQ <UI R Pid=F0 Len=56> [13:18:39R] [AN] [+--+---]
SOL,1343,2500,2418,3043,837,593,204,158,446,881,224,149
1:Fm OK0LSR To CQ <UI R Pid=F0 Len=51> [13:18:49R] [AA] [+++++++]
MGS,1060,2463,9035,-8175,4280,-32768,-32768,-32768
1:Fm OK0LSR To CQ <UI R Pid=F0 Len=67> [13:19:55R] [AA] [+++++++]
U,29327,18413834,3,1,1710,1647,1575,498,DL7NDR,14901,1322514,68,93

Satellite Data

Data Sources

SkyRoof obtains satellite date from several sources:

- SatNOGS DB is the main source of satellite data. It is a frequently updated, crowd-sourced dataset that contains detailed information about all satellites transmitting in the Ham bands;

- JE9PEL Satellite List is another dataset with information about the satellites, maintained by Mineo Wakita JE9PEL, that, in particular, includes the callsigns of the satellites. SkyRoof also mines the JE9PEL mode descriptions to fill in the signal parameters that SatNOGS leaves blank.

- LoTW - The ARRL LoTW service accepts satellite QSO only if the satellite abbreviation is one of those published on their web site. These abbreviations are stored in a file in the Data folder, you can view them in the Satellite Details window.

- AMSAT Live OSCAR Satellite Status Page accepts satellite observations with their own satellite abbreviations, these abbreviations are stored in a file in the Data folder.

Signal Parameters

To decode a satellite's telemetry, SkyRoof needs the **modulation**, **baud rate**, and **framing** of its downlink. These are resolved for each transmitter from several sources, in order of priority:

1. your manual overrides (see below);
2. the gr-satellites database;
3. the **JE9PEL** satellite list;
4. the **SatNOGS DB** transmitter description.

The first source that specifies a given parameter wins, so a higher-priority source fills in only what the lower-priority ones leave unknown. The resolved values appear in the mouse tooltip of the transmitter on the Frequency Scale.

Overriding Signal Parameters

When the automatic sources are wrong or incomplete, you can correct them in the **transmitters-override.json** file in the Data folder. Each entry is keyed by the transmitter UUID and lists only the fields to change, for example:

```
{
  "FdxJrwmqFrJnP3sd96Bip8": {
    "satellite": "SITRO-AIS-56", "norad": 59778,
    "modulation": "GMSK", "baudrate": 2400, "framing": "USP"
  }
}
```

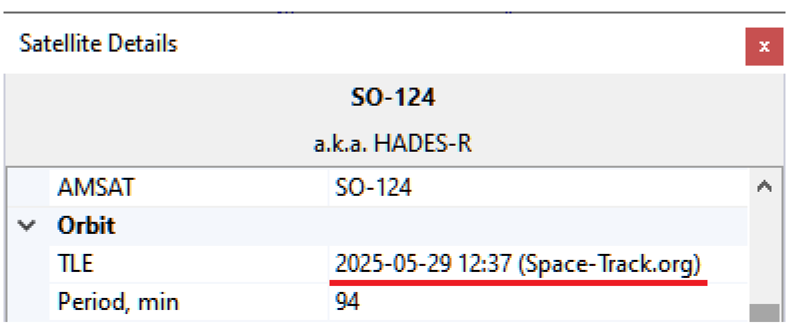
SkyRoof ships a default copy of this file and refreshes it as new corrections are published. To keep your own edits from being overwritten, add `"read_only": true` to the entry — SkyRoof then never replaces it. Entries you add that are not in the shipped file are always kept.

The telemetry definition files in the **TelemetryRegistry** folder work the same way: add `"readOnly": true` (camelCase, to match those files' key style) to a definition to keep your edits when SkyRoof updates its bundled definitions.

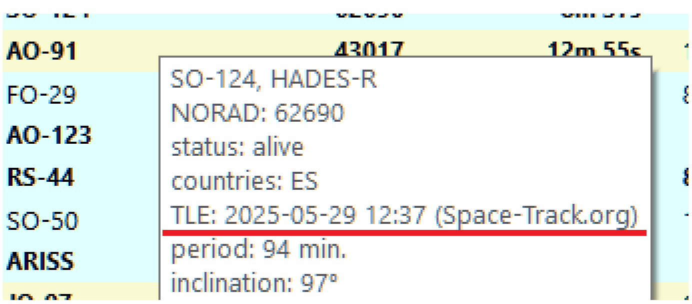
TLE

The satellite orbit elements (TLE data) are downloaded from **SatNOGS DB**.

SatNOGS obtains these data from different sources and makes the latest and most reliable data available on their web site. The source of TLE and its creation time are shown in the Satellite Details window or panel:



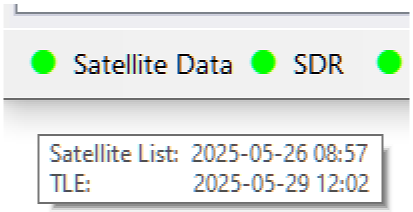
and in the mouse tooltip of the satellite:



Automatic Updates

SkyRoof automatically downloads the satellite list every 7 days, and TLE data every 24 hours.

The mouse tooltip of the Satellite Data label on the status bar shows the last download time:



The light next to the label turns yellow if the satellite data are not up to date.

Manual Updates

In addition to automatic downloads, the data may be manually downloaded at any time using the **Tools / Download All Satellite Data** and **Tools / Download Only TLE** menu commands.

Loading TLE from File

If your system is not connected to the Internet, you can load TLE data from a local file using the **Tools / Load TLE from File** menu command. Two TLE formats are supported:

- **.json** - TLE data from the SatNOGS web site, recommended (download);
- **.txt** - 3-line TLE data in a text file, available from many sources, e.g. CelesTrak (download).

Note that TLE import cannot add new satellites, it only loads orbital elements for the satellites already in the database.

AMSAT Satellite Status

AMSAT Live OSCAR Satellite Status Page is a crowd-sourced, real-time Ham satellite status page.

Posting Status Data

You can post your satellite status observations the the AMSAT web site either by filling the submission form on their site, or using the right-click menu of the satellite labels on the Frequency Scale. A valid Ham callsign must be entered in the Settings window for this function to work.

Downloading Status Data

Set the **Amsat Satellite Status / Enable** option in the Settings window to `true` to enable automatic downloads of the satellite status information from the AMSAT web site. The statuses are shown on the Current Group panel, the green and red icons represent the active and inactive status respectively.

Satellite status data are downloaded once an hour. You can manually download it at any time using the **Tools / Download AMSAT Statuses** menu command.

Doppler Tracking

The SGP4 algorithm used in SkyRoof to compute the Doppler offset produces very accurate results for the LEO satellites, typically within tens of Hertz, if it receives accurate input data. For best results, ensure that the following conditions are met.

Home Location

Make sure that your grid square is accurate. Correct it in the Settings window if necessary.

System Time

Your system clock should be accurate to a second. Get one of those little programs that run in the system tray and periodically synchronize your clock with the time servers on the Internet. NetTime is one such program.

PPM correction

Find the PPM correction factor of your SDR radio as described in the Calibrating PPM Correction section and enter it in SDR settings.

TLE Data

SkyRoof downloads the TLE data automatically every 24 hours. Some sources claim that TLE may be updated once a week, but that would not be enough for accurate Doppler tracking, especially for the satellites that perform frequent orbit corrections. When in doubt, download TLE manually as described in the Satellite Data section.

Transmitter Frequency Correction

Most satellite transmitters transmit on the frequencies that differ from the nominal values by up to a few kHz. A one-time correction described in the Frequency Control and Frequency Scale sections eliminates this error.

Data Folder

SkyRoof keeps all of its data in the **data folder**.

- Click on **Help / Data Folder** in the main menu to open this folder in File Explorer.
- To open the folder when the program is not running, type this in File Explorer:

```
%appdata%\Afreet\Products\SkyRoof`
```

Data Files

- **Settings.json** - this is the file where all user-defined settings are stored;
- **amsat_sat_names.json** - satellite names used on AMSAT Live OSCAR Satellite Status Page. The Frequency Scale section explains how to post your observations to this page;
- **lotw_sat_names.json** - the list of satellite abbreviations accepted by LoTW;
- **Satellites.json** - the satellite database compiled from the downloaded data;
- **transmitters-override.json** - hand-curated corrections to the transmitter signal parameters. Ships with a default set that SkyRoof keeps up to date; mark an entry `"read_only": true` to protect your own edits;
- **cat_info.json** - lists the CAT capabilities of a generic simplex radio;
- **wsjtx_wisdom.dat** - optimal FFT transform settings found by automatic testing.

Folders

- **Logs** - contains the log files with error messages and other information;
- **Adif** - QSO records stored in the ADIF format;
- **Downloads** - a copy of the satellite data downloaded from various sources, kept for troubleshooting;
- **Palettes** - definition of the color palettes used by the waterfall display. Add your own palette as a text file with "html" color codes. Pick the color codes at htmlcolorcodes.com;
- **sat_images** - cached satellite thumbnails shown by the Satellite Photo widget. Delete this folder to force re-download;
- **Recordings** - the default location for the audio and I/Q files saved by the Recorder panel;

- **FT4** - decoded messages saved by the FT4 Console panel, when the **Save to File** option is enabled in the settings;
- **TelemetryDecodes** - decoded telemetry frames saved by the Telemetry panel, when the **Save to File** option is enabled in the telemetry settings. One file is created per day;
- **SstvImages** - SSTV images decoded by the Telemetry panel, saved automatically as PNG files with a JSON metadata sidecar;
- **SsdvImages** - images received as SSDV packets or as raw JPEG fragments, decoded by the Telemetry panel, saved automatically as JPEG files with a JSON metadata sidecar. The sidecar also holds the received packets, which is what lets a picture be combined with its earlier receptions;
- **Codec2Voice** - Codec2 voice messages decoded by the Telemetry panel, saved automatically as WAV files with a JSON metadata sidecar;
- **TelemetryRegistry** - the definitions used to decode raw telemetry frames into named values in the Telemetry panel.

Smart Antenna Rotation

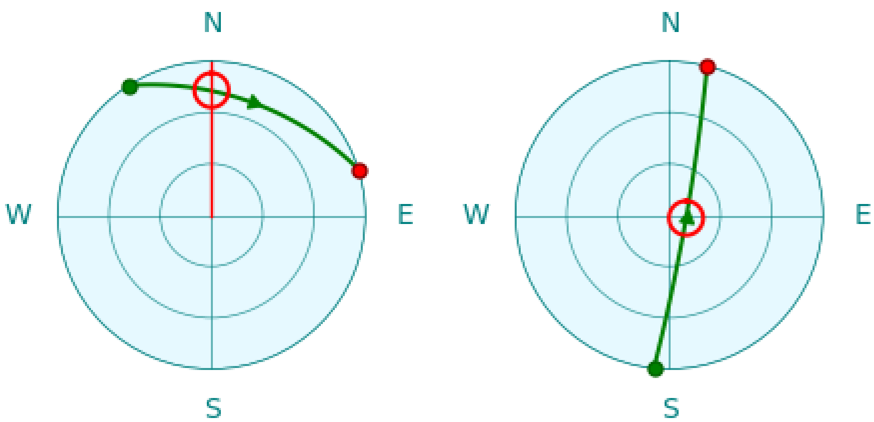
SkyRoof uses a new antenna rotation algorithm, based on dynamic programming, that globally optimizes the tracking of the satellite pass and ensures the minimum possible communication downtime. The algorithm makes full use of the extended range of azimuth and elevation available in the modern rotators.

Limitations of the Simple Approach

The satellite pass prediction algorithm computes the satellite azimuth in the range of **0° to 360°** and elevation in the range of **0° to 90°**. A simple antenna rotator just sends these azimuth and elevation values to the controller. This works most of the time, but there are cases when this requires the antenna to rotate by 180° or 360°. While the antenna is making this full circle, or half-circle, it does not point at the satellite, causing a significant downtime in the communication. The charts below show two such cases.

In the first chart the path crosses the 360° azimuth line, and the antenna needs to make a full circle counter clockwise to get to 0°.

In the second chart, when the satellite is at its highest elevation point, its azimuth quickly changes from South to North, by about 180°. This rotation may take up to a minute, causing a significant downtime.



Rotator Capabilities

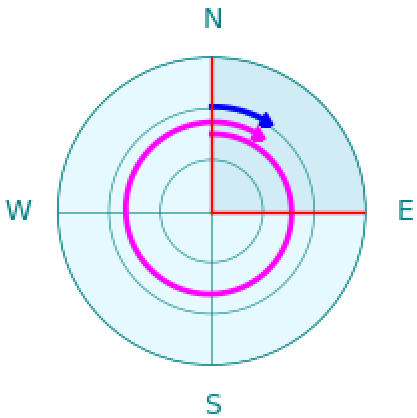
Many antenna rotators have an extended range of azimuth and elevation that goes beyond the **0° to 360°** for the azimuth and **0° to 90°** for the elevation. This allows the controller to point the antenna in the same direction in more than one way. We will use Yaesu G-5500DC, one of the most popular satellite antenna rotators, as an example. This rotator has the following rotation range:

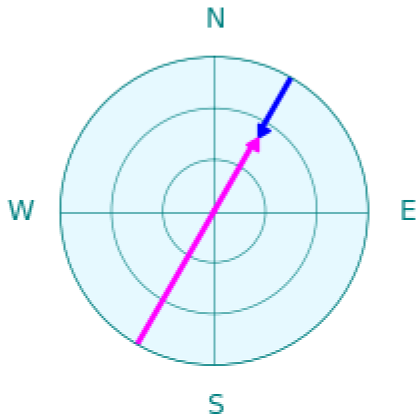
- **0° to 450°** for the azimuth;

- **0° to 180°** for the elevation.

With this azimuth range, there is a 90° segment where every point may be reached in two ways. The first chart below shows that Azimuth=30° and Azimuth=390° point in the same direction.

With this elevation range, every point may be reached in two ways, with elevation < 90° and > 90°, as shown in the second chart.





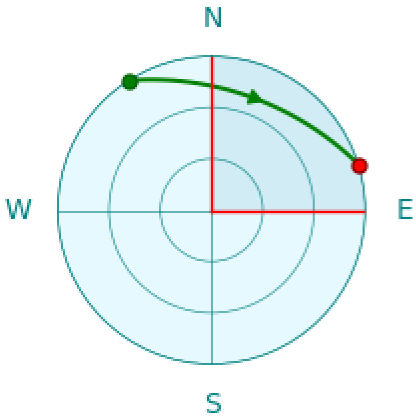
Note also that the rotation speed of G-5500DC is different in the horizontal and vertical planes. Here is what appears in its specifications:

```
Rotation time (approx.):
  Elevation (180°): 65 sec.
  Azimuth+(360°): 60 sec.
```

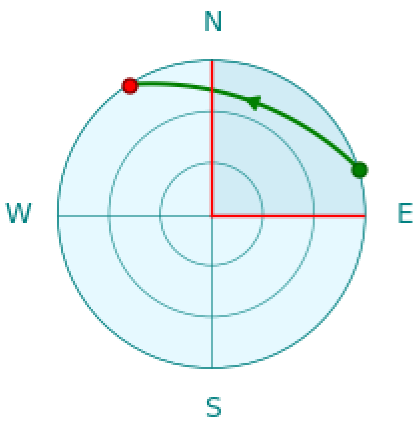
The SkyRoof algorithm assumes that vertical rotation is two times slower than horizontal.

Algorithm

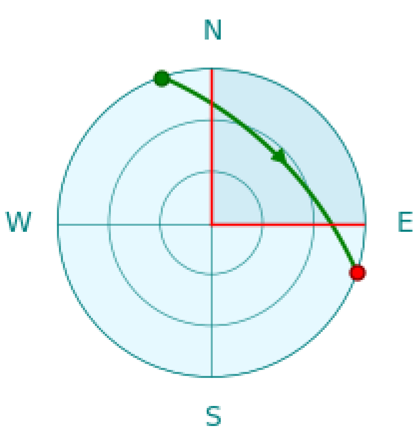
The SkyRoof algorithm that finds the optimum tracking strategy for the given satellite path is not rule-based, it uses the global optimizaton approach to find the optimal solution. The examples below show how it handles certain special cases.



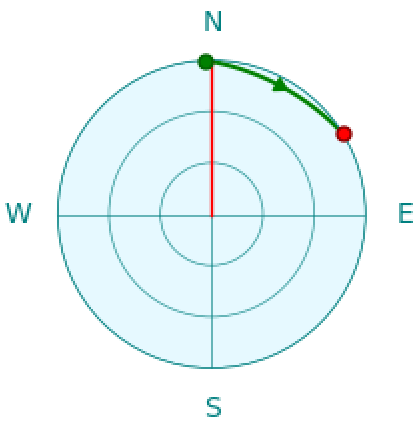
This is an easy case. The azimuth changes from 330° to 80°, so the rotation starts at 330° and, when the azimuth crosses 360°, continues in the same direction, up to 440°, which is the same direction as 80°.



The same path, but in the opposite direction. The simple approach described above, that is used in many simple antenna rotation programs, no longer works. If we start at Azimuth=80°, we hit the 0° limit and cannot continue. The algorithm in SkyRoof looks at the whole pass and chooses to start at 440° instead of 80° so that the pass is tracked without interruption.

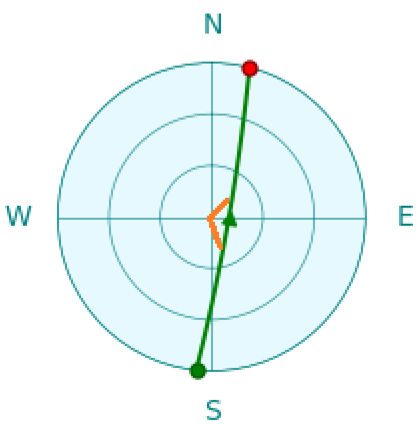


Now the path crosses both 0° and 450° lines. In this case the 90° overlap is not enough to ensure uninterrupted tracking. The dynamic algorithm still finds a solution: it uses the elevations > 90°. Tracking starts at (Az=170°, El=180°) and ends at (Az=280°, El=180°). Again, all azimuths are in the 0° - 360° range, ensuring uninterrupted tracking.

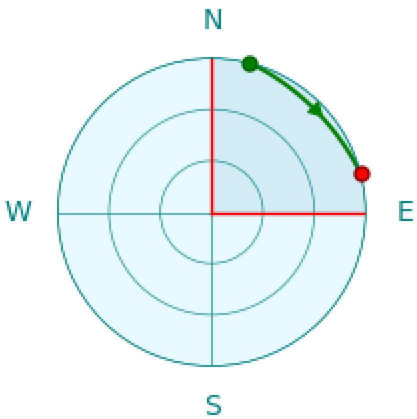


SkyRoof has a parameter in the rotator control settings that determines the steps in which the antenna is rotated, to prevent frequent starts and stops of the motors. Indeed, if the antenna beam width is, say, 60°, there is no point in rotating it in 1° steps. The default step size is 5°.

The path in the chart above crosses 360°, but it extends beyond the red line only by a couple of degrees. If the overshoot does not exceed the rotation step, the algorithm is smart enough to start the path at 0° so that 360° rotation is not required. This works even if the rotator does not have an extended azimuth range.



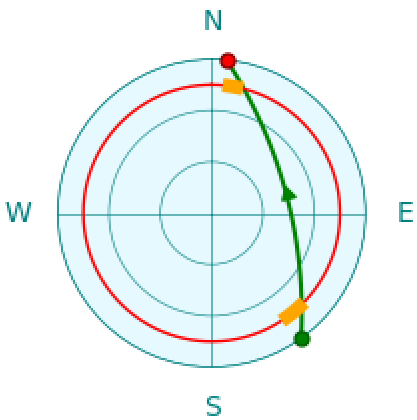
This is a high elevation pass. As we have seen before, it requires the antenna to make a fast rotation by about 180° in azimuth. The rotators that support elevations of 0° to 180° can use elevation > 90° for the second half of the path, thus eliminating the 180° azimuth change. However, to switch from the elevation < 90° to > 90°, the antenna must deviate from the path a little bit and pass through the zenith (90°), as shown with the orange lines in the chart above. Dynamic programming plans this deviation in an optimal way.



In many cases there is more than one way of tracking the same path. For example, the path above may be tracked in three different ways:

- (Az=20°, El=0°) -> (Az=70°, El=0°)
- (Az=380°, El=0°) -> (Az=430°, El=0°)
- (Az=200°, El=180°) -> (Az=250°, El=180°)

When the algorithm finds multiple optimum paths, it chooses the one which has the starting point closest to the current antenna direction.



Some antenna rotators have a turning range that does not cover the whole upper hemisphere. The chart above shows the case where the rotator has an elevation range of 15° to 90°. This could be a limitation of the hardware or just an obstructed horizon that makes lower elevation angles useless. The algorithm obeys these restrictions, it does not go beyond the red line. It follows the path shown in the chart with orange lines.

Frequently Asked Questions

Q: I downloaded SkyRoof, and my virus scanner shows an infection. Is it real?

A: Most likely, it is a false detection. A false detection is a **bug in the virus scanner**, not malicious code in SkyRoof, and the scanner is what needs to be corrected.

Modern scanners do not rely on signature detection alone. Alongside it they run machine-learning classifiers, and these are the usual source of false alarms. A signature detection means the scanner actually found known malicious code inside the file. A machine-learning verdict means nothing of the sort: it is a guess based on how the file looks from the outside and on how many people have downloaded it before you. A newly released installer that the scanner has never seen is unfamiliar by definition, and it can be flagged for that reason alone. In Microsoft Defender such verdicts are easy to recognize, because the detection name ends in `!ml`, as in `Trojan:Script/Wacatac.B!ml`.

The right way to check any download, from any author, is to scan it with several independent engines at once. There are several online virus scanning services that you can use to check the link. VirusTotal and Hybrid Analysis are just two examples.

Copy the download link and paste it in the virus scanner page. The scanner will download the file, test it with multiple antivirus programs, and show you the results.

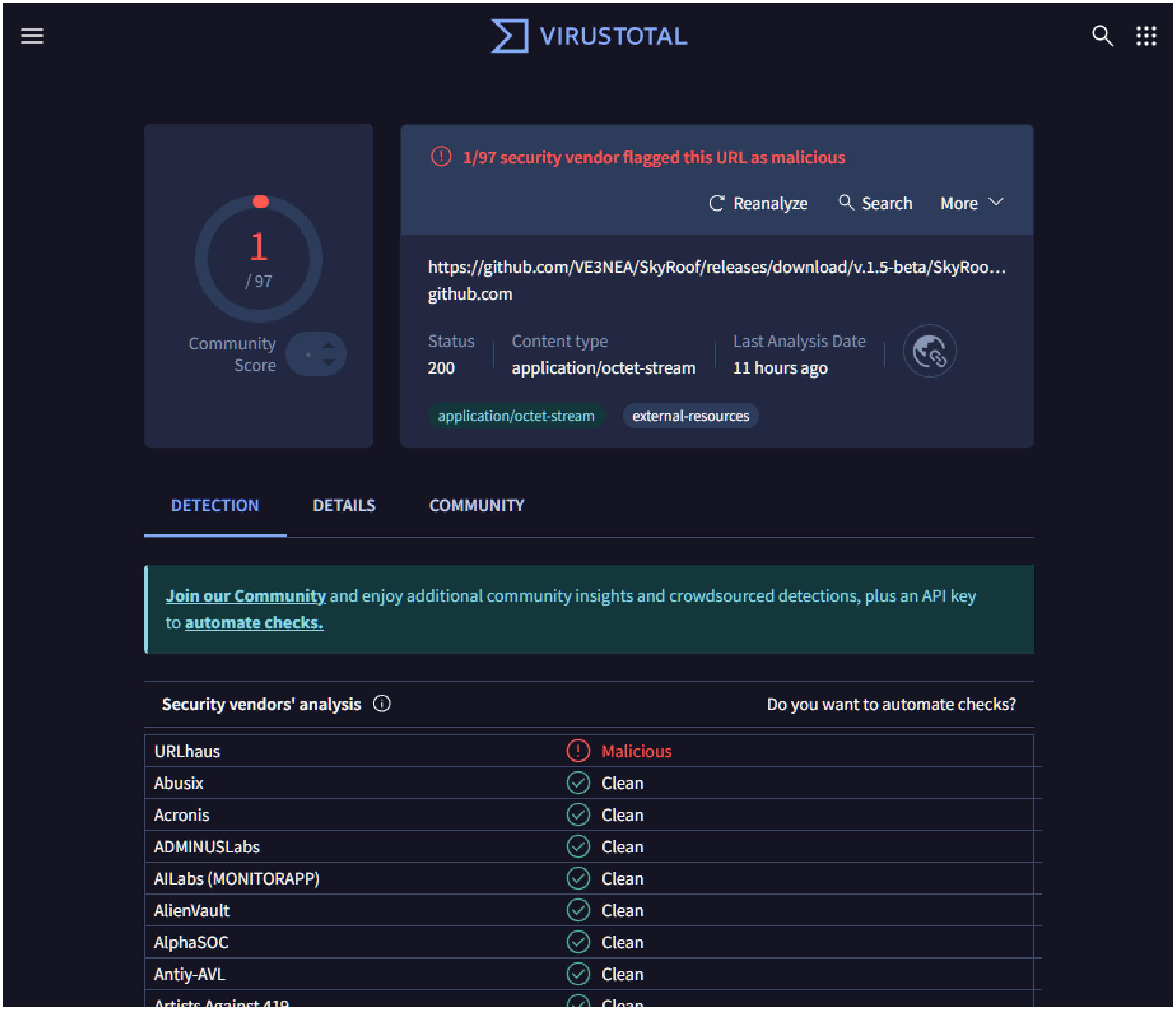
In the screenshot below the download link of SkyRoof 1.5 beta was tested with VirusTotal, and all virus scanners, except one, agreed that the file was clean. When you see something like this, you know that it was a false alarm.

If the file is clean, you can add it to the exception list of your virus scanner and safely install it. For Windows Defender follow these instructions, for other anti-virus products follow the instructions in their documentation.

The only real fix is in the scanner itself, so please report the false detection to its vendor. For Microsoft Defender, submit the file for analysis using this link:

https://www.microsoft.com/en-us/wdsi/filesubmission

On our side we do what we can to make a false detection less likely, but those measures only reduce the odds of a wrong guess. They are a work-around, not a fix.



Q: The right part of the SkyRoof toolbar does not fit in the screen, even though the screen resolution is 1980x1280.

A: This happens because your **text size** setting in Windows is too high. For example, if it is set to 200%, the effective screen width is only `1900 / 2 = 950 pixels`. To fix this, right-click on the Desktop, click on **Display Settings** and set the text size to a lower value.

Q: How can I run two instances of SkyRoof?

A: By default, only one instance of SkyRoof can run at any time, but there is a work around. Make a copy of the SkyRoof.exe file in the same folder, but with a different name, e.g., SkyRoof_2.exe. Each exe will have its own settings, its own data folder, and will run independently of the other instance.

Q: I found an error in the satellite database: one of the satellite transmitters is marked as active, while it has been inactive for years. How can I correct the error?

A: SkyRoof uses satellite data from the SatNOGS database. This database is crowd-sourced, so anyone can suggest changes. If you find an error in a satellite record (shown in the Satellite Details and Satellite Transmitters panels), click on the "SatNOGS" link in the Satellite Details panel to open the SatNOGS web site, then click on Transmitters, and select Edit from the drop-down menu. You may need to create an account with them (free) in order to submit changes.

Q: How can I add a satellite that is currently not listed in SkyRoof?

A: The only way to add a satellite to SkyRoof is to add it to the SatNOGS database. Here is how you can do this:

- navigate to the SatNOGS DB web site;
- click on **Sign Up / Log In** in the top right corner of the page;
- create a free account, or log in to an existing one;
- click on **All Satellites**;
- in the **Actions** drop-down list select **Suggest New Satellite*.

See this Wiki page for details.

Q: My video card does not have an OpenGL 3.3 driver. Is there a way to use a software-only implementation of OpenGL?

A: Some users have had success with the OpenGL 3.3 functionality implemented in the software, though this approach is very CPU-intensive. You can try this or this solution. Download the 64-bit version of the library and extract all DLL's from the archive to the SkyRoof folder.

Q: How do I prevent CTCSS from turning off when using SkyRoof with FM satellites and an Icom radio?

A: If you're experiencing an issue where the CTCSS tone turns off when transmitting or changing frequencies with SkyRoof and an Icom radio, the solution is to disable the "Auto Repeater" setting in the radio. This setting causes the radio to automatically disable repeater-related features like CTCSS when the frequency is changed, assuming you're no longer on a repeater. Disabling "Auto Repeater" prevents this behavior and keeps CTCSS enabled during frequency changes. This workaround was highlighted in a helpful YouTube video.

Q: What are all those offset settings for?

A:

- **Downlink Manual Correction:** the frequencies of most satellite transmitters slightly differ from their published values. The purpose of this setting is to compensate for the difference, you set it once for each satellite and never change it again;
- **RIT:** just as in any HF radio, it is used to quickly switch the receiver between two frequencies close to each other. A typical use case is when someone answers your CQ slightly off the frequency and you want to tune to it without changing your main operating frequency;
- **Uplink Manual Correction:** this setting allows you to precisely align your own signal received back from the satellite with your receiver frequency. It needs to be set only once per satellite, and the offset is usually very small;
- **Transponder Offset:** the position of your operating frequency within the transponder passband. This offset is included in the Corrected Downlink Frequency shown on the toolbar, the actual value of the offset is shown only on the mouse tooltip of the frequency display. There are many ways to tune within the transponder passband, such as click on the desired frequency, drag the green passband rectangle, spin the mouse wheel over the rectangle, etc. These methods work only when the transponder is selected in the Transmitter drop-down box.
- **Doppler correction:** this correction is computed and applied automatically, it keeps your RX and TX signals in sync when you tune through the transponder segment.

Q: How to troubleshoot the rotator control?

A:

1. Set up and enable rotator control as described in Setting Up Rotator Control.
    - if the Rotator light on the status bar is red, the problem is with the SkyRoof to rotctld.exe connection;
    - if the light is green, then the SkyRoof to rotcltd.exe connection is fine, the problem is with the rotctld.exe to rotor connection.
2. Add the `-vvvvv` option to the command line that starts rotctld.exe, and restart it. This option tells rotctld.exe to print detailed information to the console window (the black

window that opens when rotctld.exe is started).

3. Read the rotctld.exe documentation that will help you to understand information in the console window.

4. Possible causes of SkyRoof to rotctld.exe connection problems:

   - rotctld.exe failed to start;
   - SkyRoof is configured to use a port number different from the rotctld.exe listening port.

5. Possible causes of the rotctld.exe to rotator connection error problems:

   - the rotator is not connected to the computer or not powered on;
   - the COM port number or Baud rate specified on the command line is wrong - then the COM port fails to open;
   - the rotator model specified on the command line is wrong - then the commands sent to the rotor are rejected or have no effect.

6. Make rotator control work without SkyRoof first: use **rotctl.exe**, an interactive version of rotctld.exe, and send commands to the rotator manually, by typing them in the console window. Rotator control uses three commands, "P", "p" and "S" (set position, read position, stop rotation). Once you make rotctl.exe work with the rotator, run rotctld.exe with the same command line parameters.